

How Many Fish Can a Freshwater Aquarium Actually Hold?

Stocking capacity as a multi-constraint problem: biomass, feed input, oxygen, biofiltration, footprint, territory, social group size, and maintenance reality

Secondary research question: the apparent conflict between “do not add too many fish at once” and “your schooling species needs a larger group.”

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Scope note. This document deliberately does not begin from the assumption that the inch-per-gallon rule is always wrong, that stocking calculators are useless, that bigger filters license more fish, or that larger schools are always better. Each of those claims is examined against available evidence and labelled by evidence strength. Where the evidence does not support a confident answer, the document says so.

A note on the aquaculture literature. Much of the strongest quantitative work on stocking density, oxygen demand, ammonia production and biofilter loading comes from commercial aquaculture and from laboratory fish facilities. Those systems optimise for growth rate, survival to harvest, and cost. Home aquariums optimise for long-term individual welfare and for keeper enjoyment over years. Numbers do not transfer directly between the two. The document flags this repeatedly rather than once.

How to Read This Document

Claims throughout this document carry an evidence label. The labels are not a ranking of usefulness; a useful rule of thumb can be more practically valuable to a beginner than a precise laboratory finding. The labels describe *what kind of thing* the claim is, so that downstream use does not accidentally present a hobby convention as a measured result.

Label	Meaning	Example of the kind of claim it covers
ESTABLISHED SCIENTIFIC PRINCIPLE	Well-replicated physics, chemistry or physiology. Not seriously disputed.	Oxygen solubility in water falls as temperature rises.
STRONG EVIDENCE	Multiple independent peer-reviewed studies point the same way, with known limits.	Resting metabolic rate scales with body mass to a power near 0.79 in teleost fish.
REASONABLE INFERENCE	A defensible conclusion drawn from established principles applied to aquarium conditions, but not directly tested in home aquaria.	A heavily stocked tank has less oxygen reserve during a summer power outage.
AQUACULTURE PRACTICE	Standard operating practice in commercial fish production. Optimised for yield, not for individual welfare over a long life.	Ammonia production is estimated from feed input rather than fish count.
ANIMAL-WELFARE PRINCIPLE	A normative position from welfare science about what animals should be provided, supported by behavioural evidence.	Social species should be housed in groups that allow normal social behaviour.
COMMON AQUARIUM PRACTICE	What experienced hobbyists and retailers widely do. Often sensible, rarely formally tested.	Adding a few fish at a time to an established tank.
USEFUL RULE OF THUMB	A simplification that is wrong in detail but produces acceptable outcomes inside a narrow range of conditions.	One inch of small slender fish per gallon, for small community fish only.
MANUFACTURER RECOMMENDATION	A figure published by a company about its own product. Not independently standardised.	"Rated for aquariums up to 50 gallons."
ANECDOTAL OBSERVATION	Reported experience from keepers. Valuable for identifying questions; not proof.	"My heavily stocked tank has been fine for three years."
CONTESTED CLAIM	Credible people and sometimes credible studies disagree, or the evidence points different ways in different species.	Whether crowding reduces aggression in African rift lake cichlids.
UNSUPPORTED AQUARIUM FOLKLORE	Widely repeated, no traceable evidence, and often contradicted by basic principles.	"Bottom dwellers do not count toward stocking."

Table 0.1 — Evidence classification key used throughout this document.

Terminology warning for downstream use

Several words in this subject area are used differently by hobbyists, by welfare scientists and by aquaculture engineers. In particular: **stocking density** means fish per unit volume to an aquaculturist but usually means 'how full the tank looks' to a hobbyist; **carrying capacity** means a sustainable equilibrium in ecology but is often used in the hobby to mean 'the most fish that will survive'; and **schooling** and **shoaling** are near-synonyms in casual hobby speech but describe different behaviours in the literature. Section 1 defines each term explicitly.

Executive Research Summary

The short version of what the evidence supports, ahead of the detail. Every claim here is expanded, sourced and qualified later in the document.

1. Stocking capacity is not one number, and not one question

The single most consistent finding across the literature is that “how many fish fit” is at least six separate questions wearing one coat. A tank can have ample biological filtration and inadequate swimming room. It can have adequate swimming room and marginal night-time oxygen. It can have perfect water chemistry and a social structure that keeps one fish permanently cornered. It can be fine in every dimension except the water-change schedule the keeper can realistically sustain. The correct mental model is a set of independent ceilings, where the lowest ceiling governs. Replacing one bad formula with a better formula does not solve this, because the binding constraint changes from tank to tank and from species to species.

EVIDENCE: REASONABLE INFERENCE, CONSISTENTLY SUPPORTED ACROSS THE AQUACULTURE, WELFARE AND ETHOLOGY LITERATURES

2. Fish length is a poor proxy; mass and feed input are better

Mass rises roughly with the cube of length. Froese's meta-analysis of 3,929 weight-length relationships across 1,773 fish species found a median exponent of 3.03 in the relationship $W = aL^b$, confirming that mass scales close to the cube of length for most fishes (Froese 2006). A fish twice as long is therefore roughly eight times the mass, not twice. Metabolic rate then scales sub-linearly with mass: Clarke and Johnston's analysis of 69 teleost species found a mean scaling exponent near 0.79 (Clarke & Johnston 1999). Length-based rules compound both errors, which is why they fail hardest at the extremes of body depth and adult size.

EVIDENCE: STRONG EVIDENCE

3. Waste load tracks feed input more closely than fish count

Commercial aquaculture does not estimate ammonia production from the number of fish. It estimates it from the amount and protein content of the feed. A widely used figure derived from Ebeling and colleagues is that one kilogram of 32 percent crude protein feed generates roughly 30 grams of total ammonia nitrogen entering the water. Nitrogen enters a closed system almost entirely through the food container, and leaves through fish growth, water changes, plant uptake and denitrification. This reframes the hobby question usefully: ten lightly fed small fish and ten heavily fed small fish are not the same stocking level in any meaningful biological sense.

EVIDENCE: AQUACULTURE PRACTICE, RESTING ON ESTABLISHED PRINCIPLE

4. A mature biofilter has reserve capacity, but not unlimited reserve, and no universal adjustment time

Nitrifying communities are substrate-limited. Standing biomass is roughly proportional to the ammonia supply they have been receiving, so a filter that has been running under a light load carries a small nitrifier population. Small increases in load are usually absorbed within existing capacity because nitrifiers operate below saturation. Large increases require population growth, and ammonia-oxidising organisms grow slowly relative to heterotrophs. The literature does not support a single universal adjustment interval such as ‘two weeks.’ Reported timeframes vary with temperature, pH, alkalinity, surface area, oxygen and the size of the step change. Anyone stating a precise universal figure is stating a convention, not a measurement.

EVIDENCE: ESTABLISHED PRINCIPLE PLUS CONTESTED DETAIL

5. Group size genuinely affects welfare, and the effect is species-specific

The strongest directly relevant study for home aquaria is Saxby and colleagues (2010), who tested group size in four commonly kept ornamental species. Neon tetras and white cloud mountain minnows showed reduced aggression and darting and more shoaling in larger groups, with the largest tested group of ten showing the best composite welfare score. Tiger barbs also scored better in larger groups — and notably, the small tiger barb treatments were terminated early on ethical grounds because aggression escalated immediately. Angelfish showed no clear link between group size and welfare. That single study contains the whole argument in miniature: group size matters, it matters differently by species, and the number six is nowhere in the data.

EVIDENCE: STRONG EVIDENCE FOR THE SPECIES TESTED; DOES NOT GENERALISE TO ALL SPECIES

6. “Add more fish” and “do not add too many at once” are answers to different questions

The apparent contradiction dissolves once the two constraints are separated. “Add more fish” is usually a statement about *final group size* for a social species. “Do not add too many at once” is a statement about *rate of change* in ammonia load. A keeper adding six more neon tetras to a mature, moderately fed 29-gallon tank is making a small proportional change in nitrogen input while making a large improvement in social structure. A keeper adding six more fancy goldfish is doing something entirely different. The size of the biological step is a function of added biomass and added feed, not of the number of animals.

EVIDENCE: REASONABLE INFERENCE FROM ESTABLISHED PRINCIPLES

7. Filters, sumps and plants shift some constraints and not others

A larger filter adds ammonia-processing capacity, mechanical removal and usually circulation. It does not add swimming room, does not create territory, does not extend the tank footprint, and does not reduce the dissolved organic carbon that accumulates between water changes. A sump adds real system water volume, which genuinely buffers chemistry and adds oxygenation, but it adds no display space at all. Plants take up ammonium directly and often preferentially, add oxygen during the photoperiod, and provide cover that changes behaviour — but they respire at night, and dense planting reduces open swimming volume. Each of these lifts a specific ceiling. None of them lifts all the ceilings.

EVIDENCE: MIXED: ESTABLISHED PRINCIPLE FOR THE MECHANISMS, REASONABLE INFERENCE FOR THE AQUARIUM-SCALE MAGNITUDE

8. Zero ammonia and low nitrate do not prove appropriate stocking

Undetectable ammonia and nitrite prove that nitrification is keeping pace with nitrogen input. They say nothing about dissolved oxygen at 4 a.m., about whether the subordinate fish is eating, about whether the tank has a footprint long enough for its most active species, or about what happens when the fish reach adult size. Low nitrate has several possible causes, some of which indicate a healthy system and one of which — heavy plant uptake — can mask a nitrogen load that would otherwise be visible.

EVIDENCE: ESTABLISHED PRINCIPLE

9. The strongest practical consequence of heavy stocking is loss of margin

A heavily stocked tank can be perfectly stable in normal operation and fail quickly when something goes wrong. The same tank has less oxygen reserve during a heatwave, less biofilter reserve after a filter media mishap, less time before ammonia becomes measurable during a power outage, and less tolerance for a missed water change or an overfeeding episode. This is arguably the most under-discussed consequence of high stocking in hobby conversation, because it is invisible until the day it is not.

EVIDENCE: REASONABLE INFERENCE, STRONGLY SUPPORTED BY AQUACULTURE OPERATIONAL PRACTICE

10. What the evidence does not support

The research does not support a replacement universal formula. It does not support six as a biologically established minimum group size for all shoaling species. It does not support a fixed universal waiting period between additions. It does not support treating a filter rating as a livestock allowance. And it does not support the claim that stocking calculators are worthless — they encode useful adult-size and compatibility data, and their real limitation is the set of things they cannot observe, which is discussed at length in Section 21.

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1. Defining Stocking Capacity

The hobby uses roughly a dozen terms as though they were interchangeable. They are related, but each names a different constraint, and confusion between them is the root of most stocking arguments.

Table 1.1 — Core terminology

Term	Definition used in this document	What it actually limits
Stocking level	An informal judgement about how much livestock a tank holds relative to what a keeper considers appropriate. Not a measured quantity.	Nothing directly. It is a summary opinion, which is why two keepers can disagree about it without either being factually wrong.
Stocking density	Livestock per unit of water volume or per unit of tank footprint. In aquaculture, usually expressed as kilograms of fish per cubic metre.	Used as a control variable in production and in laboratory husbandry. In home aquaria it is rarely measured at all.
Carrying capacity	The maximum livestock load a specific system can sustain indefinitely under a specific management regime without progressive deterioration.	The system as a whole. Note the two conditionals: it is specific to the system and to the management regime. Change the water-change schedule and the carrying capacity changes.
Bioload	Hobby term for the total waste-generating burden placed on a system by its livestock and their food.	Nitrogen processing, dissolved organic accumulation, and oxygen demand. Not space and not behaviour.
Biomass	Total wet mass of living animals in the system. The aquaculture unit of account.	Oxygen demand and, indirectly through feeding rate, nitrogen input.
Waste production	The rate at which nitrogenous compounds, faecal solids and dissolved organics enter the water column.	Biofilter demand, dissolved organic carbon load, water-change frequency.
Biological filtration capacity	The rate at which the system's nitrifying community can oxidise ammonia and nitrite, at the current temperature, pH, alkalinity, oxygen level and flow.	Peak tolerable ammonia input rate. Not total fish number, and not long-term water quality.
Oxygen demand	Combined oxygen consumption of fish, invertebrates, nitrifying and heterotrophic bacteria, and plants at night.	The genuine hard ceiling in warm, heavily fed, heavily stocked systems. Frequently the first constraint reached during a heatwave.
Territorial requirement	The physical area or set of defensible resources an individual will attempt to control, which varies with species, sex, maturity and reproductive state.	How many individuals can coexist without chronic conflict. Independent of water chemistry.
Swimming-space requirement	The continuous unobstructed distance and volume a species needs to express normal locomotion.	Behavioural normality and, over time, condition. Usually governed by tank length rather than gallons.
Social-group requirement	The number and composition of conspecifics a species needs in order to express normal social behaviour and avoid chronic social stress.	Welfare. Note that this constraint pushes toward <i>more</i> animals, which is why it collides with bioload thinking.
Maintenance capacity	The water changes, feeding discipline, filter servicing and monitoring the keeper will actually perform, sustained over years.	The real-world ceiling. A stocking plan that requires more maintenance than the keeper will do is not a viable plan, however sound it looks on paper.

Table 1.1 — These twelve terms are related but not interchangeable. Most stocking disagreements are actually disagreements about which of these is being discussed.

1.1 Why these terms are related but not interchangeable

The relationships run in one direction more strongly than the other. Biomass drives oxygen demand and, through feeding rate, drives waste production, which drives biological filtration demand. That chain is reasonably tight. But biomass tells you almost nothing about territorial requirement: a single adult cichlid and a shoal of forty small tetras can have comparable biomass and completely different spatial and behavioural needs. And none of the biological quantities tell you anything about maintenance capacity, which is a fact about the person, not the tank.

A useful way to hold this: **biomass and feed input predict chemistry; species identity and tank geometry predict behaviour; the keeper predicts sustainability.** All three have to be satisfied. Satisfying one very well does not compensate for failing another.

1.2 How stocking density is discussed in other disciplines

Home aquarium keeping is the only one of these fields that does not routinely quantify stocking. That is worth noticing, because it explains why hobby rules of thumb had to fill the gap.

Table 1.2 — Stocking density across disciplines

Field	How density is expressed	What it is optimised for	Why the numbers do not transfer to home aquaria
Aquaculture (food fish)	Kilograms of fish per cubic metre; feed input in kg/day; often 70–150 kg/m ³ in intensive recirculating systems for species such as tilapia.	Yield per unit cost. Survival to harvest weight. Feed conversion efficiency.	Fish are harvested young, before the long-term costs of dense housing accumulate. Life support is engineered, monitored continuously and backed up. Mortality within an accepted range is a business cost, not a welfare failure. None of that describes a living-room aquarium.
Fisheries science	Biomass per unit area of habitat; standing stock; production per hectare.	Understanding wild population dynamics and sustainable extraction.	Wild fish are not confined. They move, disperse, and are subject to predation and recruitment. Density figures describe populations, not individuals in a box.
Laboratory animal facilities	Fish per litre. Zebrafish guidance is often framed in this unit; published work covers a wide range and regulatory guidance has proposed upper limits around 10 fish per litre.	Reproducibility, colony health, and regulatory compliance.	Racks are flow-through or heavily recirculated with continuous water exchange, so chemistry is decoupled from density in a way it is not in a home tank. Enrichment is usually minimal. The target is a standardised animal, not a displayed community.
Public aquaria / zoological husbandry	Species-specific husbandry manuals with minimum exhibit dimensions, group compositions and behavioural targets. Rarely a single density number.	Long-term individual welfare, natural behaviour, display quality, and sometimes breeding.	This is actually the closest analogue to a well-run home aquarium in philosophy, but the systems are vastly larger and professionally monitored, so the specific dimensions rarely scale down.
Home aquaria	Almost never quantified. Expressed as inches per gallon, fish counts, or a calculator percentage.	Enjoyment, appearance, and (for good keepers) long-term welfare.	Not applicable — this is the target case. The point is that no measured convention exists, which is exactly why heuristics proliferated.

Table 1.2 — The disciplines that measure stocking density carefully all measure it for goals that are not the home aquarist's goal.

The single most important transfer caution

Intensive aquaculture demonstrates that very high fish biomass per unit volume is **technically achievable** with sufficient engineering: continuous oxygenation, continuous solids removal, large purpose-built biofilters, continuous monitoring and rapid water exchange. It does not demonstrate that such densities are **appropriate**, and it certainly does not demonstrate that they are achievable in a glass box with a hang-on-back filter and a weekly bucket.

When a hobbyist cites 'but fish farms keep them much denser than this,' the honest response is that fish farms also keep them for a fraction of their natural lifespan, under continuous instrumentation, with an accepted mortality budget. The comparison is real but it does not do the work people want it to do.

EVIDENCE: AQUACULTURE PRACTICE; THE CAUTION ITSELF IS AN ANIMAL-WELFARE PRINCIPLE

2. Where “Fish Per Gallon” Came From

The inch-per-gallon rule is usually dismissed as a myth. That framing is historically unfair and practically unhelpful. It was a reasonable compression of a real constraint, aimed at a specific audience, in a period when aquarium technology looked very different. Understanding what it was designed to do explains precisely where it breaks.

2.1 The family of rules

Table 2.1 — The classic stocking rules and their internal logic

Rule	Typical statement	Underlying logic	Where it is least wrong
One inch per gallon	Total body length of all fish, in inches, should not exceed tank volume in US gallons.	Volume is a proxy for dilution capacity and, indirectly, for gas exchange in a standard-proportioned tank. Length is the only fish measurement a beginner can take without a scale.	Small, slender-bodied, similarly shaped community fish under about two inches adult length, in a rectangular tank with moderate feeding.
One centimetre per litre	The metric restatement. Roughly equivalent: 1 in/gal is about 1 cm per 1.5 L, so 1 cm/L is a meaningfully more conservative rule.	Same logic. The metric version is not simply a unit conversion of the imperial one and is generally the stricter of the two.	Same conditions, with somewhat more headroom.
One fish per gallon	One small fish per gallon of tank volume.	An even blunter compression that discards body size entirely.	Nowhere reliable. It is only defensible if ‘fish’ is implicitly assumed to mean a one-inch tetra, which no beginner knows.
Surface area rules	Commonly stated as one inch of slender-bodied fish per 12 square inches of water surface, and per 20 or more square inches for deep-bodied fish.	Explicitly about gas exchange. Oxygen enters at the air-water interface, so surface area rather than volume sets the diffusion ceiling in an unaerated tank.	Unaerated or lightly aerated tanks, and comparisons between tanks of equal volume but different footprint. Still the most physically principled of the classic rules.
Weight / biomass rules	Grams of fish per litre, or kilograms per cubic metre.	Directly indexes the quantity that actually drives oxygen demand and, with feeding rate, nitrogen input.	Scientifically the soundest basis, and standard in aquaculture. Almost unusable at home because hobbyists do not weigh fish and published weights for ornamental species are patchy.

2.2 Where did these rules come from?

An honest answer: the inch-per-gallon rule has no single traceable origin document. It appears in mid-twentieth-century beginner literature and retail advice and propagates from there. This document could not identify a primary source that derives it from data, and readers should treat any confident claim about its inventor with suspicion.

The surface-area rule is better documented in aquarium literature and is explicitly derived from gas exchange rather than dilution. Popular aquarium reference works, including Peter W. Scott's *The Complete Aquarium*, present surface-area figures on the order of 12 square inches per inch of slender fish, with a substantially larger allowance for heavy-bodied species. That distinction — slender versus deep bodied — is the tell that the rule's authors already knew the length-only version was inadequate.

EVIDENCE: COMMON AQUARIUM PRACTICE; HISTORICAL ORIGIN PARTLY UNTRACEABLE

2.3 Were they intended as beginner heuristics?

Very probably yes, and this matters. A heuristic is judged by whether it produces acceptable decisions cheaply for its intended user, not by whether it is true. Consider the counterfactual: before these rules existed, the alternative available to a 1960s beginner was not a multi-constraint model. It was no guidance at all, or a shop assistant's guess. Against that baseline, 'do not put more than about ten inches of small fish in a ten-gallon tank' prevented real harm.

The rule became a problem when three things changed: the range of species in the trade widened enormously, the range of tank shapes widened, and the hobby's welfare expectations rose from 'the fish survive' to 'the fish behave normally across a full lifespan.' A heuristic tuned for the first standard cannot be expected to satisfy the third.

2.4 What assumptions does the inch-per-gallon rule make?

Stated explicitly, the rule assumes all of the following at once:

- **Body shape is roughly constant.** That mass per unit length is similar across the fish being counted.
- **The measured length is the adult length.** Nothing in the rule distinguishes a juvenile from a mature fish.
- **Tank proportions are standard.** Volume correlates with surface area and with footprint, which is only true within the conventional rectangular range.
- **Feeding is moderate and typical.** Nitrogen input scales with fish size rather than with keeper behaviour.
- **The fish are behaviourally undemanding.** No significant territoriality, no large minimum group sizes, no strong swimming requirements.
- **Maintenance is regular.** Some baseline water-change schedule is silently assumed.
- **Actual water volume equals nominal volume.** Substrate, hardscape and fill level are ignored.

Each of those assumptions holds for a specific and once-common case: a rectangular community tank of small slender tropical fish, moderately fed and regularly maintained. That is not a coincidence. It is the case the rule was built for.

2.5 Can the rule still be useful as a starting point?

Yes, narrowly, and it is worth being honest about that rather than reflexively dismissive. Used as a *ceiling check on small slender community fish only*, and with adult length rather than purchase length, the rule produces plans that are usually conservative rather than dangerous. Its failure mode is not that it permits too much for tetras; it is that it permits catastrophically too much for anything that is deep bodied, large-growing, territorial, or heavily fed.

A more defensible version of the same instinct

The scientifically defensible core of 'fish per gallon' is this: **for a narrow, homogeneous class of fish, in a conventionally proportioned tank, under a defined feeding and maintenance regime, total body length is a crude proxy for total biomass, and total biomass is a genuine driver of oxygen demand and nitrogen input.**

Every clause in that sentence is doing work. Remove any one of them and the proxy collapses. That is the whole story of why the rule fails, and it is more useful to a listener than 'it is a myth.'

EVIDENCE: USEFUL RULE OF THUMB, WITH THE STATED RESTRICTIONS

3. Why Length-Based Rules Fail

Two fish of identical length can differ by an order of magnitude in the demand they place on a system. This section catalogues the axes along which they differ, because each one is a separate reason the rule breaks.

Table 3.1 — Eleven ways two same-length fish can differ

Variable	Why it breaks length-based counting
Body mass	Mass scales approximately with the cube of length. Two fish at the same length but different girth can differ several-fold in mass. Section 4 quantifies this.
Body depth and girth	A deep-bodied fish carries far more tissue per unit of length than a pencil-shaped one. Body depth is also the dimension that determines turning radius, which is a spatial rather than chemical constraint.
Metabolic rate	Mass-specific metabolic rate is higher in smaller fish and varies with lifestyle. Actively swimming pelagic species carry more expensive tissue than sedentary benthic species of the same mass.
Feeding rate and diet	Feed input drives nitrogen input. A fish fed a high-protein diet three times daily contributes far more nitrogen than the same fish fed a lower-protein diet once daily.
Waste production	Follows feed input and digestibility, not length. Poorly digestible feeds increase the faecal solids fraction, which loads mechanical filtration and dissolved organics rather than the biofilter directly.
Activity level	Sustained swimming raises oxygen consumption well above resting metabolic rate. Two fish of equal mass, one hovering and one cruising continuously, do not have equal oxygen demand.
Oxygen consumption	Scales with mass and temperature, and rises with activity, digestion and stress. It is the variable that most often becomes binding in warm, densely stocked tanks.
Adult size	The rule counts what is in the bag today. Section 5 covers this separately because it is the single most common beginner error.
Growth rate	Fast-growing fish change the system's load month by month, so a plan that is correct in March can be wrong by September.
Territorial behaviour	Entirely invisible to a length-based rule, and often the binding constraint in cichlid, gourami and some catfish systems.
Social requirements	Also invisible. A rule that permits four of a species that needs ten has produced a chemically acceptable and behaviourally poor outcome.

3.1 Worked contrasts

The comparisons below are chosen because they are accurate and because each isolates a different failure mode. Where a figure is approximate it is stated as approximate.

Neon tetra versus goldfish

This is the standard example and it is a good one, but the usual framing understates the case. A neon tetra reaches roughly 3–4 cm and remains slender. A common or comet goldfish is a large, deep-bodied, fast-growing coldwater cyprinid that can exceed 25–30 cm in an adequate system, with fancy varieties somewhat shorter but very deep-bodied. Under a naive inch-per-gallon reading, ten inches of neon tetra and one ten-inch goldfish are equivalent. They differ in mass by well over an order of magnitude, in absolute oxygen consumption correspondingly, and in feeding volume by more still. The failure here is primarily **mass per unit length compounded by adult size**.

Betta versus zebra danio

Both are commonly kept, both are small, and their adult lengths overlap — a male betta and a zebra danio are broadly comparable in body length. Their demands are not comparable. Zebra danios are continuously active

open-water swimmers that use tank length constantly; bettas are labyrinth-breathing, relatively sedentary, and access atmospheric oxygen directly. Meanwhile the betta is territorially intolerant of conspecific males, and the danio is socially dependent on a group. The failure here is that **length says nothing about activity, oxygen strategy, or social structure**, and those are the constraints that actually govern both species.

Corydoras versus a comparably sized cichlid

A bronze corydoras is a small armoured benthic catfish; published research work with the species reports adult individuals around 43 mm total length and roughly 1.65 g wet mass (Corydoras aeneus, University of Glasgow husbandry data reported in Killen laboratory work). Many dwarf cichlids occupy a similar length class. The corydoras is a low-aggression social forager that lives on the bottom in two dimensions; the cichlid will typically claim and defend a territory, particularly when breeding. The failure here is **behavioural carrying capacity**: identical biomass, entirely different spatial economics.

Kuhli loach versus a livebearer

A kuhli loach can reach 8–10 cm of length, but it is an eel-shaped fish of very low mass for its length, and it spends most of its time hidden. A guppy or platy of half that length carries a substantially higher mass-to-length ratio and, critically, reproduces. The failure here is **mass-to-length ratio in one direction and population growth in the other** — the loach overstates its load under a length rule and the livebearer understates it.

Fancy goldfish versus a slender schooling fish

Fancy goldfish varieties are short-bodied but extremely deep and round. Body length therefore massively understates their mass. They are also heavy feeders with a short digestive tract, they are long-lived, and their modified body shape reduces swimming efficiency. Against a slender schooling fish of the same nominal length, essentially every relevant variable differs. The failure here is **body geometry**, which is the single clearest demonstration that length alone cannot work.

EVIDENCE: REASONABLE INFERENCE FROM ESTABLISHED MORPHOLOGY AND PHYSIOLOGY; THE SPECIFIC CORYDORAS MASS FIGURE IS FROM PUBLISHED HUSBANDRY DATA

4. Fish Length Versus Fish Biomass

The mathematics here is simple enough to state in two sentences, and it changes how the whole problem looks. Length is a linear measure. The quantities that actually matter are not linear in length.

4.1 Mass scales roughly with the cube of length

The standard relationship in fisheries science is $W = aL^b$, where W is weight, L is length, a is a species-specific coefficient and b is the allometric exponent. Froese's 2006 meta-analysis in the *Journal of Applied Ichthyology* examined 3,929 such relationships covering 1,773 fish species and found a median b of 3.03, slightly above the isometric value of 3. Individual species values ranged widely, roughly 1.96 to 3.94 across the dataset, with the great majority falling between about 2.5 and 3.5.

In plain terms: for most fish, doubling length multiplies mass by roughly eight. A 2-inch fish is not twice the biological load of a 1-inch fish of the same shape. It is closer to eight times.

EVIDENCE: STRONG EVIDENCE &MDASH; FROESE 2006, META-ANALYSIS

Table 4.1 — What a length multiple actually implies

Length multiple	Approximate mass multiple ($b = 3$)	Approximate resting metabolic multiple ($\text{mass}^{0.79}$)	Plain-language reading
1.0x	1x	1x	Reference fish.
1.5x	about 3.4x	about 2.6x	A modest-looking length increase is already a large load increase.
2.0x	about 8x	about 5.1x	This is the case that breaks inch-per-gallon most visibly.
3.0x	about 27x	about 13.9x	A fish three times as long is not 'three fish' by any measure that matters.
5.0x	about 125x	about 48x	The gap between a juvenile and an adult large-growing species.

Table 4.1 — Illustrative only, assuming constant body shape and $b = 3$. Real species deviate. The purpose is to show the shape of the relationship, not to provide a calculation tool.

4.2 Metabolic rate does not scale linearly with mass either

Having established that mass rises faster than length, the second correction runs the other way. Metabolic rate rises more slowly than mass. Clarke and Johnston's 1999 analysis in the *Journal of Animal Ecology* compiled 138 studies covering 69 teleost species across a temperature range of about 40 degrees C and found that resting metabolic rate related to body mass by $R = aM^b$ with a mean scaling exponent of 0.79 (standard error 0.11). The overall between-species relationship gave an exponent near 0.80. Individual species exponents varied widely, from roughly 0.40 to 1.29, so the mean is a first-order description rather than a law.

The practical consequence is that a gram of small fish consumes more oxygen than a gram of large fish. Ten one-gram fish have a higher combined resting oxygen demand than one ten-gram fish. This partially — only partially — offsets the cube-law penalty on large fish, and it is the reason that 'many small fish' is not automatically the lighter option it appears to be.

EVIDENCE: STRONG EVIDENCE &MDASH; CLARKE & JOHNSTON 1999

4.3 Temperature multiplies the whole picture

The same analysis found the relationship between resting oxygen consumption and temperature to be curvilinear, and reported that a typical tropical fish at 30 degrees C requires approximately six times as much oxygen for resting metabolism as a polar fish at 0 degrees C. Within the narrower band that concerns aquarists, the paper reported a mean Q_{10} of about 1.83 across species over 0–30 degrees C, with within-species acclimation studies showing a median Q_{10} nearer 2.40. A Q_{10} of 2 means metabolic rate roughly doubles for every 10 degree C rise.

This is why the same stocking level behaves differently in summer, and it is compounded by the fact that warmer water holds less oxygen. Section 15 returns to this.

EVIDENCE: STRONG EVIDENCE

The three-step chain worth remembering

Step 1. Mass rises with roughly the cube of length (Froese 2006).

Step 2. Resting oxygen demand rises with roughly mass to the power 0.79 (Clarke & Johnston 1999).

Step 3. Both are multiplied by temperature, and by activity, digestion and stress on top of resting values.

Net effect: a length-based rule mis-estimates load in one direction for large fish and in the other direction for very small fish, and ignores the temperature multiplier entirely.

5. Adult Size: Stocking for the Fish You Will Have

The most consequential beginner error in stocking is not choosing the wrong number. It is choosing the right number for the wrong fish — the juvenile in the bag rather than the adult it becomes.

5.1 What changes as a fish matures

Table 5.1 — Juvenile to adult: what actually changes

Dimension	What changes between juvenile and adult	Why the stocking plan has to change with it
Body mass	Increases with roughly the cube of length. A fish sold at 2 cm and maturing at 8 cm has gained roughly 60-fold in mass.	Oxygen demand, feed requirement and nitrogen output all scale with this, not with the number of animals.
Feeding rate	Absolute food intake rises with mass, even though intake as a percentage of body weight typically falls.	Nitrogen input to the system rises steadily over months, often without the keeper noticing they are feeding more.
Oxygen demand	Rises with mass, though less than proportionally.	A tank that was comfortably oxygenated with juveniles can become marginal at night once the same fish are adult.
Territorial behaviour	Frequently absent or minimal in juveniles and appears at sexual maturity. This is why cichlid and gourami tanks often go wrong at months four to eight rather than in week one.	The behavioural ceiling can drop sharply at a point in time, with no warning from any water test.
Swimming space	A larger fish needs a longer turning radius and a longer sustained swimming path.	The footprint that suited juveniles may not permit normal locomotion for adults.
Social structure	Some species shoal as juveniles and become territorial or pair-forming as adults. Others do the reverse.	The correct group size can genuinely change over the life of the same group of fish.
Longevity	Many commonly sold species live far longer than buyers expect. Goldfish and many catfish and cichlids are decade-scale animals.	The plan has to be sustainable for the whole span, not for the first year.

5.2 “It fits now” versus “it is appropriately stocked”

These are different claims and they are routinely conflated. “It fits now” is a statement about the present physical and chemical state of the tank. “It is appropriately stocked” is a claim about a trajectory: that the system will remain within all its constraints as the animals grow, mature, possibly breed, and age.

The honest version of the upgrade argument is worth stating carefully, because the hobby tends to treat it as either obviously fine or obviously irresponsible. Growing juveniles out in a smaller system and moving them to a larger one is a legitimate practice with a real precondition: the larger system has to actually exist by the time it is needed. The failure mode is not the plan, it is the plan without a date, a budget and a location. Section 25 treats this as a scenario.

EVIDENCE: ANIMAL-WELFARE PRINCIPLE; THE PRACTICE ITSELF IS COMMON AQUARIUM PRACTICE

Practical framing for the eventual episode

The useful question is not ‘how many fish fit?’ but **‘how many of these fish, at their adult size, in this footprint, with this feeding and this maintenance, for the next several years?’**

Every clause narrows the answer, and each one is a place where a stocking calculator or a rule of thumb loses information.

6. Bioload: What the Word Actually Denotes

“Bioload” is hobby vocabulary with no precise scientific definition, which is not a criticism — it is a useful compression. But it compresses several distinct waste streams, and the distinctions matter when diagnosing a problem.

6.1 The components of what hobbyists call bioload

Table 6.1 — Bioload decomposed

Component	What it is	What it loads	Main driver
Dissolved ammonia / ammonium	Nitrogen excreted directly, mostly across the gills.	The nitrifying biofilter. This is the acutely toxic fraction.	Protein intake and protein catabolism, hence feed input.
Urea and other nitrogen forms	A smaller fraction of nitrogen excretion; also amino acids and mucus protein.	Heterotrophic bacteria, which mineralise it toward ammonia.	Feed input and mucus turnover.
Faecal solids	Undigested feed and sloughed gut material.	Mechanical filtration and, once broken down, the biofilter and oxygen demand.	Feed quantity and, importantly, feed digestibility.
Uneaten food	Feed that never enters a fish at all.	Everything at once, and at the worst efficiency. It is pure input with no biological benefit.	Keeper behaviour. This is the most controllable variable in the whole system.
Dissolved organic carbon	Soluble organics from waste, food, mucus, decaying plant matter.	Heterotrophic bacterial respiration, hence oxygen. Accumulates between water changes and is not measured by standard hobby test kits.	Total organic input and water-change frequency.
Phosphate	From feed, principally.	Not toxic at aquarium levels but drives algae growth in combination with light and nitrogen.	Feed input.
Carbon dioxide	Respiratory output from fish, bacteria and plants at night.	Depresses pH and, at high concentrations, impairs the fish's own ammonia excretion and oxygen uptake.	Total respiring biomass and gas exchange rate.

6.2 Is waste production driven by fish count or by biomass and feed?

By biomass and feed, and the aquaculture industry settles this question by how it does its engineering. Recirculating aquaculture system design sizes biofilters from feed loading, not from fish numbers. A commonly cited working figure derived from Ebeling and colleagues is that adding 1 kg of feed at 32 percent crude protein to a system results in approximately 30 g of ammonia nitrogen entering the water. Fish count does not appear in that calculation at all.

This is the single most useful reframing in the whole document for the hobby audience, because it explains why so much stocking advice feels contradictory. Fish count is a poor proxy for nitrogen input. Biomass is a decent proxy, because feeding rate is generally set as a fraction of biomass. Feed input is the actual driver.

EVIDENCE: AQUACULTURE PRACTICE BUILT ON ESTABLISHED NITROGEN MASS BALANCE

Why this matters for the schooling-fish question

If waste load tracked fish *count*, then adding eight small tetras would be a large step and adding one large fish would be a small one. Because waste load tracks *biomass and feed*, the reverse is often closer to the truth.

Eight neon tetras at roughly 0.2–0.3 g each add on the order of two grams of biomass. A single adult fancy goldfish can weigh a hundred times that. Yet hobby advice routinely treats 'eight fish' as the alarming addition. This is the central mechanical insight behind resolving the 'add the whole school' question in Section 26.

6.3 The full list of contributors to waste load

- **Fish mass** — sets the baseline against which feeding is calculated and against which oxygen is consumed.
- **Feed input** — the actual nitrogen and phosphorus source. Very nearly all of the nitrogen in a closed aquarium arrived in the food container.
- **Protein content** — nitrogen is carried in protein. Higher protein feeds mean more nitrogen per gram fed.
- **Digestibility** — determines the split between dissolved nitrogen excreted across the gills and particulate nitrogen deposited as faeces. Lower digestibility shifts load toward solids.
- **Feeding frequency** — affects the peak-to-average ratio of ammonia production. Post-feeding ammonia excretion rises measurably.
- **Metabolic rate** — sets the baseline rate of protein catabolism.
- **Temperature** — raises metabolic rate, raises feeding, raises ammonia production, and simultaneously lowers oxygen solubility.
- **Species** — differs in digestive efficiency, diet type and activity level.
- **Growth** — growing fish retain a larger fraction of ingested nitrogen in tissue. A growing population is, counter-intuitively, more nitrogen-efficient per gram fed than a static adult population.
- **Reproduction** — adds biomass without the keeper buying anything, which is why livebearer tanks drift into overstocking silently.
- **Uneaten food** — contributes nitrogen at zero biological benefit and is the fastest way to make a well-planned tank behave badly.

7. Food Input as the Real Driver

If the document had to reduce to one operational insight it would be this: in a closed aquarium, nitrogen enters through the food and leaves through growth, water changes, plant uptake and gas. Feeding is the tap.

7.1 The nitrogen mass balance

Studies of nitrogen budgets in commercial recirculating systems give a consistent picture. In one documented commercial facility, roughly 35 percent of nitrogen from feed was assimilated into fish flesh and approximately 65 percent was unassimilated or excreted in various forms. The excreted nitrogen was reported at about 62 g per kilogram of fish produced. The exact split varies with species, feed formulation, feeding practice and growth stage, but the structure is stable: a substantial minority of fed nitrogen becomes fish, and the majority enters the water.

EVIDENCE: AQUACULTURE PRACTICE / STRONG EVIDENCE FOR THE MASS-BALANCE STRUCTURE; SPECIFIC PERCENTAGES ARE FACILITY-SPECIFIC

Table 7.1 — The fate of nitrogen entering an aquarium

Stage	What happens to fed nitrogen	Aquarium-relevant consequence
Eaten and assimilated	Retained as fish tissue. Roughly a third in well-run production systems, typically less in adult non-growing fish.	This is the only nitrogen that leaves the water column without processing. It also means a tank of growing juveniles is temporarily more forgiving than the same biomass of static adults.
Eaten and excreted as ammonia	Released across the gills, continuously.	The direct load on nitrification. Rises after feeding.
Eaten and excreted as faeces	Deposited as solids, then mineralised by heterotrophs.	Loads mechanical filtration first, then oxygen demand and the biofilter. Removing solids promptly genuinely reduces downstream load.
Not eaten	Decomposes in place.	Full nitrogen contribution, zero biological return, and a disproportionate oxygen demand while it breaks down.

7.2 Why ten lightly fed fish and ten heavily fed fish are not the same stocking level

Consider two 20-gallon tanks, each with ten similarly sized small community fish. Tank A is fed once daily, a small quantity fully consumed within a minute. Tank B is fed three times daily, generously, with visible food reaching the substrate.

- Tank B receives several times the nitrogen input, so its steady-state nitrate rises faster between water changes.
- Tank B accumulates dissolved organic carbon faster, raising heterotrophic bacterial oxygen consumption. This is invisible on a standard test kit.
- Tank B's biofilter must sustain a higher ammonia oxidation rate, so it carries a larger nitrifier population and consumes more oxygen and alkalinity doing so.
- Tank B's fish are more likely to be in positive energy balance, so they grow faster, which increases biomass, which increases everything else. This is a reinforcing loop, not a one-off difference.
- Tank B has less margin for a missed water change, because its accumulation rate is higher.

Both tanks contain 'ten fish.' They are not at the same stocking level in any sense that a biologist would recognise. Any stocking framework that indexes on fish count is blind to the single largest lever the keeper controls.

EVIDENCE: REASONABLE INFERENCE FROM ESTABLISHED NITROGEN AND OXYGEN MASS BALANCE

Implication for the 'can I add more fish?' question

A keeper who wants to add a group of schooling fish to a mature tank has two independent levers, and the hobby usually only discusses one of them.

Lever 1 (widely discussed): add fewer fish, or add them in stages.

Lever 2 (rarely discussed): reduce feeding temporarily during the transition, so total nitrogen input rises less than the biomass increase would suggest. Fish in good condition tolerate a short period of reduced feeding without difficulty, and the biofilter is given time to expand against a smaller step change.

Lever 2 is genuinely useful, low-risk and under-discussed. It is revisited in Section 26.

8. How Fish Actually Produce Ammonia

A persistent hobby misconception is that fish waste is the visible material on the substrate. The acutely dangerous nitrogen fraction is invisible and is produced continuously, whether or not anything is visible on the sand.

8.1 Nitrogen excretion in teleost fish

Teleost fish are predominantly ammonotelic: they excrete nitrogen mainly as ammonia rather than converting it to urea or uric acid, because producing ammonia is energetically cheaper and the surrounding water provides effectively unlimited dilution. The great majority of that ammonia leaves across the gills rather than through the kidney.

The mechanism is well characterised. Ammonia moves down its partial pressure gradient from blood to water across the gill epithelium, partly by simple diffusion through membranes and partly by facilitated diffusion through Rhesus (Rh) glycoprotein channels, whose expression can be induced under ammonia loading. Reviews by Wright and Wood and colleagues established the Rh-protein model as the current understanding of branchial ammonia excretion.

EVIDENCE: ESTABLISHED SCIENTIFIC PRINCIPLE

Table 8.1 — Routes of nitrogen excretion

Route	Approximate share of nitrogen excretion	Aquarium relevance
Gills, as ammonia	The dominant route in freshwater teleosts. A detailed study of juvenile rainbow trout partitioned nitrogenous waste as 53–68 percent ammonia-N.	Continuous, invisible, and the reason a spotlessly clean-looking tank can still have an ammonia problem.
Urea	A minority fraction. The same trout study reported 6–10 percent urea-N; broader reviews of teleosts give a range on the order of 10–30 percent of non-ammonia nitrogen as urea.	Mineralised toward ammonia by heterotrophic bacteria, so it eventually reaches the biofilter anyway, with a lag.
Amino acids and protein	Smaller but non-trivial. The trout study found 4–10 percent as amino acids via the gills and 3–11 percent as protein, probably via body mucus.	Contributes to dissolved organic load, which drives heterotrophic bacterial oxygen consumption.
Faeces	Particulate nitrogen, highly variable with feed digestibility.	The visible fraction. Loads mechanical filtration and, after breakdown, everything else.

Table 8.1 — Percentages are from rainbow trout under specific fed and fasted conditions and should be read as illustrative of the structure, not as universal constants for tropical ornamental species.

8.2 Why this matters for stocking judgements

- **Ammonia production is continuous.** It does not stop between feedings, though it does peak after them. A tank's biofilter faces a baseline load plus post-feeding peaks.
- **Visible cleanliness is not a stocking indicator.** A tank with a gravel vacuum run yesterday and a heavy fish load still has a heavy dissolved nitrogen load.
- **Ammonia excretion depends on the fish's own water chemistry.** Because excretion is driven by a partial pressure gradient, high environmental ammonia or high dissolved carbon dioxide can impair the fish's ability to offload nitrogen. This is one mechanism by which overstocking becomes self-reinforcing under stress.
- **pH determines toxicity, not production.** The ammonia/ammonium equilibrium shifts toward the un-ionised, far more toxic NH_3 form as pH and temperature rise. The same total ammonia nitrogen reading is a different hazard at pH 6.5 than at pH 8.2.

EVIDENCE: ESTABLISHED SCIENTIFIC PRINCIPLE

9. Biological Filtration Capacity

The question that matters for stocking is not 'is the tank cycled?' but 'how much more ammonia per day can this particular nitrifying community oxidise, and how fast can it grow if asked to do more?'

9.1 Fixed, reserve, or adaptive capacity?

All three descriptions are partly right, which is why the hobby argues about it. The accurate composite answer is that an established biofilter has **a modest instantaneous reserve sitting on top of an adaptive capacity that expands slowly**.

1. **It is not fixed.** Nitrifying populations are substrate-limited. Standing biomass adjusts to the ammonia supply available. A filter carrying a light load carries a small population; increase the supply sustainably and the population grows.
2. **There is real short-term reserve.** Nitrifiers do not normally operate at their maximum rate. At typical aquarium ammonia concentrations, the community is operating well below saturation, so it can immediately process somewhat more than it currently is. This is why modest additions to a mature tank usually produce no measurable ammonia at all.
3. **Expansion is slow relative to the disturbance.** Ammonia-oxidising organisms are slow growers compared with heterotrophic bacteria. In documented commercial biofilter start-ups, ammonia oxidisers became detectable over a period of weeks and nitrite oxidisers developed more slowly still. Established aquarium systems respond faster than a cold start-up because a seed population already exists, but the underlying growth rates are the same.

EVIDENCE: ESTABLISHED PRINCIPLE FOR THE MECHANISM; TIMEFRAMES ARE SYSTEM-SPECIFIC AND GENUINELY CONTESTED

9.2 What governs nitrification rate

Table 9.1 — Determinants of nitrification rate

Factor	Effect on nitrification	Practical note for aquarists
Ammonia availability	Substrate concentration drives rate up to saturation, and also sets the population the filter can sustain.	A filter cannot maintain capacity it is not being asked to use. This is why a tank that has been lightly stocked for a year does not have a hidden large reserve waiting.
Surface area	Sets the ceiling on attached biofilm. More colonisable area supports a larger population.	Media surface area is real, but only the fraction with adequate flow and oxygen is fully productive. Clogged or dead-zone media contributes less than its specification suggests.
Dissolved oxygen	Nitrification is strictly aerobic and oxygen-hungry; the reaction consumes oxygen stoichiometrically.	In a heavily loaded tank, nitrification and fish respiration compete for the same oxygen. This links the biofilter constraint and the oxygen constraint, which are usually discussed as if separate.
Temperature	Rates rise with temperature over the normal aquarium range, up to a point.	Warmer tanks nitrify faster, which partly offsets the higher ammonia production — but oxygen solubility falls at the same time, so the net effect is not favourable.
pH	Nitrification slows substantially as pH falls; performance is markedly impaired at low pH and can approach a standstill in strongly acidic water.	Soft, acidic blackwater-style tanks may rely more on plant uptake and less on nitrification than their keepers assume.
Alkalinity (KH)	Nitrification consumes alkalinity. Without replacement, KH is drawn down over time.	A heavily stocked tank consumes KH faster. Once KH is exhausted, pH becomes unstable and nitrification itself is threatened. This is a genuine and under-recognised stocking constraint in soft-water areas.
Flow	Delivers substrate and oxygen to the biofilm and removes products.	Media in a low-flow region of a filter contributes far less than its surface area implies.

Why no honest source gives a single adjustment timeframe

The time for a biofilter to accommodate a given increase in load depends on: the size of the step change relative to the existing load; the existing population size; temperature; pH; alkalinity; available surface area; oxygen; and how much of the new nitrogen is intercepted by plants before it reaches the filter.

Those variables span an enormous range across real aquariums. Any single number — ‘two weeks,’ ‘ten days,’ ‘72 hours’ — is a convention chosen for communicability, not a measured constant. That is not a criticism of conventions; it is a reason to describe them accurately as conventions.

EVIDENCE: CONTESTED CLAIM &MDASH; NO UNIVERSAL CONSTANT IS SUPPORTABLE

10. Adding Fish to an Established Aquarium

What actually happens, mechanically, when new livestock enters a cycled system — and what determines whether ammonia or nitrite ever becomes measurable.

10.1 Does every addition cause a spike?

No. The common experience of adding a few small fish to a mature tank and seeing no detectable ammonia is real and is explained by the reserve capacity described in Section 9. Whether a spike occurs depends on whether the increase in ammonia production rate exceeds the community's current spare oxidation capacity, and on how quickly the population can close the gap.

It is also worth noting a measurement caveat: hobby liquid test kits have limited resolution at low concentrations. 'No detectable ammonia' means below the kit's practical detection limit, not necessarily zero. A brief, small excursion can occur and resolve without ever appearing on a test.

EVIDENCE: REASONABLE INFERENCE; WELL SUPPORTED BY COMMON AQUARIUM PRACTICE

10.2 What determines whether ammonia or nitrite becomes measurable

Table 10.1 — What governs whether an addition is visible on a test kit

Factor	Increases the risk of a measurable excursion	Reduces the risk
Size of the step change	A large proportional increase in feed and biomass relative to the existing load.	A small proportional increase. This is the key variable and it is proportional, not absolute.
Existing load history	A lightly stocked, lightly fed tank has a correspondingly small nitrifier population and therefore less absolute reserve.	A tank already carrying a moderate load has a larger standing population and more absolute spare capacity.
Filter maturity	A recently cycled system, or one recently disturbed by media replacement or a medication course.	A long-established, undisturbed filter with well-developed biofilm.
Colonised surfaces beyond the filter	A bare tank with minimal substrate and hardscape concentrates nitrification in the filter alone.	Substrate, hardscape, plant surfaces and equipment all carry biofilm and add distributed nitrification capacity. This is genuinely significant and frequently underestimated.
Plant mass and growth	Few or slow-growing plants.	Substantial actively growing plant mass, particularly fast-growing stems and floaters, intercepts ammonium directly. See Section 18.
Feeding during the transition	Continuing or increasing feeding immediately after the addition.	Holding feeding steady or reducing it briefly, so nitrogen input rises less than biomass.
Temperature and pH	Low pH slows nitrification; low temperature slows population growth.	Warm, near-neutral to slightly alkaline water with adequate KH supports faster nitrification and faster recovery.
Handling and transport stress	Stressed fish excrete more ammonia acutely, and transport water added to the tank carries its own nitrogen load.	Careful acclimation, and not adding transport water to the display.

10.3 The role of surfaces other than the filter

Hobby discussion tends to treat the filter as the location of the biofilter. In reality every wetted surface with adequate oxygen carries nitrifying biofilm: substrate grains, hardscape, plant leaves, heater and tubing surfaces, and the glass itself. Substrate in particular offers very large surface area. This distributed capacity is a major reason established, aquascaped tanks are more forgiving than bare systems of the same volume, and it is why a tank can

survive a filter failure for longer than a purely filter-centric model predicts.

EVIDENCE: ESTABLISHED PRINCIPLE; THE AQUARIUM-SCALE MAGNITUDE IS A REASONABLE INFERENCE

11. “Only Add Two or Three Fish at a Time”

One of the most repeated pieces of aquarium advice. This section asks directly whether any of its specific numbers can be traced to evidence.

11.1 Searching for the evidence base

This document could not identify peer-reviewed or aquaculture-derived support for any of the specific figures commonly given: two fish, three fish, a 10 percent increase, a 20 percent increase, one week, or two weeks. These appear in hobby literature, retail guidance and community advice without derivation.

That is not the same as saying the advice is wrong. The *principle* behind it — that a nitrifying community adjusts to increased load over time and that smaller step changes are less likely to overshoot — is well founded and follows directly from Section 9. What is unsupported is the precision. ‘Add a few and wait a while’ is sound. ‘Add exactly three and wait exactly fourteen days’ presents a convention with the confidence of a measurement.

EVIDENCE: USEFUL RULE OF THUMB; THE SPECIFIC NUMBERS ARE UNSUPPORTED

Table 11.1 — Common versions of the staged-addition rule

Version of the advice	Underlying principle it encodes	Why the specific number is not defensible
“Add two or three fish”	Keep the step change in ammonia production small relative to existing capacity.	Counts animals, not load. Three adult goldfish and three neon tetras are not comparable steps by any relevant measure. The number is the wrong unit.
“Increase by 10 percent” / “20 percent”	Proportional framing, which is genuinely better because it scales with the existing system.	The percentage is of what? Fish count, biomass, or feed? Only the latter two are biologically meaningful, and neither is measured by hobbyists. The framing is right and the operationalisation is missing.
“Wait one week”	Allow time for nitrifier population growth and for the keeper to observe the new fish.	One week may be more than enough for a small addition to a mature system and insufficient for a large addition to a young one.
“Wait two weeks”	Same, more conservatively. Also happens to align with common quarantine practice, which may be part of why it persists.	Same objection. The conservatism is defensible; the specificity is not.

11.2 A more defensible restatement

What the principle actually supports

1. Keep each addition small *relative to the tank’s existing feed and biomass load*, not small in animal count.
2. Prefer a longer interval when the system is young, lightly loaded, sparsely planted, cool, or acidic — all of which slow nitrifier response.
3. Prefer a shorter interval, or a single larger addition, when the system is mature, well planted, warm, well buffered and already carrying a moderate load.
4. Test after the addition rather than assuming the interval guarantees safety. The test result is information; the calendar is not.
5. Recognise that the interval is doing two jobs at once — biofilter accommodation and disease observation — and the second job may be the more important one for a keeper without a quarantine tank.

Point 5 deserves emphasis because it may be the honest answer to why the advice has survived. Staggering additions gives the keeper time to spot disease in one batch before committing the next. That is a real benefit, and it

is not a biofilter argument at all.

EVIDENCE: REASONABLE INFERENCE

12. Schooling Fish Versus Bioload: The Central Tension

This is one of the two pillars of the document. A keeper is told in the same week that they must not add too many fish at once and that their four tetras need to be a group of ten. Both pieces of advice are often correct. They are answers to different questions.

12.1 Separating the two claims

Table 12.1 — The two claims are about different variables

	“Do not add too many fish at once”	“This species needs a larger group”
What it is about	The rate of change in nitrogen input relative to biofilter capacity.	The final composition of the community, for welfare reasons.
Correct unit	Added biomass and added feed input, as a proportion of existing load.	Number of conspecifics, and sometimes their sex ratio and size range.
Timescale of the risk	Days. An ammonia or nitrite excursion resolves or kills within days.	Weeks to years. Chronic social stress is cumulative and slow.
What happens if ignored	Acute water quality event, potentially lethal.	Chronic stress, suppressed behaviour, poor feeding, possibly reduced disease resistance. Rarely immediately lethal.
Who is usually right	Correct whenever the addition is large relative to the system.	Correct whenever the species is genuinely social and the group is genuinely undersized.
When they actually conflict	Only when the required group size represents a large proportional increase in the system’s total load.	For small schooling fish in an adequately sized mature tank, they very often do not conflict at all.

12.2 Why the conflict is usually smaller than it sounds

Run the numbers, approximately. A neon tetra weighs on the order of 0.2 to 0.3 g. Adding six of them adds roughly one and a half to two grams of biomass, and a correspondingly small increment of feed. In a 29-gallon tank already holding, say, a dozen small fish and some invertebrates, that is a modest proportional increase in nitrogen input — and the welfare gain from completing an undersized shoal can be substantial.

Now run it for a different case. A keeper with two juvenile fancy goldfish in a 20-gallon tank is told goldfish are social and should be kept in groups. Adding four more is a completely different proposition: large adult biomass, heavy feeding, high oxygen demand, and a tank already at or beyond its spatial ceiling. Here the two pieces of advice genuinely conflict, and the resolution is not to add fish — it is that the system is the wrong size for the plan.

EVIDENCE: REASONABLE INFERENCE FROM ESTABLISHED MASS-BALANCE AND FROM GROUP-SIZE WELFARE EVIDENCE

The resolution, stated plainly

“Add more fish” is good advice when the species is genuinely social, the group is genuinely undersized, and **the tank has spare capacity in every other dimension** — footprint, oxygen, biofiltration, and maintenance.

“Add more fish” is bad advice when any of those conditions fails. In that case the correct answer is not a smaller group; it is either a larger tank, or a different species, or rehoming the existing group into a suitable setup.

The advice that is almost never correct is ‘keep the undersized group as it is and accept it,’ because that resolves the biological question by permanently accepting the welfare cost.

12.3 The behavioural mechanisms that make group size matter

- **Predator dilution.** Individual risk of being the one taken falls as group size rises. Work on banded killifish showed individual predation risk declining with school size in close accordance with the dilution hypothesis. Captive fish retain the behavioural machinery even without predators present.
- **Collective vigilance (the ‘many eyes’ effect).** Minnow studies by Magurran and colleagues found that larger shoals detected an approaching model predator sooner and were able to resume and sustain foraging longer, while small shoals reacted late and then stopped feeding abruptly. Larger groups balanced feeding against watching more efficiently.
- **Foraging benefit.** Work on bluntnose minnows found that in the absence of a predator, foraging latency fell and foraging rate rose with increasing shoal size. Fish in larger shoals found and ate food faster.
- **Aggression distribution.** In some species, aggression that would be concentrated on one or two individuals in a small group is spread across many in a larger one. This is not universal and is discussed critically in Section 17.
- **Social facilitation and confidence.** Fish in appropriate groups spend more time in open water and less time hiding, which is why an undersized shoal often looks ‘shy’ and a complete one looks ‘active.’ The behavioural difference is real, not aesthetic.

EVIDENCE: STRONG EVIDENCE IN BEHAVIOURAL ECOLOGY; EXTRAPOLATION TO SPECIFIC AQUARIUM SPECIES IS A REASONABLE INFERENCE

12.4 The key aquarium-specific study

Saxby, Adams, Snellgrove, Wilson and Sloman (2010), published in *Applied Animal Behaviour Science*, is the most directly relevant controlled work for home aquaria. It examined four commonly kept ornamental species held in 10-litre tanks at varying group sizes, scoring darting, aggression, shoaling and latency to feed, and combining them via principal components analysis into a composite welfare indication.

Table 12.2 — Saxby et al. (2010): group size and welfare in four ornamental species

Species	Group sizes tested	Reported findings
Neon tetra (<i>Paracheirodon innesi</i>)	1, 2, 5 and 10 per tank	Reduced aggression and darting, and more time shoaling, in larger groups. Composite analysis indicated improved welfare in the larger groups. Aggression was highest in the groups of two and five.
White cloud mountain minnow (<i>Tanichthys albonubes</i>)	1, 2, 5 and 10 per tank	Same pattern as neon tetras: reduced aggression and darting, increased shoaling, improved composite welfare in larger groups.
Angelfish (<i>Pterophyllum scalare</i>)	1, 2, 3 and 5 per tank	Increased shoaling in larger groups, but no clear link between group size and the composite welfare measure. Aggression was not reduced by larger numbers. Later work reports longer time spent near a neighbour and lower latency to feed in larger groups, alongside increased darting and marginally increased aggression.
Tiger barb (<i>Puntigrus tetrazona</i> , published as <i>Barbus tetrazona</i>)	Intended 1, 5 and 8 per tank	Increased shoaling in larger groups and improved composite welfare. Notably, small-group treatments were terminated on ethical grounds because aggression escalated immediately when very small numbers were placed together.

Table 12.2 — Source: Saxby A, Adams L, Snellgrove D, Wilson RW, Sloman KA (2010), *Applied Animal Behaviour Science* 125(3–4): 195–205. Group sizes and densities as reported. The study’s own conclusion is that group-size effects are species-specific.

Three things about this study deserve to survive into any downstream content. First, the effect is real and measurable. Second, it is species-specific — the angelfish result is genuinely different from the tetra result, and that difference is informative rather than inconvenient. Third, the largest group tested for the small shoaling species was ten, and ten scored best. The study therefore does not establish an upper bound, and it certainly does not establish six as a threshold.

EVIDENCE: STRONG EVIDENCE FOR THE SPECIES TESTED

12.5 Mixed-species context changes the picture again

Follow-up work by Sloman and colleagues (2011) examined the same species in mixed assemblages and reported that tiger barbs were less aggressive toward smaller white cloud mountain minnows and neon tetras when angelfish were present, apparently because they spent more time avoiding the larger fish. This is a striking result for stocking practice: the welfare consequence of a given group size can depend on what else is in the tank. It also connects directly to the dither-fish concept in Section 32.

EVIDENCE: STRONG EVIDENCE FOR THE SPECIFIC ASSEMBLAGE TESTED; DO NOT GENERALISE

13. Schooling, Shoaling and Social Grouping

Hobby usage treats these as synonyms. The literature does not, and the distinction turns out to have practical consequences for group-size advice.

Table 13.1 — Three terms the hobby uses interchangeably

Term	Definition in the behavioural literature	Practical consequence for stocking
Shoaling	Any social aggregation in which individuals remain close to one another for social reasons. Members need not be aligned or moving in a coordinated way.	The broader category. Most aquarium 'schooling fish' are shoaling most of the time. Group size still matters, but tight coordination is not the welfare target.
Schooling	A shoal in which individuals are polarised — aligned in the same direction and moving with coordinated speed and heading.	Usually a response to a stimulus such as perceived threat or strong current. A tank in which fish school constantly may be signalling chronic disturbance rather than good husbandry.
Social grouping	The broadest term. Covers species that associate socially without forming shoals in the technical sense, including loose foraging aggregations and mixed-age groups.	The most accurate description of what many popular aquarium species actually do, including corydoras and many loaches.

13.1 Why the distinction matters

A keeper who has been told their fish are 'schooling fish' may worry that their group of eight is not schooling and conclude the group is too small. In many cases the fish are shoaling normally and simply have no reason to polarise. Conversely, a keeper who sees tight, constant, high-speed schooling may take it as a sign of health when it can indicate a stressor — a new environment, excessive flow, an external disturbance, or too little cover.

The relevant welfare indicators from the aquarium literature are not 'are they schooling?' but the ones Saxby and colleagues actually measured: time spent in proximity to conspecifics, frequency of darting, frequency of aggression, and latency to feed.

EVIDENCE: ESTABLISHED SCIENTIFIC PRINCIPLE FOR THE DEFINITIONS; THE PRACTICAL READING IS A REASONABLE INFERENCE

13.2 Group-size recommendations should follow social ecology

Species differ in what their group actually does. Bronze corydoras have been observed in the wild in mixed-age, mixed-size groups, sometimes as small as three individuals, and they communicate through direct tactile interaction — nudging — which spreads information about threats and maintains proximity after a flight response. That is a very different social system from a pelagic characin shoal that relies on visual alignment. It would be surprising if a single number served both well.

EVIDENCE: STRONG EVIDENCE FOR THE CORYDORAS FIELD AND LABORATORY OBSERVATIONS

14. Minimum Group Sizes: Where Did “Six” Come From?

“Keep at least six” is possibly the most repeated number in the freshwater hobby. This section asks whether it can be traced to anything.

14.1 The honest answer about the number six

This document could not identify a scientific source establishing six as a threshold for shoaling fish in general, or for any specific commonly kept species. It appears in hobby literature and retail guidance as a convention. Where controlled work exists, the tested group sizes do not cluster around six: Saxby and colleagues tested 1, 2, 5 and 10 for the small shoaling species, and 10 performed best. Six is not a data point in that study at all.

EVIDENCE: UNSUPPORTED AS A UNIVERSAL BIOLOGICAL THRESHOLD; DEFENSIBLE AS A CONSERVATIVE FLOOR AND USEFUL RULE OF THUMB

Two things can be true simultaneously, and both should survive into downstream content:

1. **Six is not a scientifically established minimum.** Nothing in the literature identifies a discontinuity at six individuals for any species.
2. **Six is a reasonable conservative floor for retail and beginner guidance.** It is memorable, it is above the group sizes at which problems were observed in controlled work, it is affordable, and it fits in tanks people actually own. As a heuristic aimed at preventing the common failure of buying two or three, it does useful work.

The problem is not the number. The problem is presenting it as a biological fact and, worse, as a ceiling — treating six as ‘done’ when the evidence for the species tested points toward larger groups being better.

14.2 Do some species benefit from more?

Table 14.1 — What group-size evidence does and does not support

Group-size framing	What the evidence supports	Important caveat
Larger is better for small shoaling characins and cyprinids	Saxby et al. found the largest group tested (10) produced the best composite welfare score for neon tetras and white cloud mountain minnows. Behavioural ecology consistently finds vigilance and foraging benefits rising with shoal size.	The studies did not test where the benefit plateaus, and the benefit is constrained by tank capacity. ‘More is better’ is not ‘more is unlimited.’
Larger groups reduce concentrated aggression in some barbs	Tiger barb small-group treatments in Saxby et al. were terminated because aggression escalated immediately; larger groups scored better.	The mechanism is aggression distribution, which requires a group large enough that no single individual is the persistent target. What counts as large enough is not established numerically.
Some species form loose social groups rather than structured shoals	Bronze corydoras are documented in wild mixed-age groups as small as three, with low aggression and tactile communication.	This is not licence to keep three. Wild groups exist in a landscape where fish can join and leave larger aggregations; a closed tank offers no such option.
Some species need much larger groups than the hobby default	For strongly shoal-dependent, high-activity open-water species, hobby sources widely recommend 10, 12 or 20-plus.	Species-specific recommendations of this kind are mostly experienced hobbyist consensus rather than controlled findings. They should be presented as such.
Some species are worse in groups	Male bettas, and males of a number of territorial species, do not form compatible same-sex groups in aquarium-scale space.	‘More conspecifics’ is not a general welfare improvement. It is species-specific in both directions.

A defensible way to state group-size guidance

Rather than 'keep at least six,' a more accurate framing is: **'this is a social species; plan the tank around the group you intend to keep, not the other way round.'**

That reframing does the important work — it stops people buying three — without asserting a threshold the evidence does not support, and without implying that six is a finish line.

It also puts the constraint in the right order. Group size is a species requirement; tank size is the keeper's choice. The species requirement should determine the tank, not be trimmed to fit it.

EVIDENCE: ANIMAL-WELFARE PRINCIPLE

15. When “Add More Fish” Is Correct — and When It Is Not

Two symmetrical sections, because the advice is genuinely correct in some situations and genuinely harmful in others, and the difference is predictable.

15.1 When increasing group size is likely to improve welfare

Table 15.1 — Conditions under which adding conspecifics is likely to help

Situation	Expected outcome	Evidence strength
A social species currently kept below the group size at which normal behaviour is observed, in a tank with spare capacity	Reduced darting, reduced intraspecific aggression, more time shoaling, shorter latency to feed.	Strong evidence for the species tested in Saxby et al.
A shy shoal that spends most of its time hiding despite adequate cover	More time in open water, more natural activity, better feeding confidence. Group-size effects on vigilance and foraging are well documented in behavioural ecology.	Strong evidence for the mechanism; reasonable inference for the individual tank.
A group in which one or two individuals are the persistent target of aggression, in a species where aggression distributes	Aggression spread across more individuals rather than concentrated on one.	Contested. It works in some species and not others. Verify the species before recommending it.
A recently reduced group, after losses	Restoration of normal social behaviour. An undersized remnant group is a common and often unrecognised welfare problem.	Reasonable inference.
A species where the literature reports welfare benefits at higher numbers than the tank currently holds, and the footprint supports it	Improved composite behaviour.	Species-dependent. Check per species rather than generalising.

15.2 When “add more fish” is irresponsible advice

The advice becomes harmful the moment it is given without checking the other constraints. A socially incomplete group is a real problem, but it does not follow that adding fish to an unsuitable aquarium is the solution — that trades a chronic welfare problem for an acute one, or simply adds a second problem to the first.

Table 15.2 — Conditions under which adding conspecifics is the wrong answer

Condition already present	Why adding fish makes it worse	What the actual answer usually is
Tank footprint is too small for the species	More individuals in the same inadequate space intensifies competition for the limited usable area. The spatial ceiling does not move.	A longer tank, or a different species suited to this one.
Oxygen is already marginal, especially overnight or in summer	Every added fish adds respiration, and added feeding adds bacterial respiration on top.	Improve gas exchange first, verify overnight conditions, then reassess.
Nitrate rises quickly between water changes	Indicates nitrogen input is already high relative to export. More fish steepens the curve.	Increase export (water changes, plant mass) or reduce input (feeding) before adding load.
Filtration or circulation is inadequate	Ammonia capacity and solids handling are already at their limit.	Fix the filtration, then reassess. Adding fish first is the wrong order.
Territory is limited and already contested	Additional individuals compete for the same defensible resources, which typically escalates rather than distributes conflict in territorial species.	Add structure and sight breaks; reduce numbers of the territorial species; or rehome.
Fish are already showing stress signs	Adding conspecifics adds an acclimation disturbance to a group that is already struggling, and may add pathogens.	Identify and fix the existing stressor first. Adding animals is never the first response to visible stress.

Condition already present	Why adding fish makes it worse	What the actual answer usually is
The species' social needs are being misunderstood	Not all species benefit from more conspecifics. For territorial and male-intolerant species, more is actively worse.	Verify the species' actual social system before acting on generic 'fish like company' advice.

The honest hard case

A keeper has four of a shoaling species in a tank that cannot support ten. There is no comfortable answer, and pretending otherwise does the listener a disservice.

The available options are: upgrade the tank; rehome the group to someone who can house it properly and restock with a species suited to the current tank; or accept an acknowledged welfare compromise and mitigate it as far as possible with heavy cover, dense planting, careful tankmate selection and excellent water quality.

The third option is not ideal and should not be presented as though it were. But it is what many keepers will actually do, and giving them mitigation guidance is more useful than giving them only a verdict. This case appears as Scenario T in Section 28.

EVIDENCE: ANIMAL-WELFARE PRINCIPLE

16. Group Size Versus Tank Size and Geometry

“Keep ten” without a tank-size qualifier is incomplete advice. Group-size recommendations and footprint recommendations are the same recommendation stated from two ends.

16.1 What group size interacts with

- **Tank length.** The dimension that determines continuous swimming path. For active shoaling species this matters more than volume.
- **Tank width (front to back).** Determines turning room and the usable bottom area for benthic species. Frequently the neglected dimension.
- **Open swimming space.** Not the same as volume. A heavily aquascaped 40-gallon tank may offer less open water than a sparsely furnished 29-gallon.
- **Hardscape.** Adds surface area and sight breaks, both of which can raise the number of individuals that coexist comfortably — while reducing open volume.
- **Cover.** Plants and shade allow fish to modulate their own exposure. Groups in tanks with good cover typically spend *more* time in the open, not less, because retreat is available.
- **Current.** Species adapted to flowing water use current as a behavioural resource; species from still water find strong flow a stressor. The same tank can be understocked for one and overstocked for the other.
- **Territorial zones.** The number of defensible sites (caves, overhangs, plant thickets) can set the number of territorial individuals more directly than volume does.

16.2 Why the two recommendations must be given together

A recommendation of the form ‘keep at least ten’ assumes a tank capable of holding ten. Given to someone with a 10-gallon tank, it produces one of two bad outcomes: they overstock, or they conclude the advice is unrealistic and ignore all of it. Either way the fish lose.

The more useful construction pairs the two: *‘this species is social and does best in groups of around ten or more, which in practice means a tank of at least this footprint. If that is not available, this is not the right species for this tank.’* That framing is longer and it is the only version that is actionable.

EVIDENCE: ANIMAL-WELFARE PRINCIPLE

17. Swimming Space, Footprint and Surface Area

Two related arguments for why the shape of a tank matters at least as much as its stated volume — one behavioural, one physical.

17.1 Why length can matter more than gallons

Volume is a poor descriptor of the space a fish actually uses. Fish move horizontally far more than vertically. Continuous swimming distance, turning radius, and the ability to move away from a conspecific are all functions of the longest horizontal dimension, and none of them improve when a tank is made taller.

For active open-water species — danios, larger rasboras, many barbs, hillstream-type fishes — a long, shallow tank is functionally larger than a tall tank of the same volume. For deep-bodied species such as angelfish, height genuinely does matter, because their body form requires vertical clearance. This is one of the few cases where tall tanks have a defensible advantage, and it is worth stating precisely rather than dismissing tall tanks generally.

EVIDENCE: REASONABLE INFERENCE; WIDELY SUPPORTED BY COMMON AQUARIUM PRACTICE

Table 17.1 — Tank proportions and what they suit

Tank form	What it favours	What it penalises
Long and shallow	Active horizontal swimmers; benthic species (more bottom area); gas exchange (more surface per unit volume); distributing territorial fish along a length.	Deep-bodied species that need vertical clearance; tall plant arrangements; certain aquascaping styles.
Tall and narrow	Deep-bodied species such as angelfish; vertically structured aquascapes; species that use midwater column height.	Horizontal swimming distance; bottom area for benthic species; surface area for gas exchange relative to volume.
Cube	Aquascaping and viewing from multiple angles; species that use structure rather than open water.	Continuous swimming length. A cube offers the least usable linear distance per unit of volume.
Breeder-proportioned (wide front to back)	Benthic species; territorial species that need separated ground-level zones; species that use the bottom in two dimensions such as corydoras.	Little. This is often the most useful proportion per gallon for freshwater community and benthic stocking.

17.2 Surface area and gas exchange

The historical surface-area stocking rules were built on a real physical principle: oxygen enters water principally at the air-water interface, so in an unaerated tank the rate of oxygen replenishment scales with surface area rather than with volume. Aquarium reference works give figures on the order of 12 square inches of surface per inch of slender-bodied fish and roughly 20 or more square inches per inch of deep-bodied fish. The distinction between the two body types is itself evidence that the authors understood the length-only version was insufficient.

EVIDENCE: ESTABLISHED PRINCIPLE FOR THE PHYSICS; THE SPECIFIC FIGURES ARE COMMON AQUARIUM PRACTICE

17.3 How modern equipment changes the relationship

Surface agitation from a filter outflow, a powerhead or an air stone increases the effective rate of gas exchange substantially by continuously renewing the surface film and increasing the effective interface area. In a well-agitated tank the raw surface-area figure is less limiting than it was in the 1960s, which is a genuine reason the old rules are more conservative than modern practice requires.

But two things do not change. First, agitation can bring water toward saturation; it cannot exceed saturation, and saturation falls with temperature. Second, surface area still matters for swimming and territory independently of gas exchange, and it correlates with footprint — the dimension that governs the behavioural ceiling. The surface-area rule remains the most physically principled of the classic heuristics even though the exact figures are now

conservative.

One caveat frequently missed: in a planted tank running injected carbon dioxide, aggressive surface agitation drives carbon dioxide out of solution. Keepers in that situation are balancing gas exchange against plant carbon availability, and heavy stocking narrows the window in which both can be satisfied.

EVIDENCE: ESTABLISHED PRINCIPLE

18. Oxygen as a Stocking Limit

The constraint most likely to be binding and least likely to be measured. A tank can process ammonia perfectly and still be oxygen-limited at four in the morning in August.

18.1 The physical starting point

Oxygen solubility in freshwater falls as temperature rises. At sea level, air-saturated freshwater holds approximately 14.6 mg/L at 0 degrees C, about 9.1 mg/L at 20 degrees C, about 8.2–8.3 mg/L at 25 degrees C, and about 7.0 mg/L at 35 degrees C. A tropical community tank at 26 degrees C is therefore working with roughly 8 mg/L at full saturation, and real tanks sit below saturation because respiration is continuous.

EVIDENCE: ESTABLISHED SCIENTIFIC PRINCIPLE

Table 18.1 — Oxygen solubility and temperature

Temperature	Approximate saturation DO in freshwater at sea level	Relative to 20 degrees C
0 deg C (32 deg F)	about 14.6 mg/L	160 percent
20 deg C (68 deg F)	about 9.1 mg/L	100 percent
25 deg C (77 deg F)	about 8.2–8.3 mg/L	about 91 percent
30 deg C (86 deg F)	about 7.5 mg/L	about 83 percent
35 deg C (95 deg F)	about 7.0 mg/L	about 77 percent

Table 18.1 — Values are approximate, for sea level and zero salinity. Altitude reduces these figures. Sources include FAO aquaculture reference material and standard environmental water-quality references.

18.2 What consumes oxygen in an aquarium

- **Fish respiration** — scales with biomass, temperature, activity, digestion and stress. Rises after feeding.
- **Invertebrate respiration** — smaller per individual but non-zero, and populations can be large.
- **Nitrifying bacteria** — nitrification is an aerobic process with a substantial stoichiometric oxygen demand per unit of ammonia oxidised.
- **Heterotrophic bacteria** — often the largest and least visible consumer in a heavily fed tank, breaking down dissolved and particulate organics. This is the mechanism by which overfeeding causes oxygen problems that look like nothing on a nitrogen test kit.
- **Plants at night** — plants respire continuously and only photosynthesise under light. A densely planted tank is a net oxygen producer by day and a net consumer at night.
- **Decomposing organic matter** — uneaten food, dead plant material, mulm, and any unnoticed mortality.

18.3 The nightly minimum is what matters

Aquaculture experience is explicit on this point: the response of stocked animals is governed primarily by the lowest dissolved oxygen concentration reached overnight, not by the daytime average. Pond management guidance recommends monitoring particularly between midnight and dawn. In a planted aquarium the same daily cycle operates — oxygen peaks late in the photoperiod and troughs shortly before lights come on.

The practical consequence for stocking is that a keeper who tests in the evening is testing at the best moment of the day. Any oxygen assessment that informs a stocking decision should be made in the early morning.

EVIDENCE: AQUACULTURE PRACTICE, ON ESTABLISHED PRINCIPLE

18.4 Thresholds, and why they are not simple

General guidance for aquatic communities commonly treats about 5 mg/L as a conservative floor, with values below that adversely affecting biological communities and values around 2 mg/L or below being lethal to most fishes. Aquaculture practice is more nuanced: warm-water species often survive dissolved oxygen as low as around 1 mg/L, and cold-water species around 2.5–3.5 mg/L — but survival is not the target. Fish at low dissolved oxygen are stressed, more susceptible to disease and grow poorly. A common operational guideline in culture systems is not to allow dissolved oxygen to fall below 50 percent of saturation.

Note that 50 percent of saturation at 26 degrees C is roughly 4 mg/L — a number that is difficult to guarantee overnight even in well-managed pond systems. The gap between 'survives' and 'thrives' is exactly where stocking judgement lives.

EVIDENCE: AQUACULTURE PRACTICE WITH STRONG SUPPORTING EVIDENCE; SPECIFIC THRESHOLDS ARE SPECIES-DEPENDENT

Why a tank can pass every chemistry test and still be oxygen-limited

Standard hobby test kits measure ammonia, nitrite, nitrate, pH and hardness. None of them measures dissolved oxygen, and almost no hobbyist owns a dissolved oxygen meter.

A tank can therefore show zero ammonia, zero nitrite and low nitrate — because the biofilter is coping and water changes are frequent — while running at a dangerously low overnight oxygen minimum driven by fish respiration, bacterial respiration and plant respiration combined.

The observable signs are behavioural: fish at the surface in the early morning, elevated gill movement, listlessness, reduced feeding, and clustering near filter outflows. These are discussed in Section 24.

19. Temperature and Stocking

Temperature is the variable that moves every other constraint at once, and it moves most of them in the same unhelpful direction.

Table 19.1 — What a temperature rise does to a stocked aquarium

What temperature affects	Direction as temperature rises	Net effect on stocking capacity
Oxygen solubility	Falls. Roughly 9.1 mg/L at 20 deg C to about 7.5 mg/L at 30 deg C.	Reduces supply.
Fish metabolic rate	Rises. Clarke & Johnston reported a mean Q_{10} around 1.83 across species, with within-species acclimation studies nearer 2.40 — roughly a doubling of resting metabolism per 10 degree C rise.	Increases demand.
Fish oxygen demand	Rises with metabolic rate, and rises further with the increased activity and feeding that accompany warmth.	Increases demand, on top of the metabolic effect.
Feeding and therefore ammonia production	Rises. Warmer fish eat more and excrete more nitrogen.	Increases biofilter load.
Bacterial metabolism	Rises. Both nitrifying and heterotrophic activity increase.	Mixed: faster nitrification helps, but higher heterotrophic respiration consumes more oxygen.
Ammonia toxicity	Rises. The equilibrium shifts toward the un-ionised NH_3 form as temperature increases, at any given pH.	The same total ammonia reading is more dangerous.

19.1 Why a stocking level that works at 74 degrees F behaves differently in a heatwave

The mechanisms stack rather than offset. Supply falls while demand rises in fish, rises in bacteria, and rises through increased feeding. Ammonia toxicity rises at the same time. A tank running comfortably at 23–24 degrees C can be genuinely marginal at 30–31 degrees C during a summer heat event, without anything about the stocking having changed.

This is one of the strongest practical arguments for keeping a safety margin rather than stocking to the apparent limit. A tank stocked to its normal-conditions ceiling has no room for the weather. Section 34 develops this into a general principle.

EVIDENCE: REASONABLE INFERENCE FROM ESTABLISHED PRINCIPLES, EACH OF WHICH IS INDIVIDUALLY WELL SUPPORTED

Practical implications during a heat event

Increase surface agitation. This is the fastest and most effective intervention and it costs nothing.

Reduce feeding. Lower nitrogen input, lower post-feeding oxygen demand, and less organic material for heterotrophs.

Do not add livestock during a heat event, regardless of what a stocking calculator says. The calculator is not modelling the weather.

Watch behaviour in the early morning, which is when the oxygen minimum occurs.

Note that lowering temperature deliberately is a legitimate tool for reducing metabolic demand, though it should be done gradually rather than abruptly.

20. Territory and Behavioural Carrying Capacity

A tank can have flawless water chemistry and still be overstocked, because the constraint being exceeded is not chemical. This is the constraint that no test kit measures and no calculator can fully model.

20.1 What territoriality actually claims

Territorial fish do not defend volume. They defend resources and the space around them: a cave, an overhang, a patch of substrate, a flat stone, a plant thicket, a section of the bottom, a stretch of the surface film. This is why adding water volume often does nothing for a territorial conflict while adding hardscape and sight breaks can transform it.

- **Breeding territories.** Frequently the trigger. A previously peaceful pair can claim and defend a substantial area once spawning begins, and the tank's effective size for everyone else drops overnight.
- **Pair bonding.** In many cichlids, pair formation converts two individuals into a coordinated defensive unit, which is behaviourally quite different from two separate fish.
- **Line of sight.** Aggression in many territorial species is triggered visually. Interrupting sight lines with rock, wood or planting can reduce conflict without changing any dimension of the tank. Aquarium literature on cichlid keeping treats sight-line disruption as a primary tool.
- **Cave and shelter competition.** Where shelters are scarce relative to the number of individuals that want one, conflict is structural. The number of defensible sites can be the operative stocking limit.
- **Zonal territory.** Bottom, midwater and surface territories are partly independent, which is the legitimate core of the 'niche' argument in Section 21.

EVIDENCE: ESTABLISHED PRINCIPLE IN BEHAVIOURAL ECOLOGY; THE SPECIFIC AQUARIUM TECHNIQUES ARE COMMON AQUARIUM PRACTICE WITH GOOD SUPPORTING RATIONALE

20.2 Habitat complexity cuts both ways

This is more nuanced than hobby advice usually allows, and the research literature is explicit about the tension. Structural complexity can delimit individual territories and allow subordinate individuals to avoid conflict. But physical structures can also *raise the value of a resource* and encourage active defence by whoever owns it — an effect documented in mice, zebrafish and cichlids. Work in rainbow trout found that enrichment with cobbles and dividers reduced individual territory sizes as habitat visibility decreased, which supported a higher density of territory holders.

So the accurate statement is not 'add caves to reduce aggression.' It is: **structure changes the geometry of conflict. It usually helps when it breaks sight lines and provides retreat. It can hurt when it creates a small number of high-value contested resources.** One cave among six cichlids is a fight. Ten caves among six cichlids is a community.

EVIDENCE: STRONG EVIDENCE FOR THE MECHANISM; CONTESTED IN APPLICATION

20.3 The trade-off with open swimming volume

Adding hardscape and dense planting improves behavioural capacity and simultaneously reduces open swimming space and actual water volume. For a territorial or shy community the trade is usually worth making. For a tank of active open-water shoaling fish it may not be. This is a real design tension and the resolution depends on which species is present — another reason a single stocking formula cannot exist.

21. Aggression, Crowding and Social Stress

The relationship between density and aggression is genuinely bidirectional depending on species and context. This is one of the few places in the document where the honest answer is that the evidence points different ways.

21.1 The two opposing patterns

Table 21.1 — Density and aggression: three documented patterns

Pattern	Where it is observed	Proposed mechanism	Important caveat
More crowding produces more aggression	Territorial species with defensible resources; low-density housing of some species; situations where escape is impossible.	Encounter rate rises with density while escape options fall, so contests occur more often and resolve less cleanly.	This is the intuitive pattern and is frequently correct, but it is not universal.
Higher density distributes and reduces aggression	Documented in Nile tilapia under aquaculture intensification, where social structure shifts from dominance hierarchies toward shoaling as density rises. Also the basis for the mbuna 'controlled crowding' practice.	No individual can successfully monopolise a territory, so hierarchy formation is disrupted and any single individual receives a smaller share of total aggression.	Critically important: in the tilapia work, high-density fish were less aggressive but also darker in skin colour, a proxy for stress , less active, and showed gene expression consistent with a reactive stress-coping style and activation of the stress axis.
Very low density produces more aggression	Documented in zebrafish: fish held at 1 fish per litre showed higher aggression, higher water cortisol, and altered vertical distribution compared with 4–16 fish per litre.	Shoaling species held below their social threshold form dominance relationships instead of shoals.	This is the finding that most directly supports 'add more fish' as welfare advice, and it is worth noting the study still found an upper working range rather than 'more is always better.'

The crowding finding that matters most

Transcriptomic work on Nile tilapia found that fish reared at high density were less aggressive than those at low density — and also darker, less active, with larger brains, and with hypothalamic gene expression consistent with a reactive stress-coping style and activation of the hypothalamic-pituitary-interrenal axis.

That is the whole argument about crowding in one result. Reduced aggression is not the same thing as improved welfare. A crowded fish that has stopped fighting may have stopped because it is suppressed, not because it is content.

Any downstream use of the 'overstock mbuna to reduce aggression' practice should carry this caveat. The practice may genuinely reduce visible violence. Whether it improves welfare is a separate and much less settled question.

EVIDENCE: STRONG EVIDENCE FOR THE TILAPIA FINDING; THE EXTENSION TO RIFT LAKE CICHLID KEEPING IS A CONTESTED CLAIM

21.2 The zebrafish density literature

Zebrafish are the best-studied ornamental-adjacent species, and the findings are instructive precisely because they are not simple. One nine-week study found fish at 1 fish per litre showed elevated aggression and elevated water cortisol relative to 4, 8, 12 and 16 fish per litre, concluding that zebrafish can be held between 4 and 16 fish per litre without compromising the welfare indicators measured, and should not be held at 1 fish per litre. A separate study rearing fish at 1, 3 or 6 fish per litre similarly found the lowest density produced the highest aggression and cortisol. Meanwhile other work found lower cortisol release at 5 fish per litre than at 10, 20 or 40, and a systematic review

concluded that growth, reproduction, stress response and behaviour are all affected by density and tank size, with at least 5 fish per litre suggested to minimise physiological stress. European regulatory guidance has proposed a maximum around 10 fish per litre.

The pattern across that body of work is a **window**, not a ceiling or a floor: too few is bad, too many is bad, and the acceptable band is wide and somewhat inconsistent between studies. Reviewers have explicitly noted that the evidence is sometimes contradictory. That honest irregularity is more useful for the hobby than a clean number would be, because it matches what keepers actually observe.

EVIDENCE: STRONG EVIDENCE THAT DENSITY MATTERS; CONTESTED ON SPECIFIC OPTIMAL VALUES

21.3 Chronic social stress and its consequences

Chronic stress in fish is well documented in the aquaculture and welfare literature. The plausible pathway from social stress to poor outcomes runs through sustained activation of the stress axis, which in fish as in other vertebrates is associated with metabolic costs and immune modulation.

Table 21.2 — Chronic social stress: sources and consequences

Source of chronic social stress	Plausible consequences	Confidence
Undersized groups of social species	Elevated cortisol, elevated aggression, reduced shoaling, altered spatial distribution — all directly observed in zebrafish at 1 fish/L and in the small-group treatments of Saxby et al.	Well supported for the species tested.
Incompatible tankmates	Persistent avoidance behaviour, suppressed feeding in subordinates, physical damage.	Well supported as a phenomenon; magnitude is situation-specific.
Lack of cover or retreat	Inability to modulate exposure. Enrichment studies consistently find cover reduces stress responses, though not in every experiment.	Generally supported; some studies find no effect over short exposure periods.
Sustained aggression toward a specific individual	Reduced feeding, poor condition, and in the aquaculture literature, impaired growth.	Well supported.
Competition at feeding	Unequal food access, divergent growth within a group, and persistent hunger in subordinates.	Well supported.

A necessary caution on causal claims: it is well established that chronic stress is associated with immune modulation and increased disease susceptibility across vertebrates. It does not follow that any specific outbreak in any specific tank was caused by social stress. The honest framing is that chronic stress raises risk, not that it explains individual events.

EVIDENCE: ANIMAL-WELFARE PRINCIPLE WITH STRONG SUPPORTING EVIDENCE; INDIVIDUAL CAUSAL ATTRIBUTION IS NOT SUPPORTABLE

22. Niches, Bottom Dwellers, Shrimp and Snails

Three closely related claims that all take a partial truth and extend it past where it holds.

22.1 Does occupying different zones improve use of space?

Yes, genuinely, and this is a legitimate aquascaping and stocking principle. Surface-dwelling, midwater and benthic species use partly separate physical space and partly separate defensible territory. A community distributed across three zones experiences less spatial competition than the same number of fish all competing for the midwater.

The principle is real. The error is extending it from **space** to **biology**.

22.2 “Bottom dwellers do not count”

Table 22.1 — The bottom-dweller question, answered directly

Question	Answer	Explanation
Do bottom dwellers consume oxygen?	Yes.	Every fish respire continuously regardless of which part of the tank it occupies. Benthic species are often less active than open-water species, which lowers their demand somewhat, but it does not approach zero. Corydoras and some loaches additionally take atmospheric air, which changes their tolerance of low oxygen but not their contribution to system oxygen demand.
Do they produce waste?	Yes.	They eat, therefore they excrete nitrogen. A common misconception is that they ‘eat the waste,’ which is doubly wrong: most commonly kept bottom dwellers require their own dedicated food, and reprocessing organic matter does not remove nitrogen from the system.
Do they compete for territory?	Some do, considerably.	Many plecos and some loaches are territorial, particularly around caves and at maturity. Corydoras are notably low-aggression, but they compete for foraging area, which makes bottom footprint their operative constraint.
Do they contribute to biomass?	Yes, and sometimes a great deal.	A common pleco is one of the highest-biomass, highest-waste fish in the freshwater hobby at adult size. Counting it as ‘a cleaner’ rather than as a large fish is among the most consequential errors in beginner stocking.

Restated cleanly: **using a different swimming zone reduces spatial and territorial competition. It does not reduce oxygen demand, nitrogen output, or biomass by any amount.**

EVIDENCE: ESTABLISHED PRINCIPLE; THE CLAIM THAT BOTTOM DWELLERS DO NOT COUNT IS UNSUPPORTED AQUARIUM FOLKLORE

22.3 Shrimp and snails

Invertebrates are frequently treated as free. They are not free, but they are genuinely cheaper per animal than fish, and both halves of that sentence matter.

Table 22.2 — Invertebrate contributions to stocking

Invertebrate	Biomass and waste contribution	Stocking considerations
Dwarf shrimp (<i>Neocaridina</i> , <i>Caridina</i>)	Individually very small — on the order of a fraction of a gram at adult size. Low individual contribution.	The catch is population growth. A stable <i>Neocaridina</i> colony can reach several hundred individuals in a small tank, at which point the aggregate biomass and feeding demand are no longer negligible. Shrimp are also more sensitive than most fish to ammonia, nitrite, copper and dissolved oxygen depletion, so they are often the first casualties of an overstocked system rather than an inert addition to it.

Invertebrate	Biomass and waste contribution	Stocking considerations
Mystery / apple snails (<i>Pomacea</i> spp.)	Substantial. An adult mystery snail is a large-bodied animal with a high feeding rate and visible waste output.	These should be counted as meaningful livestock. Several adult mystery snails in a small tank represent a real bioload increase, comparable to adding fish of similar mass.
Nerite snails	Moderate and individually small. Do not reproduce in freshwater, so populations remain fixed at whatever the keeper stocked.	The population stability is their main stocking advantage. Load is predictable and does not compound.
Pest snails (<i>Physa</i> , ramshorn, Malaysian trumpet)	Individually tiny; collectively significant when populations are large.	Population size tracks available food, which means a large pest snail population is usually a <i>symptom</i> of overfeeding rather than an independent cause of high bioload. Treating the population as the problem often misses the actual driver.
Amano shrimp, filter shrimp, larger species	Larger individual biomass than dwarf shrimp; correspondingly higher individual feeding and waste.	Count them, at roughly the scale you would count a small fish of similar mass.

How to frame invertebrates honestly

Shrimp and snails are **livestock and ecosystem members**, not equipment and not a solution to a waste problem. They have their own welfare requirements, their own environmental sensitivities, and their own contribution to system load.

The accurate statement is that **individual invertebrates contribute much less than individual fish of the same nominal size, and populations of them can contribute a great deal**. 'They do not count' is wrong; 'they count the same as fish' is also wrong.

A related point worth making: a large snail population does not 'clean' a tank in any nitrogen-removal sense. Detritivores convert particulate organic matter into dissolved nitrogen faster, which makes it more available to the biofilter and to plants. That is genuinely useful for cycling organic matter through the system, and it removes nothing.

EVIDENCE: REASONABLE INFERENCE; THE INDIVIDUAL MASS FIGURES ARE APPROXIMATE

23. Plants and Stocking Capacity

Live plants genuinely change a system's nitrogen and oxygen dynamics. They do not make stocking unlimited, and the mechanism by which they help is often misunderstood.

23.1 What plants actually do

Table 23.1 — What live plants contribute and what they do not

Mechanism	What it does	Limits
Ammonium uptake	Aquatic plants take up ammonium directly and generally preferentially over nitrate, because assimilating ammonium is energetically cheaper than reducing nitrate first. Split-chamber work on aquatic macrophytes indicates leaf uptake from the water column is a real and often preferred route rather than root uptake only.	Uptake rate is limited by growth rate, light, carbon availability and other nutrients. Uptake stops in darkness. Plants cannot buffer an acute ammonia event on their own.
Nitrate uptake	Plants assimilate nitrate once ammonium is depleted, which is why well-planted tanks often show low nitrate.	This is the mechanism by which planting can mask a nitrogen load. Low nitrate in a heavily planted tank tells you less about stocking than low nitrate in an unplanted one.
Oxygen production	Photosynthesis during the photoperiod. A densely planted, well-lit tank can approach or exceed saturation during the day.	Only during the photoperiod. Section 18 covers the night-time reversal.
Night respiration	Plants respire continuously. At night a dense plant mass is a net oxygen consumer.	This is a genuine stocking constraint, not a footnote. Heavy planting plus heavy stocking plus warm water is the combination most likely to produce a pre-dawn oxygen minimum.
Surface area for biofilm	Plant surfaces host nitrifying and heterotrophic biofilm, adding distributed processing capacity.	Modest relative to dedicated media and substrate, but real.
Cover and sight breaks	Changes behaviour: more time in the open, less aggression in many species, refuge for subordinates and for fry.	Reduces open swimming volume. For fry-producing livebearers, more cover means more surviving fry, which is a stocking consequence in itself.

EVIDENCE: ESTABLISHED PRINCIPLE FOR THE MECHANISMS; THE AQUARIUM-SCALE MAGNITUDE IS A REASONABLE INFERENCE AND VARIES ENORMOUSLY

23.2 Fast-growing versus slow-growing plants

Nutrient uptake is a function of growth. A plant that is not growing is not removing nitrogen, whatever its leaf area. This is why the practical distinction is between fast and slow growers rather than between 'planted' and 'unplanted.'

Table 23.2 — Plant growth rate and nutrient uptake

Plant type	Typical nutrient uptake rate	Role in the stocking picture
Floating plants	Often the highest, because they have unrestricted access to atmospheric carbon dioxide and to strong light.	The most effective nutrient sink available to a low-tech tank. Also shades the tank, which changes behaviour, and reduces surface gas exchange where coverage is heavy — a real trade-off in a heavily stocked system.
Fast-growing stem plants	High when actively growing and regularly trimmed.	Nutrient removal happens when the trimmings leave the tank. Uncut stems that eventually decay return their nitrogen. Export requires physical removal.
Epiphytes (anubias, java fern, bucephalandra)	Low. These are slow growers.	Valuable for cover, structure and sight breaks — behavioural benefits — but they should not be counted on for meaningful nutrient export.

Plant type	Typical nutrient uptake rate	Role in the stocking picture
Carpeting plants	Variable; often demanding of light and carbon rather than high in net nitrogen removal.	Primarily aesthetic and structural in stocking terms.
Mosses	Low to moderate.	Excellent cover for shrimp and fry, which has population consequences.

The correct summary of the plant question

Plants can **raise the biological ceiling** by intercepting ammonium before it reaches the filter, exporting nitrogen when trimmings are removed, and adding daytime oxygen.

Plants do **not** raise the oxygen ceiling at night, the spatial ceiling, the territorial ceiling, or the maintenance ceiling. Dense planting reduces open swimming volume and reduces actual water volume.

So 'I have lots of plants, can I keep more fish?' has a genuinely two-part answer: *yes for nitrogen, not necessarily for anything else, and the night-time oxygen picture is actually worse.*

24. Filtration Versus Stocking

“A bigger filter lets you keep more fish” is partly true in a narrow sense and misleading as a general claim. The distinction rests on separating two things the word ‘capacity’ conflates.

24.1 Ammonia processing capacity is not aquarium carrying capacity

Table 24.1 — What more filtration does and does not do

A larger or additional filter genuinely improves...	...and cannot fix
Ammonia and nitrite oxidation capacity, by providing more colonisable surface area with adequate flow and oxygen.	Dissolved oxygen availability. A filter does not create oxygen; at best it improves gas exchange through surface agitation, which is a circulation effect rather than a filtration one.
Mechanical removal of solids, which reduces downstream oxygen demand and the accumulation of decomposing material.	Open swimming space. This is fixed by the glass.
Circulation, which reduces dead zones and distributes heat, oxygen and nutrients.	Territory. Adding filtration does not add defensible sites, and strong flow can actually reduce usable space for some species.
Redundancy. Two filters mean a single mechanical failure is not a total failure — a real and underrated benefit.	Nitrate and dissolved organic accumulation. Nitrification produces nitrate; it does not remove nitrogen from the system. Only water changes, plant export and denitrification do that.
Resilience during maintenance, because media can be cleaned in stages.	Social stress and inappropriate group sizes.
Buffering against feeding spikes, because there is more standing nitrifier population to absorb a peak.	Adult size mismatches. No filter makes a tank longer.

EVIDENCE: ESTABLISHED PRINCIPLE

24.2 Filter ratings

Ratings of the form ‘for aquariums up to 50 gallons’ are manufacturer marketing figures. There is no independent industry standard defining what fish load, feeding regime, temperature or maintenance interval a given rating assumes. Two filters bearing the same rating can differ substantially in media volume, flow rate, and the proportion of their media that receives adequate flow.

Table 24.2 — What a filter rating does not tell you

Does the rating account for...	Answer
Fish biomass?	Not in any published, verifiable way. The rating is expressed in tank volume, which does not determine load.
Feeding rate?	No. Yet feeding is the primary driver of the ammonia the filter must process.
Plants?	No. A heavily planted tank places less demand on the filter for nitrogen processing than an unplanted one.
Maintenance frequency?	No. A filter serviced regularly performs very differently from one clogged with accumulated solids.
Flow through media versus bypass?	Not disclosed in a comparable way. Turnover rate is usually published; the proportion of media receiving useful flow is not.
Media surface area?	Sometimes stated, but not in a standardised unit that permits comparison between manufacturers.
Temperature?	No, despite temperature affecting both ammonia production and nitrification rate.

The clearest way to put it**A filter rating is a rating of the filter, not a licence for the livestock.**

A filter rated for 50 gallons installed on a 20-gallon tank gives that tank better ammonia processing headroom, better solids handling and more circulation. Those are real benefits and worth having. What it does not give the tank is 50 gallons of swimming room, 50 gallons of oxygen reserve, or 50 gallons of dilution against nitrate accumulation.

This is Scenario C in Section 28, and it is one of the most common beginner questions in the hobby.

EVIDENCE: MANUFACTURER RECOMMENDATION; THE CAUTION IS ESTABLISHED PRINCIPLE

25. Water Changes, Nitrate and Other Indicators

How maintenance interacts with sustainable stocking, and why the most commonly cited stocking indicator is the weakest one.

25.1 Water changes as an export mechanism

Nitrification converts ammonia to nitrate. It does not remove nitrogen. In a closed system without substantial plant export or denitrification, nitrogen accumulates as nitrate until water changes remove it. Alongside nitrate, water changes export dissolved organic carbon, phosphate and accumulated dissolved metabolites, and replace minerals and alkalinity consumed by nitrification.

The relationship between stocking and maintenance is therefore direct: **a higher stocking level requires a higher export rate to reach the same steady-state concentration.** That can be achieved through larger changes, more frequent changes, or more plant export.

EVIDENCE: ESTABLISHED PRINCIPLE

25.2 What increased maintenance can and cannot compensate for

Table 25.1 — The reach and the limits of maintenance

More frequent or larger water changes can compensate for...	...and cannot compensate for
Nitrate accumulation.	Inadequate footprint or swimming space.
Dissolved organic carbon build-up, and its associated oxygen demand.	Territorial conflict.
Alkalinity and mineral depletion from nitrification.	Undersized social groups.
Phosphate accumulation and associated algae pressure.	Night-time oxygen minima. A water change is a moment in time; the oxygen trough happens at 4 a.m.
Accumulated dissolved metabolites not measured by hobby kits.	Adult size mismatch.
Some degree of overfeeding, at the cost of the keeper's time.	Chronic stress from incompatible tankmates.

The left column is genuinely powerful. A keeper doing large weekly water changes can sustain a stocking level that would degrade badly under monthly changes. The right column is why 'I change water twice a week' does not settle a stocking argument.

25.3 Nitrate as a stocking indicator

"My nitrate is under 10 ppm, so I can add more fish" is one of the most common reasoning errors in the hobby. Low nitrate is consistent with several quite different situations.

Table 25.2 — Six reasons nitrate might read low

Why nitrate might be low	What it tells you about stocking
Genuinely low stocking and low feeding	The reading is informative here. Nitrogen input is low.
Heavy plant uptake	Nitrogen input may be substantial and simply intercepted. The reading tells you about the plants, not the fish.
Frequent or large water changes	Export is high. The reading tells you about the keeper's schedule.
Denitrification in deep substrate or in low-oxygen zones within filter media	Real in some systems, difficult to quantify, and not something a hobbyist can verify.

Why nitrate might be low	What it tells you about stocking
Recent water change before testing	The reading is a snapshot near the bottom of the sawtooth. The trend between changes is far more informative than any single value.
Test kit inaccuracy	Hobby nitrate kits are notoriously imprecise and require vigorous shaking of one reagent bottle. A low reading may simply be wrong.

What low nitrate does not prove

Low nitrate is **consistent with** adequate biological capacity. It does **not prove** adequate dissolved oxygen, adequate territory, adequate swimming space, appropriate group sizes, compatible behaviour, or appropriate planning for adult size.

A more useful question than 'what is my nitrate?' is '**how fast does my nitrate rise between water changes?**' The rate of accumulation is a direct measure of net nitrogen input minus export, which is exactly the quantity that stocking affects. A single number taken at an arbitrary point in the cycle is close to uninformative.

EVIDENCE: ESTABLISHED PRINCIPLE

25.4 TDS and conductivity

Total dissolved solids and its close proxy conductivity are sometimes proposed as accumulation indicators. They deserve a cautious treatment because they are frequently over-interpreted.

- **What TDS can indicate.** It responds to the total ionic content of the water. A rising TDS between water changes reflects accumulation of dissolved material, some of which is waste-derived. Tracked over time in one tank, a rising trend is a genuine signal that something is accumulating.
- **What TDS cannot tell you.** It is completely non-specific. It cannot distinguish nitrate from calcium from fertiliser from evaporation concentration from remineralised RO water. A high TDS reading in a hard-water area says nothing about stocking at all.
- **The evaporation confound.** Topping off with tap water rather than pure water concentrates minerals over time, which raises TDS entirely independently of livestock.
- **Where it is genuinely useful.** In shrimp keeping and in soft-water specialist tanks, where the keeper is deliberately controlling mineral content and knows the baseline. As a general stocking indicator for a typical community tank, it is weak.

EVIDENCE: REASONABLE INFERENCE; TDS AS A STOCKING METRIC IS CONTESTED AND GENERALLY OVER-CLAIMED

26. Stocking Calculators

Neither the dismissal nor the endorsement is warranted. Calculators do specific things well and other things not at all, and knowing which is which is more useful than a verdict.

26.1 What typical calculators consider

Popular aquarium stocking calculators generally take some subset of the following inputs: tank dimensions or volume; species selected from a database; number of each species; filtration model or turnover rate; and sometimes an assumed water-change frequency. They then return a percentage of capacity, along with warnings about compatibility, minimum group sizes, adult size, temperature and pH overlap, and tank-length requirements.

Table 26.1 — Calculator inputs and how well they are handled

Input a calculator typically uses	What it does with it	Quality of that handling
Tank dimensions	Derives volume and often checks minimum length against species requirements.	Good. Length checking is one of the most valuable things calculators do and one of the things beginners most often get wrong.
Species identity	Looks up adult size, temperament, group requirements, water parameter ranges and swimming zone.	Genuinely valuable. This encodes species knowledge a beginner does not have and cannot easily acquire.
Adult size	Uses adult rather than purchase size in the load calculation.	Very good, and a real advantage over inch-per-gallon as commonly applied.
Filtration	Adjusts an assumed filtration capacity term.	Weak. Turnover rate is a poor proxy for biological capacity, and the underlying model is usually undisclosed.
Water-change assumptions	Sometimes offered as a user-selectable schedule that shifts the resulting percentage.	Directionally correct and quantitatively unverifiable.
Territoriality and compatibility	Flags known problem combinations.	Useful as a warning system. Cannot account for individual variation.
Minimum school size	Warns when a social species is stocked below a database threshold.	Valuable, though the thresholds are hobby conventions rather than research findings — see Section 14.

26.2 What calculators cannot know

The limitation is not that the algorithms are bad. It is that the most important variables are unobservable from a web form.

Table 26.2 — What no calculator can observe

Variable	Why it is invisible to a calculator	How much it matters
Individual aggression	Fish have individual personalities. Betta welfare research explicitly notes behavioural variation between individuals of the same species.	Can be decisive. One unusually aggressive individual can make a theoretically sound stocking plan unworkable.
Aquascape and sight lines	The calculator sees dimensions, not layout. Structure and sight breaks substantially change behavioural capacity.	High for territorial species; moderate otherwise.
Plant mass and growth rate	Not an input in most calculators, and even where it is, actual uptake depends on light, carbon and trimming discipline.	Moderate to high for nitrogen; the night oxygen effect is invisible.
Actual flow pattern	Turnover rate does not describe circulation. Dead zones are common and invisible.	Moderate.

Variable	Why it is invisible to a calculator	How much it matters
Actual feeding	The single largest driver of nitrogen input, and never an input.	Very high. This is the biggest single gap.
Maintenance discipline	A user can select a schedule; the calculator cannot know whether it will be followed for three years.	Very high over the long term.
Filter maturity and disturbance history	A recently disturbed filter behaves very differently from a settled one.	High for the addition decision specifically.
Fish age and condition	Juveniles, adults and old fish differ in load, behaviour and resilience.	Moderate to high.
Local water chemistry	Source water hardness, alkalinity and pH determine nitrification performance and species suitability.	High, particularly in very soft or very hard water areas.
Existing social dynamics	An established hierarchy responds differently to new arrivals than a newly formed group.	High for staged additions — see Section 27.

A fair verdict on calculators

Calculators are **good at species knowledge**: adult size, minimum tank length, temperament, group requirements, parameter compatibility. For a beginner, this is the information most likely to prevent a serious mistake, and getting it from a calculator is far better than not getting it.

Calculators are **weak at system modelling**: filtration capacity, feeding, maintenance, aquascape, and every behavioural variable that depends on the individual animals.

The right use is therefore as a **screening and warning tool** rather than as a verdict. A calculator saying 'this species needs a longer tank' or 'these two species are incompatible' is worth acting on. A calculator saying '104 percent' is a soft signal to look more closely, not a measurement of anything.

It follows that both 'calculators are accurate' and 'calculators are useless' are wrong, and for the same reason: they treat a tool with a specific competence as though it had a general one.

EVIDENCE: REASONABLE INFERENCE; NO INDEPENDENT VALIDATION STUDIES OF AQUARIUM STOCKING CALCULATORS WERE LOCATED

27. Alternative Metrics and the Multi-Constraint Model

If length is the wrong measure, what are the alternatives, and why has none of them replaced it?

27.1 Stocking by biomass

Biomass-based stocking — grams per litre, kilograms per cubic metre — is the aquaculture standard and is scientifically the soundest basis, because biomass is what actually drives oxygen demand and sets the feeding rate that drives nitrogen input. Intensive recirculating systems commonly operate at densities such as 80–100 kg per cubic metre for species like tilapia and seabream, with continuous life support.

Why it has not transferred to the hobby is straightforward. Hobbyists do not weigh their fish and reasonably will not start. Published weight data for ornamental freshwater species is patchy compared with food-fish species. And even if a keeper knew their total biomass, they would have no validated hobby-scale reference figure to compare it against, because the aquaculture figures optimise for a different goal. The metric is right and the infrastructure to use it does not exist.

EVIDENCE: AQUACULTURE PRACTICE; HOBBY APPLICATION IS IMPRACTICAL

27.2 Stocking by surface area

Covered in Section 17. It remains the most physically principled of the classic hobby rules because it indexes a real limiting process. Its specific figures are conservative for a modern aerated tank, and it ignores everything behavioural. As a cross-check when comparing two tanks of equal volume but different footprint, it still earns its place.

EVIDENCE: USEFUL RULE OF THUMB

27.3 Stocking by metabolic demand

Theoretically the most defensible approach of all: estimate total oxygen consumption at the tank's operating temperature and compare it against the system's oxygen delivery capacity, and separately estimate nitrogen input from actual feed weight and protein content and compare it against measured biofilter performance. The physiology to do this exists — the mass and temperature scaling relationships from Clarke and Johnston provide the framework.

It is impractical for hobbyists for the obvious reasons: it requires weighing fish, weighing food, knowing feed composition, measuring dissolved oxygen continuously, and species-specific metabolic constants that mostly do not exist for ornamental fish. It is worth mentioning in any downstream content only to make a specific point — that the scientifically correct approach is known and simply not usable, which is precisely why heuristics persist.

EVIDENCE: SCIENTIFICALLY SOUND, PRACTICALLY UNAVAILABLE

27.4 The practical multi-constraint model

Based on the material in Sections 1 through 26, the following framework is offered not as a formula but as a checklist of ceilings. The governing constraint is whichever is lowest, and it differs between tanks.

Table 27.1 — The twelve ceilings

#	Constraint	What sets it	How the keeper can assess it	Can it be raised?
1	Adult biomass	Species adult size times number of individuals.	Look up adult sizes; project forward, not from purchase size.	No, other than by changing the stock or the tank.
2	Feed input / waste production	Quantity and protein content of food entering the system.	Track nitrate rise between water changes.	Yes — the most controllable variable in the system.

#	Constraint	What sets it	How the keeper can assess it	Can it be raised?
3	Oxygen demand	Biomass, temperature, activity, bacterial and plant respiration.	Observe behaviour at dawn; increase surface agitation and observe whether behaviour changes.	Partly — agitation and lower temperature help; saturation is a hard ceiling.
4	Biological filtration	Nitrifier population, surface area, oxygen, temperature, pH, alkalinity, flow.	Test ammonia and nitrite after any change in load.	Yes, slowly, and only up to the limits set by oxygen and alkalinity.
5	Tank footprint / swimming room	Length and width; open space after hardscape.	Compare species requirements against the longest horizontal dimension.	No. This is the hardest ceiling in the list.
6	Territorial capacity	Number and distribution of defensible resources; sight lines; species temperament; maturity.	Observe over weeks, particularly after maturity and around spawning.	Partly — layout changes can help substantially.
7	Social group needs	Species social ecology and current group composition.	Observe shoaling, darting, aggression and latency to feed.	Yes, by adding conspecifics — but only if constraints 1 to 6 permit.
8	Maintenance capacity	What the keeper will actually sustain over years.	Honest self-assessment against the last twelve months, not against intentions.	Yes, but only by the keeper genuinely changing behaviour.
9	Plant mass and growth	Species, light, carbon, nutrients, trimming discipline.	Observe growth rate and whether trimmings are actually removed.	Yes.
10	Temperature	Species requirements, room conditions, season.	Monitor, particularly the summer maximum.	Partly, and it interacts with everything else.
11	Compatibility	Species temperament, size disparity, dietary overlap, predation risk.	Research before purchase; observe after.	Only by changing the stock list.
12	Growth trajectory	Species growth rate and current age.	Project the tank six and twenty-four months forward.	No — but it can be planned for.

Table 27.1 — Offered as a diagnostic checklist, not a formula. The list is not final and is not claimed to be exhaustive; it is the set of constraints for which this document found supporting evidence.

How to use the model

The purpose of the model is not to produce a number. It is to answer a different and more tractable question: **which of these twelve is the lowest ceiling in this particular tank, and what would raising it require?**

That reframing has a practical advantage. A keeper who knows their binding constraint is footprint knows that no amount of filtration, water changing or planting will fix it. A keeper whose binding constraint is feeding discipline has a problem they can solve this week, for free.

It also explains why two experienced keepers can look at the same tank and disagree. They may be assessing different ceilings, and both may be right about the one they are looking at.

28. “Overstocked” Has Six Different Meanings

Much of the hobby's stocking argument is people using one word for six different conditions. Naming them separately makes most of the disagreement disappear.

Table 28.1 — Six distinct forms of overstocking

Type of overstocking	Definition	How it presents	How to confirm it	What actually fixes it
Biologically overstocked	Nitrogen input exceeds the system's processing and export capacity.	Measurable ammonia or nitrite; nitrate climbing rapidly between water changes.	Test kit. This is the one form that standard testing detects reliably.	Reduce feeding, increase export, expand biofiltration, or reduce stock.
Oxygen-limited	Oxygen demand approaches or exceeds supply, usually at the pre-dawn minimum.	Surface gasping in early morning, elevated gill rate, listlessness, reduced feeding, clustering at flow outlets.	Dissolved oxygen measurement, ideally before dawn. Otherwise behavioural observation at that hour.	Increase surface agitation, reduce temperature, reduce feeding, reduce stock.
Spatially overstocked	Insufficient open swimming space or footprint for the species present.	Restricted movement, fish unable to swim continuously, constant near-encounters, inability to rest undisturbed.	Compare species requirements against tank length and open area. No test kit applies.	A larger or longer tank, or different species. Nothing else.
Behaviourally overstocked	Too many individuals relative to available territory or defensible resources.	Persistent chasing, fin damage, one or more individuals permanently excluded, fish hiding continuously.	Sustained observation over weeks, particularly after maturity.	Add structure and sight breaks, reduce numbers of the territorial species, or increase footprint.
Socially mis-stocked	Group sizes or compositions that do not permit normal social behaviour — in either direction.	Shyness, hiding, poor feeding, elevated intraspecific aggression, reduced shoaling.	Behavioural observation against the species' known social ecology.	Adjust group size or composition — which may mean adding fish, removing fish, or rehoming.
Maintenance-limited	The stocking level requires more upkeep than the keeper reliably provides.	Progressive drift: rising nitrate, accumulating detritus, algae, declining condition over months.	Honest review of the last several months against the schedule the stocking requires.	Either reduce stocking to match realistic maintenance, or genuinely change the maintenance.

28.1 A tank can be fine in one dimension and poor in another

This is the practical payoff of the taxonomy. Some real combinations:

- Zero ammonia, zero nitrite, nitrate at 15 ppm, and a betta with torn fins from a persistently nipping tankmate. **Biologically fine, behaviourally overstocked.**
- A well-maintained tank with six corydoras in a tall 20-gallon column. **Biologically fine, spatially overstocked** — the bottom footprint is the constraint, not the volume.
- A pristine 55-gallon with a single neon tetra. **Biologically understocked, socially mis-stocked.**
- A densely planted, heavily stocked tank with zero nitrate and fish at the surface every morning. **Chemically excellent, oxygen-limited.**
- A beautifully stocked tank owned by someone working eighty-hour weeks. **Biologically and behaviourally sound, maintenance-limited** — and this will not show up on any test for several months.

EVIDENCE: REASONABLE INFERENCE; THE TAXONOMY IS A FRAMEWORK PROPOSED BY THIS DOCUMENT RATHER THAN A CITATION FROM THE LITERATURE

29. Observable Signs and What Tests Can Tell You

No single sign is diagnostic. The value is in the pattern, and in knowing which instrument answers which question.

29.1 Observable indicators

Table 29.1 — Observable signs and their differential diagnoses

Sign	What it can indicate	Alternative explanations to rule out
Persistent chasing or aggression	Behavioural or social overstocking; inadequate territory; wrong group size.	Normal breeding behaviour; a single problem individual; recent rearrangement; species that are simply boisterous.
Fish unable to rest undisturbed	Spatial or behavioural overstocking; inadequate cover.	Excessive lighting; strong flow; a nocturnal species being active at night is normal, not a symptom.
Visible crowding at rest	Spatial overstocking, or insufficient resting sites.	Some species genuinely aggregate to rest; shoaling species cluster normally.
Rapid detritus accumulation	High input relative to export; overfeeding; inadequate mechanical filtration.	Poor flow creating a collection point; recent substrate disturbance.
Surface gasping, especially before dawn	Oxygen limitation. This is the most important single behavioural sign in the whole list.	Gill damage; ammonia or nitrite toxicity impairing oxygen uptake; labyrinth fish surfacing normally; some species are natural surface dwellers.
Elevated gill movement	Oxygen limitation; ammonia or nitrite exposure; gill parasites; high dissolved carbon dioxide.	Recent activity; recent feeding; temperature at the top of the species range.
Nitrate rising quickly between changes	High net nitrogen input — a genuine and quantifiable stocking signal.	Nitrate in the source water; test kit error; recent overfeeding episode.
Poor growth in juveniles	Chronic stress, competition at feeding, or water quality.	Genetics; inadequate diet; species that simply grow slowly.
Persistent hiding or shyness	Social mis-stockng; inadequate group size; inadequate cover; an intimidating tankmate.	Recent introduction; naturally cryptic species; daytime observation of a nocturnal fish.
Competition or scrambling at feeding	Insufficient food distribution; too many individuals for the feeding method; subordinates excluded.	Normal feeding excitement, which is not the same thing. Look for whether specific individuals consistently fail to eat.

The differential column matters. Treating any one of these as proof of overstocking produces false positives, and a keeper who reduces stock in response to a sign with a different cause has not solved their problem.

29.2 What each measurement can and cannot tell you

Table 29.2 — Measurements and their limits

Measurement	What it tells you	What it cannot tell you
Ammonia	Whether nitrogen input currently exceeds oxidation capacity. The most important acute safety test.	Anything about oxygen, space, behaviour, or the future. A zero reading is necessary and nowhere near sufficient.
Nitrite	Whether the second nitrification step is keeping pace. Often the lagging indicator after a load increase.	Same limits as ammonia.
Nitrate	Net accumulation of nitrogen. Most informative as a <i>trend between water changes</i> , not as a single value.	Whether the tank is appropriately stocked. See Section 25.3.
Dissolved oxygen	The constraint that is most often binding and least often measured. Meaningful only if measured at the pre-dawn minimum.	Nothing about nitrogen, space or behaviour. Also, almost no hobbyist owns a meter, which is the practical problem.

Measurement	What it tells you	What it cannot tell you
pH	Determines ammonia toxicity at any given total ammonia reading, and affects nitrification rate.	Load. pH is a chemistry variable, not a stocking measure.
KH / alkalinity	The buffering reserve that nitrification consumes. A falling KH in a stocked tank is a genuine early warning that is widely ignored.	Anything directly about fish numbers — but a rapidly depleting KH does indicate a high nitrification rate.
GH	Mineral content relevant to species suitability and to invertebrate health.	Stocking level. It is a suitability measure, not a load measure.
Temperature	Multiplies oxygen demand, ammonia production and ammonia toxicity while reducing oxygen supply.	Load, directly. But it changes how much load the system can carry.
TDS / conductivity	Non-specific accumulation, useful only as a trend within one known system.	What is accumulating, or whether it matters. See Section 25.4.

The measurement gap at the heart of the hobby

The standard hobby test kit measures the constraint that is **easiest to detect and most likely already fine** in an established tank (nitrogen), and does not measure the constraints most likely to be binding (oxygen, space, behaviour, maintenance sustainability).

This is a large part of why 'my parameters are perfect' is such a common response to a stocking concern. The parameters genuinely are perfect. They are also measuring the wrong thing for the question being asked.

30. Stocking a New Tank Versus an Established One

A freshly cycled tank and a two-year-old tank of identical dimensions are not the same system, and the difference is larger than most stocking advice acknowledges.

30.1 What maturity actually provides

Table 30.1 — New versus established systems

Attribute	Newly cycled tank	Long-established tank
Nitrifier population	Sized to the ammonia dose used during cycling, and only just. It has never been tested against a real, variable, feed-driven load.	Sized to a sustained real load, with biofilm developed across all wetted surfaces rather than concentrated in the filter.
Distributed biofilm	Minimal on substrate, hardscape and glass. Nitrification is concentrated in the filter media.	Extensive. Substrate and hardscape carry substantial capacity, which makes the system far more resilient to filter disruption.
Microbial community diversity	Narrow. Commercial biofilter studies show community composition continuing to shift for months after start-up.	Broad and stable, including heterotrophs that mineralise organics, and in some systems archaeal and comammox ammonia oxidisers alongside bacterial ones.
Plant establishment	Plants are usually still rooting and may be melting back rather than growing. Nutrient uptake is low or negative during this phase.	Established plants actively growing and taking up ammonium and nitrate.
Buffering and mineral equilibrium	Whatever the source water provided.	Some systems develop mineral equilibria from substrate and hardscape; others have drawn down KH and need attention.
Keeper knowledge of the system	None. The keeper does not yet know how fast nitrate rises, where detritus collects, or how the fish behave.	Substantial. This is a genuine and underrated safety factor.

EVIDENCE: STRONG EVIDENCE FOR THE MICROBIAL DEVELOPMENT PATTERN; REASONABLE INFERENCE FOR AQUARIUM-SCALE CONSEQUENCES

30.2 Does fishless cycling to a given ammonia level predict fish load?

This deserves a careful answer because the reasoning is superficially sound and the conclusion is not reliable. The argument runs: if my tank processed 2 ppm of ammonia in 24 hours, then it can handle any fish load producing less than 2 ppm of ammonia per day, so I can stock fully immediately.

Several things are wrong with it.

- 1. The unit is wrong in a subtle way.** Processing a single 2 ppm dose in 24 hours demonstrates a certain oxidation capacity under one specific condition. Fish produce ammonia continuously with post-feeding peaks, and add faecal solids, dissolved organics and oxygen demand that a pure ammonia dose does not. The test does not replicate the load it is being used to predict.
- 2. The keeper cannot convert fish to ppm.** Even granting the capacity figure, translating ‘a dozen fish’ into ppm of ammonia per day requires knowing their biomass and feed input — the very quantities hobbyists do not measure. The calculation has a known left side and an unknown right side.
- 3. Nitrifier populations decline without substrate.** If dosing stops or drops before fish are added, the population begins to fall back toward whatever the ammonia supply supports. A capacity measured on Monday is not necessarily present on Friday.
- 4. Ammonia capacity is only one of twelve ceilings.** Even a perfectly accurate biofilter capacity figure says nothing about oxygen, footprint, territory, group size, adult size or maintenance. Section 27 applies in full.

5. **Full immediate stocking removes all observation opportunity.** A keeper who stocks everything at once has no way to identify which addition caused a problem, and no staged disease observation.

What fishless cycling to a given level *does* establish is genuinely useful: that a functioning nitrifying community exists, that it is capable of processing a non-trivial ammonia load, and that both nitrification steps are working. That is a meaningful safety improvement over stocking an uncycled tank. It is not a stocking permit.

EVIDENCE: REASONABLE INFERENCE; THE STRONG VERSION OF THE CLAIM IS A CONTESTED CLAIM NOT SUPPORTED BY THE MECHANISM

30.3 Evaluating whether an established tank can take more

A practical assessment sequence, ordered so that cheap observations come before expensive ones:

1. **Nitrogen stability.** Ammonia and nitrite consistently undetectable, including in the day or two after a feeding.
2. **Nitrate trend.** Measure just after a water change and again just before the next one. The rise, not the value, is the signal. A slow rise indicates spare capacity; a steep one indicates the system is already working hard.
3. **Alkalinity trend.** If KH is falling noticeably between water changes, nitrification is consuming buffer at a rate that will eventually destabilise pH. That is a load signal.
4. **Early-morning behaviour.** Observe before the lights come on. Fish at the surface, elevated gill rates or clustering at the filter outflow indicate the oxygen ceiling is closer than it appears.
5. **Existing social dynamics.** Is every individual feeding? Is any fish consistently excluded or hiding? Adding to a group with an existing problem usually compounds it.
6. **Spatial honesty.** Is there genuinely unused space, or does the tank only look empty because the fish are hiding?
7. **Adult size projection.** Are the current occupants fully grown? If not, the tank is already committed to a load it has not yet reached.
8. **Maintenance honesty.** Has the current schedule actually been sustained over the past several months, or is it aspirational?

A tank that passes all eight can probably take a moderate addition. A tank that fails any one of them has an answer to the question before the livestock is chosen.

31. Adding a Full School at Once Versus Staged Introduction

The second pillar of the document. The tension is real, both approaches have genuine advantages, and the correct answer depends on identifiable conditions rather than on a universal rule.

31.1 The two sides, laid out honestly

Table 31.1 — Whole group versus staged introduction

	Add the whole group at once	Add in stages
Biofilter effect	A single larger step change. Whether it matters depends on the added biomass and feed relative to the existing load — which for small shoaling fish is usually small.	Several small step changes, each more easily absorbed by existing reserve capacity.
Social outcome	The group arrives complete. No period of undersized-group stress, and no established incumbency among conspecifics.	The group is undersized for a period, potentially weeks. Early arrivals may establish position before later ones arrive.
Territorial incumbency	All individuals are equally new to the space, which levels the playing field. This is a real advantage for species where residency confers an advantage in contests.	Earlier arrivals hold residency. In territorial species this can produce sustained aggression toward later additions.
Disease risk	One exposure event. If the batch is healthy, the tank is exposed once. If it is not, everything arrives at once.	Multiple exposure events, but each smaller, and problems in one batch can be caught before the next is committed.
Observation	Harder to attribute a problem to a specific cause.	Easier. Each addition is a separate, observable event.
Acclimation stress	Fish arrive with familiar conspecifics, which for shoaling species is itself a buffer against novel-environment stress.	Later arrivals face both a novel environment and an established group.
Practicality	One purchase, one quarantine cycle, one acclimation.	Repeated purchases, repeated quarantine, repeated disturbance of the display.

31.2 When adding eight to twelve small schooling fish at once is reasonable

- The tank is **mature** — established filter, substrate and hardscape biofilm, no recent disturbance to media or medication course.
- The addition represents a **small proportional increase in biomass and feed**. Eight neon tetras at roughly 0.2–0.3 g each add on the order of two grams; in a tank already carrying tens of grams of fish, that is a modest step.
- The tank is **well planted**, providing direct ammonium uptake and cover for the new arrivals.
- The keeper is willing to **reduce feeding temporarily**, so nitrogen input rises less than biomass does.
- The keeper will **test ammonia and nitrite daily** for several days and can perform a water change if anything appears.
- The species is **genuinely social** and the welfare benefit of a complete group is real.
- The group has been **quarantined together** and is known to be healthy — which removes the single strongest argument for staging.

31.3 When it is risky

- The tank is **newly cycled** or the filter has recently been disturbed.
- The addition is a **large proportional increase** — which is about mass and feed, not count. Eight of a large-growing species is a completely different proposition from eight small tetras.

- The tank is **already near another ceiling**: warm, oxygen-marginal, spatially tight, or maintained at the limit of the keeper's schedule.
- The fish are **not quarantined** and the tank contains established livestock the keeper values.
- The species is **territorial rather than social**, in which case simultaneous introduction has fewer advantages and the numbers themselves may be the problem.
- The keeper **cannot test or perform a water change** for the next few days.

EVIDENCE: REASONABLE INFERENCE FROM ESTABLISHED PRINCIPLES; NO CONTROLLED STUDY OF AQUARIUM INTRODUCTION STRATEGY WAS LOCATED

31.4 Territorial incumbency

Prior residency conferring an advantage in territorial contests is a well-documented phenomenon in the behavioural ecology literature. The aquarium consequence is that staged introduction is not neutral: it systematically favours earlier arrivals, and in territorial species this can produce persistent aggression toward later additions.

Table 31.2 — Territorial incumbency by group

Group	How much incumbency matters	Practical handling
Cichlids	Considerable. Established territory holders frequently attack later arrivals. Territories and pair bonds form quickly and are defended actively.	Where a group is intended, adding all individuals simultaneously to a rearranged tank is common aquarium practice specifically to disrupt existing claims. Rearranging hardscape at the time of introduction resets sight lines and territory boundaries.
Gouramis	Significant, particularly among males and particularly for surface-territory holders.	Simultaneous introduction where possible; abundant sight breaks and surface cover; careful attention to sex ratio.
Some barbs	Moderate. Hierarchy formation in an established group can direct aggression at newcomers.	Larger groups distribute this better than small ones. Adding several at once rather than singly is preferable.
Livebearers	Moderate, and mainly expressed as male harassment of females rather than as classic territoriality.	Sex ratio matters more than arrival order. See Section 33.
Small shoaling characins and rasboras	Low. These species generally accept conspecific additions readily, which is one reason the whole-group question is less fraught for them.	Either approach works. The decision can be made on biofilter and disease grounds rather than social ones.
Corydoras and similar low-aggression benthic groups	Low. Wild groups are mixed-age and mixed-size, and aggression levels are documented as very low.	Additions are generally well tolerated. The constraint is bottom footprint, not social acceptance.

EVIDENCE: ESTABLISHED PRINCIPLE FOR PRIOR-RESIDENCY EFFECTS; THE SPECIES-LEVEL HANDLING IS COMMON AQUARIUM PRACTICE

31.5 Three introduction strategies compared

Table 31.3 — Introduction strategies

Strategy	Best suited to	Main risk
All at once	Small social species; mature well-planted tanks; groups quarantined together; territorial species where simultaneous introduction levels residency.	A single larger biological step and a single larger disease exposure.
Half now, half later	A reasonable compromise for medium-sized additions to moderately mature tanks. Limits both the biological step and the duration of the undersized-group period.	Still creates some incumbency, and still requires two quarantine cycles if done properly.

Strategy	Best suited to	Main risk
Small batches over weeks	Newly cycled tanks; large or high-waste species; keepers without a quarantine tank who need staged disease observation.	Prolonged period of undersized social group; repeated disturbance; strongest incumbency effects; and in practice, the risk that the plan is abandoned halfway and the group is never completed.

The last risk in that table deserves attention because it is common and rarely mentioned. A keeper who intends to complete a shoal over six weeks and gets distracted after the second batch has produced exactly the welfare problem the staging was meant to avoid, and has done so quietly.

31.6 Quarantine changes the calculation

If a group is quarantined together for the standard observation period and emerges healthy, the strongest argument for staged introduction — disease observation — has already been satisfied. What remains is only the biofilter argument, which for small shoaling fish is usually modest.

This is the cleanest resolution of the central tension in the document. **Quarantine converts the whole-school question from a disease question into a nitrogen question, and the nitrogen question is usually the easier one to answer.** A keeper with a quarantine tank has substantially more freedom to add complete social groups than a keeper without one, and this is an under-appreciated benefit of quarantine that has nothing to do with disease control itself.

A secondary benefit: the quarantine period is itself a biofilter opportunity. Feeding can be moderated in the display tank during that time, and any planned filter maintenance completed, so that the display is in its best condition on the day the group arrives.

EVIDENCE: REASONABLE INFERENCE

32. Growth, Reproduction, Sex Ratios and Predation

Four ways a stocking plan changes after it has been made, without the keeper buying anything.

32.1 Projecting future bioload

Section 5 covered adult size in principle. In practice the projection matters most for the categories below, where the gap between purchase size and adult size is largest or the timeline is longest.

Table 32.1 — Where forward projection matters most

Category	Why the projection matters	Common failure
Juvenile shoals	A group bought at half adult size will roughly multiply its mass several fold. The cube relationship applies within a species just as between species.	Stocking a group to the tank's current comfortable capacity when the fish are half grown.
Goldfish	Large adult size, deep body, heavy feeding, high waste output and decade-scale longevity. Section 35 covers this specifically.	Bought as small juveniles in tanks and bowls that will never be adequate.
Cichlids	Growth plus the onset of territoriality and pair formation at maturity. Two constraints change at once.	A peaceful juvenile group that becomes a violent adult one over a few months.
Plecos and large catfish	Common plecos are among the largest-growing and highest-waste fish routinely sold to beginners, and are frequently sold as 'cleaners.'	Bought as a functional item rather than as a large fish that will eventually dominate the tank's bioload.
Livebearers	Growth is modest; population growth is not. See below.	The tank is stocked correctly on day one and is overstocked by month four without a single purchase.
Other large-growing species	Bala sharks, tinfoil barbs, iridescent sharks, many loaches and various catfish are sold small and reach sizes incompatible with typical home tanks.	Purchase based on the size in the bag.

32.2 Reproduction as a stocking variable

Livebearers are the clearest case. Guppies, platies, mollies and swordtails give birth to free-swimming young, females can store sperm and produce successive broods from a single mating, and in a well-planted tank with cover a meaningful fraction of fry survive. The result is a population that compounds.

- **Fry survival is largely a function of cover.** A bare tank with adult fish present may see very few fry survive; a densely planted tank with floating cover can see many. The same species in two different aquascapes produces very different population trajectories.
- **Population growth changes the load slowly and then quickly.** Fry contribute little biomass initially and a great deal by the time they mature, at which point they begin reproducing themselves.
- **The keeper's plan for the offspring is part of the stocking plan.** If there is no plan — a shop that will take them, another tank, a predator that will control numbers — then the tank is being stocked on an implicit assumption that the population will somehow stop growing.
- **Maintenance rises with population, not with intention.** Feeding increases, waste increases, and water changes must increase in step.

EVIDENCE: ESTABLISHED PRINCIPLE; THE AQUARIUM-SCALE DYNAMICS ARE COMMON AQUARIUM PRACTICE

32.3 Sex ratios

For several groups, the ratio of males to females matters more than the total number, because the welfare problem is harassment rather than crowding.

Table 32.2 — Sex ratio as a stocking constraint

Group	Why ratio matters	Common guidance and its status
Livebearers	Males court females persistently. A single female with multiple males can be harassed continuously, with real welfare and condition consequences.	Keeping several females per male, or single-sex groups, is widespread hobby practice. The underlying phenomenon of male harassment in poeciliids is well documented; the specific ratios are conventions.
Gouramis	Males can be territorial toward each other and persistent toward females.	Hobby guidance commonly recommends one male with multiple females, or single specimens of the more aggressive species. Common aquarium practice.
Cichlids	Varies enormously by species and mating system. Pair-bonding species, harem species and mouthbrooders have completely different ratio requirements.	Genuinely species-specific. Generic advice is unhelpful here; the mating system has to be looked up per species.
Bettas	Male-male combat is the defining constraint. Female groups are also not trivially peaceful and require specific conditions.	Single males, or carefully managed setups. See Section 34.

32.4 Predation and compatibility

Predator-prey compatibility constrains stocking entirely independently of bioload. A tank can have ample biological, spatial and social capacity for both species and still be an unsuitable combination, because one will eat the other. The relevant question is not size difference in the abstract but whether the smaller fish fits in the larger fish's mouth — now or at adult size.

This is a case where forward projection matters again. A community that is compatible with a juvenile predator may not be compatible with the adult, and the change can be abrupt.

EVIDENCE: ESTABLISHED PRINCIPLE

33. Tank Dimensions, Actual Volume and Sumps

Nominal gallons is a poor descriptor of a tank's stocking suitability, for three separate reasons: shape, real volume, and where the water actually is.

33.1 Equal volume, unequal suitability

Table 33.1 — Common tank formats compared

Tank	Typical footprint	What it suits	What it does not suit
20-gallon long	About 30 x 12 inches	Small shoaling fish that use length; corydoras and other benthic groups; good surface area for gas exchange relative to volume.	Tall aquascapes; deep-bodied species needing vertical clearance.
20-gallon high	About 24 x 12 inches	Vertically structured scapes; species that use the water column; smaller footprint for limited furniture space.	Active horizontal swimmers; benthic groups needing bottom area; the reduced surface area relative to volume is a real gas-exchange disadvantage.
29-gallon	About 30 x 12 inches, taller than a 20-long	A common community shape offering both length and height.	Species needing substantial bottom area relative to volume.
40-gallon breeder	About 36 x 18 inches	Widely regarded as one of the best freshwater proportions available: long, unusually wide, good bottom area, good surface area. Excellent for benthic groups and for distributing territorial fish.	Little. Its main drawback is depth front-to-back for maintenance access.
Cube tanks	Equal length and width	Aquascaping, multi-angle viewing, structure-oriented species.	The least continuous swimming length per unit volume of any common shape. Poor for active shoaling species.

Table 33.1 — Dimensions are approximate and vary between manufacturers. The point is the proportion, not the exact figure.

The general principle: **for most freshwater community stocking, footprint is a better predictor of suitability than volume, and length is the most important single dimension.** A 40 breeder and a tall 40-gallon column hold the same water and are not remotely the same tank.

EVIDENCE: REASONABLE INFERENCE; STRONGLY SUPPORTED BY COMMON AQUARIUM PRACTICE

33.2 Actual water volume

A tank labelled 10 gallons does not contain 10 gallons in service. Substrate displaces water, hardscape displaces water, equipment displaces water, and the fill level sits below the rim. A commonly cited figure is that a nominal 10-gallon tank holds roughly 8.5 to 9 gallons once set up, and the proportional loss is larger in small tanks with deep substrate.

Does it matter for stocking? Modestly and unevenly. It matters more in small tanks, where a 15 percent volume loss is a meaningful reduction in dilution capacity and thermal stability. It matters less in large tanks. And it matters not at all for the constraints that are usually binding — footprint, territory, group size — which are unaffected by how much gravel is in the tank. Substrate depth does, however, add colonisable surface area, which partly offsets the volume loss on the biological side.

EVIDENCE: REASONABLE INFERENCE

33.3 Sumps and total system volume

A sump adds genuine water to the system, and this is qualitatively different from adding a larger canister filter.

Table 33.2 — What a sump adds and what it does not

A sump genuinely provides...	...but does not provide
Real additional water volume, which slows the rate of change of every dissolved parameter. This is genuine buffering, not a filtration effect.	Any additional display swimming space. Not one inch.
Substantially more space for biological media, mechanical media and chemical media than a hang-on or canister filter.	Additional territory. Fish cannot use the sump.
Additional gas exchange surface, and often a turbulent return that improves oxygenation.	Relief from social constraints or group-size requirements.
Somewhere to place heaters and equipment, which frees display space visually and physically.	Any change to the tank footprint, which is where the behavioural constraints live.
A place to add top-off water and dose without disturbing the display.	Protection against the failure modes a sump introduces — overflow, siphon failure, return pump failure, evaporation concentration.

Large filters versus sumps versus larger tanks

These three are frequently discussed as though they were interchangeable ways of ‘increasing capacity.’ They are not.

A larger filter adds processing capacity and circulation. No water, no space.

A sump adds processing capacity, circulation *and* real system water volume. Still no display space.

A larger tank adds all of the above *and* footprint, swimming distance and territory — which are the constraints that filtration and system volume cannot touch.

So the honest answer to ‘does a sump let me keep more fish?’ is: **yes for the biological and chemical ceilings, no for the spatial and behavioural ones.** Whether that translates into more fish depends entirely on which ceiling was binding in the first place.

EVIDENCE: ESTABLISHED PRINCIPLE

34. Body Shape, Activity Level and Benthic Species

Three morphological and behavioural variables that determine how much of a tank a given fish actually needs, independent of its mass.

34.1 Body shape

Table 34.1 — Body form and its consequences

Body form	Consequence for biomass	Consequence for space
Deep-bodied and laterally compressed (angelfish, discus, fancy goldfish)	High mass per unit length. Length-based rules badly understate their load.	Need vertical clearance. This is the one case where tank height is a genuine requirement rather than a preference. Turning radius is also larger relative to length.
Long and slender (danios, many rasboras, kuhli loaches)	Low mass per unit length. Length-based rules overstate their load.	Need horizontal distance. Tank length is the operative dimension.
Bottom-oriented and dorsoventrally flattened (corydoras, many loaches, plecos)	Variable. Small corydoras are light; a common pleco is very heavy.	Need bottom area rather than volume. Their space is measured in square inches of substrate, not gallons.
Large-finned varieties (long-finned bettas, veiltail livebearers, fancy goldfish)	The fins add little mass.	Fins reduce swimming efficiency and manoeuvrability, increase drag, and make the fish more vulnerable to nipping and to strong flow. A long-finned individual may need lower flow and more open space than its short-finned equivalent.
Elongated predatory forms (gar-like, pike-like)	Substantial mass at adult size.	Need continuous straight-line length. These are among the worst matches for home aquarium proportions.

EVIDENCE: REASONABLE INFERENCE FROM ESTABLISHED MORPHOLOGY

34.2 Activity level

Sustained swimming raises oxygen consumption above resting metabolic rate, so an active species of a given mass has a higher oxygen demand than a sedentary one. Metabolic scaling work has found that lifestyle affects scaling relationships, with more active pelagic forms differing from benthic ones — likely related to differing quantities of energetically expensive swimming tissue.

The space consequence is larger than the oxygen one. A continuously cruising species that cannot swim continuously is being denied its primary behaviour, and no water change fixes that. This is why active swimmers require disproportionately more space than their mass suggests, and why the tank-length recommendations for danios and similar species are so much larger than their body size implies.

EVIDENCE: STRONG EVIDENCE FOR THE METABOLIC COMPONENT; REASONABLE INFERENCE FOR THE SPACE REQUIREMENT

34.3 Benthic species and the footprint question

For bottom-dwelling species, the operative dimension is the usable substrate area, not the water volume. Bronze corydoras shoal in two dimensions on the bottom, which is precisely why researchers film them from above. A tall tank offers a bottom-dwelling group no more room than a short tank of the same footprint, however many gallons it holds.

Table 34.2 — Benthic groups and their binding constraints

Benthic group	Dominant constraint	Notes
Corydoras	Bottom footprint and group size. Very low aggression.	Also require soft substrate for barbel health, and access to the surface for facultative air breathing. Wild groups are mixed-age and have been observed as small as three, but captive welfare guidance points toward larger groups.
Kuhli and other small loaches	Bottom area, hiding places, and group size.	Very low mass for their length, so they contribute far less bioload than a length-based rule suggests. Their real requirement is shelter density and a group to feel secure enough to be visible at all.
Larger loaches (clown loach and similar)	Adult size, which is very large, plus group size, which is also large.	One of the most severe mismatches between purchase size and adult requirement in the hobby.
Plecos and large suckermouth catfish	Adult biomass, waste production and territory.	Common plecos are large-growing, high-waste fish. Bristlenose and other small species are a genuinely different proposition and should not be conflated with them.

35. Species-Specific Stocking: Which Constraint Dominates

The purpose of this section is not to provide numbers. It is to identify, for each common group, which of the twelve ceilings from Section 27 is usually the binding one — because that is what determines whether a given intervention will help.

Table 35.1 — Dominant stocking constraint by group

Group	Dominant constraint	Secondary constraints	The mistake this group invites
Small schooling tetras	Social group size.	Tank length for the more active species; biological load is usually minor.	Buying three or four. The biological cost of a proper group is small; the welfare cost of an undersized one is not.
Rasboras	Social group size, and water chemistry suitability for the softer-water species.	Length for larger and more active species.	Treating all rasboras as interchangeable. Adult sizes and activity levels vary considerably across the group.
Barbs	Social group size <i>and</i> aggression distribution together.	Tank length; compatibility with long-finned tankmates.	Keeping too few, which concentrates aggression. Saxby et al. terminated small tiger barb groups on ethical grounds for exactly this reason.
Danios	Tank length and swimming space.	Group size; temperature suitability, as several species prefer cooler water than a typical tropical community.	Keeping active swimmers in short or cube tanks that meet the volume requirement and not the length one.
Corydoras	Bottom footprint and group size.	Substrate type; surface access.	Counting them as free because they are on the bottom, and housing them over sharp gravel.
Kuhli loaches	Shelter density and group size.	Bottom area. Bioload is low relative to their length.	Assuming they are absent when they are simply hidden, and understocking the group as a result.
Bettas	Territory and conspecific intolerance.	Flow tolerance; enrichment; tankmate selection. See Section 36.	Assuming a single fish means minimal requirements.
Gouramis	Territory, particularly surface territory, and sex ratio.	Tank length; surface cover; compatibility.	Keeping multiple males of territorial species in insufficient space.
Livebearers	Population growth and sex ratio.	Waste load, which is moderate to high for mollies in particular; water hardness suitability.	Stocking correctly on day one and not planning for month six.
Dwarf cichlids	Territory and number of defensible sites.	Water chemistry; compatibility; pair dynamics.	Assuming small size means small space requirement. Territory does not scale down with body length.
Goldfish	Adult biomass, waste output and oxygen demand together.	Longevity; sociality; temperature.	Every part of the standard goldfish purchase. See Section 37.
Shrimp	Water quality sensitivity and, eventually, population size.	Copper and medication sensitivity; predation by tankmates.	Treating them as zero-load. They are low-load and highly sensitive, which is a different thing.
Snails	Highly variable by species — see Section 22.3.	Population dynamics for reproducing species; feeding requirements for nerites and mystery snails.	Treating a large mystery snail as equivalent to a pest snail.

Table 35.1 — Deliberately expressed as constraints rather than as fish-per-gallon figures. Which constraint dominates determines which intervention will actually help.

EVIDENCE: REASONABLE INFERENCE; LARGELY CONSISTENT WITH EXPERIENCED HOBBYIST CONSENSUS

36. Three Species Cases in Detail

36.1 Bettas: why one fish is not a small requirement

A single betta appears to be the easiest possible stocking question and is in fact a good illustration of why gallons-per-fish reasoning fails. The binding constraints for a betta are behavioural and environmental, not chemical.

Recent welfare research is directly relevant. Clark-Shen, Taniel-Adam, Gajanur and Brown (2024), published in *Animal Welfare*, housed male Siamese fighting fish across five conditions ranging from a 1.5-litre jar to a 19.3-litre aquarium, with and without furnishings. Fish in larger tanks were significantly more active, swimming on average 92 seconds more during observation periods than those in jars. The authors recommended a minimum of 5.6 litres for display and sale, and larger tanks for home keeping, with gravel, live plants and refuges in all cases. They also noted that existing guidance on betta tank size is vague and more concerned with water quality than with other welfare considerations — which is precisely the failure this document is documenting.

Earlier work reported in the same journal quantified pairs of male bettas in bare or enriched aquaria of 10, 38, 208 and 378 litres and found fewer approaches and aggressive displays, and more bouts of foraging, in larger aquaria. Both studies also observed marked individual variation in behaviour, which is a direct argument against any single number.

EVIDENCE: STRONG EVIDENCE FOR THE DIRECTION OF THE EFFECT; THE SPECIFIC MINIMUM FIGURES ARE THE AUTHORS' RECOMMENDATIONS FOR RETAIL DISPLAY RATHER THAN OPTIMAL HOME KEEPING

- **Territory.** A male betta will treat the whole tank as its territory and will not tolerate another male. This is a hard compatibility constraint, not a space one.
- **Swimming.** The research finding that fish in larger tanks swim more is the clearest evidence available that small containers suppress normal behaviour rather than merely constraining it.
- **Enrichment.** Both studies point toward furnishings mattering alongside volume. A larger barren tank is not automatically better than a smaller furnished one on every measure.
- **Flow.** Long-finned varieties struggle in strong current, which constrains filtration choices and interacts with gas exchange.
- **Tankmates.** Compatibility is individual as much as it is species-level, which no calculator can predict.

36.2 Goldfish: why they break every simple rule

Goldfish are the canonical counterexample to inch-per-gallon, and it is worth being precise about why rather than repeating the conclusion.

- **Adult size.** Single-tailed varieties reach sizes far beyond what any beginner tank accommodates. Fancy varieties are shorter but very deep-bodied.
- **Biomass per unit length.** Deep-bodied fancy varieties carry extremely high mass for their length, so the cube relationship works maximally against a length-based rule.
- **Feeding.** Heavy feeders with a short digestive tract and correspondingly high waste output per unit of food consumed.
- **Oxygen.** High absolute demand from high biomass, and they are typically kept at cooler temperatures — which helps, since cooler water holds more oxygen and slows metabolism.
- **Waste.** Among the highest-output freshwater aquarium fish per individual.
- **Longevity.** Decade-scale animals. A stocking plan for goldfish is a ten-year commitment, not a one-year one.
- **Sociality.** They are social and generally do better with conspecifics, which multiplies every constraint above.

A note on the frequently cited figure that goldfish need 20 gallons each: this document could not identify a research derivation for that number. It is a hobby convention. The *reasoning* behind conventions of that kind — large adult size, very high waste, high oxygen demand, long life and sociality — is thoroughly sound. The specific figure is a convention and should be presented as one.

EVIDENCE: REASONABLE INFERENCE FOR THE REASONING; THE SPECIFIC GALLONAGE IS A USEFUL RULE OF THUMB WITHOUT A TRACEABLE DERIVATION

36.3 Small schooling fish: separating count from load

The mirror image of the goldfish case, and it deserves equal emphasis because the hobby is much better at warning about too many fish than at recognising when a large number is appropriate.

Twenty small tetras in a well-planted, well-filtered 40-gallon breeder is a large fish *count* and a modest biological *load*. Their combined mass may be on the order of five grams. Their combined feed requirement is small. They use midwater space efficiently and do not defend territory. And a group of that size expresses shoaling behaviour far more completely than a group of six.

Meanwhile a single adult common pleco in the same tank represents substantially more biomass, more waste, more oxygen demand and more territorial complication than all twenty tetras combined. The count says one; the load says otherwise.

The count/load distinction, stated for downstream use

Count matters for social welfare, for spatial competition and for territorial conflict.

Biomass and feed input matter for oxygen demand, nitrogen production and biofilter loading.

These two are frequently uncorrelated and occasionally anti-correlated. A tank can be high-count and low-load, or low-count and high-load. Almost every confusing piece of stocking advice in the hobby comes from a failure to say which of the two is being discussed.

EVIDENCE: REASONABLE INFERENCE

37. Understocking, Social Facilitation and Dither Fish

37.1 Can a tank be understocked?

Biologically, low stocking is essentially always fine. There is no biological penalty to having fewer fish than a system could support — water quality is better, oxygen reserve is greater, and the margin for error is larger. Claims that a tank ‘needs’ more fish for biological reasons do not hold up; a lightly loaded biofilter simply maintains a smaller population appropriate to its supply.

Socially, however, a tank absolutely can be understocked, and this is the sense in which the term is meaningful. A schooling species kept below its functional group size experiences measurable welfare costs, as documented in Saxby et al. for neon tetras, white cloud mountain minnows and tiger barbs, and in the zebrafish literature at very low densities.

The two senses of understocked

Biologically understocked is not a problem. It is a safety margin.

Socially understocked is a genuine welfare problem, and it is invisible to every measurement a hobbyist takes.

A tank can be both at once: a single neon tetra in a 55-gallon tank is biologically understocked, which is fine, and socially understocked, which is not.

EVIDENCE: STRONG EVIDENCE FOR THE SOCIAL SENSE; THE BIOLOGICAL SENSE IS ESTABLISHED PRINCIPLE

37.2 “More fish makes them feel safe”

This is broadly supported for shoaling species, with mechanisms identified in Section 12.3: predator dilution, collective vigilance, and improved foraging. Minnow work found larger shoals detected an approaching predator sooner and sustained feeding longer; bluntnose minnow work found foraging latency fell and foraging rate rose with shoal size even in the absence of a predator.

It should not be universalised. It does not apply to solitary or territorial species, where additional conspecifics are a threat rather than a comfort. It also does not apply without limit within shoaling species — larger shoals increase competition for resources, and the zebrafish density literature found costs at the high end as well as the low end. The accurate statement is that **for social species, group size below a species-specific threshold imposes welfare costs, and above some point the benefits plateau while other constraints begin to bind.**

EVIDENCE: STRONG EVIDENCE FOR SHOALING SPECIES; UNSUPPORTED AS A UNIVERSAL CLAIM

37.3 Dither fish

“Dither fish” is an aquarium term for active, visible, typically shoaling fish added to a tank specifically to signal to shyer or more nervous species that the environment is safe. The logic rests on a real behavioural principle: fish use conspecific and heterospecific behaviour as information about predation risk, and the visible presence of calm, feeding fish in open water is evidence that no predator is present.

- **Where the concept is applied.** Traditionally with cichlids, particularly shy or cave-dwelling species; with anabantoids; and with nervous or newly imported wild-caught fish.
- **Does it work?** The underlying mechanism — social facilitation and the use of others' behaviour as risk information — is well established in behavioural ecology. Direct aquarium evidence is thinner but suggestive. Sloman and colleagues (2011) found that tiger barbs were less aggressive toward smaller tankmates when angelfish were present, apparently because they spent more time avoiding the larger fish. That is a related but distinct effect — a larger fish redirecting attention rather than reassuring a shy one — and it shows that

mixed-species composition genuinely changes behaviour.

- **The trade-offs.** Dither fish are fish. They add biomass, feed requirement, oxygen demand and nitrogen output. They must themselves be compatible — both able to avoid predation by the target species and unlikely to harass it. And they need their own appropriate group size, since most dither species are themselves shoaling fish. Adding three dither fish to solve a shyness problem creates a second social problem.
- **Where it can backfire.** Adding active, fast-feeding fish to a tank with a slow or timid feeder can worsen feeding competition rather than improve confidence.

EVIDENCE: COMMON AQUARIUM PRACTICE RESTING ON ESTABLISHED BEHAVIOURAL PRINCIPLE; DIRECT AQUARIUM EVIDENCE IS LIMITED

38. Disease Risk, Water-Quality Swings and Safety Margin

Possibly the most practically important section in the document, and the one least represented in ordinary hobby conversation.

38.1 Stocking density and disease

Higher density is associated with increased disease risk through several plausible pathways: increased contact rate between individuals facilitating pathogen transmission; increased physical contact and injury in territorial species; chronic stress with its associated immune modulation; and higher organic loading supporting larger opportunistic bacterial populations.

The evidence should be characterised carefully. The individual mechanisms are each well supported. What is much harder to support is any specific quantitative claim about a home aquarium — there is no threshold density above which disease becomes likely, and attributing any given outbreak to stocking density is not defensible. The honest framing is that **heavy stocking raises risk across several independent channels simultaneously**, which is a reason for caution rather than a prediction.

EVIDENCE: ANIMAL-WELFARE PRINCIPLE WITH MECHANISTIC SUPPORT; SPECIFIC AQUARIUM-SCALE QUANTIFICATION IS NOT AVAILABLE

38.2 Loss of margin: the strongest practical argument

A heavily stocked tank and a moderately stocked tank can look identical on every measurement during normal operation. They diverge sharply when something goes wrong, and the divergence is not gradual.

Table 38.1 — How stocking level determines behaviour under disruption

Disruption	Moderately stocked tank	Heavily stocked tank	Why the difference
Filter failure or power outage	Oxygen depletes slowly; distributed biofilm on substrate and hardscape continues working; hours to days of margin.	Oxygen depletes quickly because demand is high relative to the reservoir; biofilter oxygen starvation begins sooner; margin measured in hours.	Both oxygen consumption and ammonia production scale with the load, while the water volume does not.
Heatwave	Solubility falls but demand remains modest; usually survivable with increased agitation.	Falling supply meets rising demand from a high baseline. This is the classic summer loss event.	Section 19. Every mechanism moves the wrong way at once.
Missed water change	Nitrate and organics rise slowly. A week's delay is inconsequential.	Accumulation is fast. A week's delay can be a measurable water quality decline.	Accumulation rate is proportional to net input, which is proportional to the load.
Overfeeding episode	Absorbed by biofilter reserve and diluted by the water volume.	May exceed the remaining reserve, and the added organics drive an oxygen demand spike on top of a high baseline.	Reserve capacity is what is left over after the current load, and there is less of it.
Filter media replacement or medication course	Distributed biofilm and reserve capacity absorb the disturbance.	The system may have been depending on the full filter capacity, so disturbing it produces a measurable excursion.	A heavily loaded system is running closer to its ceiling by definition.
A single mortality going unnoticed	Decomposition of one small fish in a large water volume is a minor event.	Higher biomass means larger individual animals and a smaller relative water volume; decomposition is a proportionally larger insult.	Scale.

Safety margin as a stocking principle

Every constraint in Section 27 has a **reserve** as well as a **ceiling**: oxygen reserve, biofilter reserve, dilution reserve, maintenance slack, and the keeper's own time.

A tank stocked to its apparent ceiling under normal conditions has consumed all of its reserves simultaneously. It is not fragile in one dimension; it is fragile in every dimension at once, which is why problems in heavily stocked tanks tend to arrive as cascades rather than as isolated events.

This may be the single most useful practical argument against stocking to the limit, and it is an argument that does not require anyone to agree about what the limit is.

EVIDENCE: REASONABLE INFERENCE; STRONGLY CONSISTENT WITH AQUACULTURE OPERATIONAL PRACTICE

39. Maintenance Capacity, and “Can” Versus “Should”

39.1 The keeper is part of the system

Every constraint discussed so far has been a property of the tank, the water or the animals. This one is a property of the person, and it is the one that determines whether any of the others hold over time.

Consider a stocking level that is genuinely sustainable with 50 percent water changes every two days. That is a technically valid configuration. It is also a configuration that fails the first time the keeper travels, gets ill, has a busy month at work, or simply loses enthusiasm. The biological argument says the tank works. The practical argument says the tank works *as long as a specific human behaviour continues without interruption for years*, which is a very different claim.

- **Assess against history, not intention.** The relevant question is what maintenance was actually performed over the last six to twelve months, not what schedule the keeper intends to adopt.
- **Account for life changes.** A stocking plan should survive a two-week absence, a house move, or a period of illness.
- **Recognise the compounding effect.** Heavier stocking requires more maintenance, and more maintenance requirement makes lapses more likely and more consequential.
- **Note that this constraint is genuinely raisable.** Automation, python-style water changers, larger plant mass and a more disciplined feeding routine all raise it. But raising it requires actual behaviour change, not a purchase.

EVIDENCE: REASONABLE INFERENCE; CONSISTENT WITH COMMON AQUARIUM PRACTICE

39.2 “Can” versus “should”

This is the conceptual distinction that most of the document has been building toward, and it may be the most valuable thing to carry into downstream content.

Table 39.1 — Two different questions

	How many fish a tank CAN keep alive	How many fish it SHOULD contain
The question being asked	Will the animals survive?	Will the animals live well, across their full lifespan?
Constraints considered	Ammonia and nitrite below lethal levels; oxygen above lethal levels; no fatal aggression.	All twelve constraints in Section 27, including behaviour, social structure, adult size and maintenance sustainability.
Timeframe	Now, and the near term.	Years, including growth, maturity, reproduction and ageing.
What evidence answers it	Test kits and mortality.	Behavioural observation, species knowledge, and honest forward projection.
Where the answer comes from	Aquaculture and physiology can answer this well, because survival is measurable.	Welfare science and ethology can inform it, but it is partly a values question and should be stated as one.
Typical result	A higher number, often considerably higher.	A lower number, with more conditions attached.

Why this distinction resolves so many arguments

A great many stocking disputes are between one person answering the 'can' question and another answering the 'should' question. The keeper who says 'my heavily stocked tank has been fine for three years' is usually reporting accurately on the 'can' question. The keeper objecting is usually raising the 'should' question. Neither is lying.

Being explicit about which question is on the table converts an unresolvable argument into a discussable one. It also makes the values component visible: **where the line falls between 'survives' and 'lives well' is partly a judgement, and it is more honest to say so than to present a preference as a measurement.**

This document takes the position that the 'should' question is the one worth answering for home aquarists, since the entire point of the hobby is animals living well over years rather than animals not dying. But that is stated as a position, not smuggled in as a fact.

EVIDENCE: ANIMAL-WELFARE PRINCIPLE; THE FRAMING ITSELF IS A VALUES POSITION AND IS LABELLED AS SUCH

40. Scenario Library

Each scenario is answered in a fixed structure so that a retrieval system can locate the relevant field directly. The scenarios lettered A to T correspond to the most frequently encountered hobby questions; scenarios U onward were identified during research as equally common and equally confusing.

A note that applies to all of them: in almost every case the correct first response is a question, not an answer. The information a keeper provides unprompted is rarely the information needed to answer their question.

Scenario A — “I have a 20-gallon tank. How many fish can I keep?”

Aspect	Assessment
What we know	Volume only. That is close to the least informative fact about a tank.
What we do not know	Dimensions and therefore footprint. Which species. Adult sizes. Filtration. Whether it is planted. Feeding habits. Maintenance schedule. Source water chemistry. Temperature. Whether it is cycled.
Follow-up questions to ask	Is it a 20-long or a 20-high? What species are you considering, and what are their adult sizes and group requirements? Is it planted? What filtration? How often will you change water, honestly?
Likely limiting factor	Unanswerable as stated. Once species are named, the limiting factor is usually either footprint (for active or benthic species) or social group requirements (for small shoaling species) rather than biological capacity.
What to measure	Nothing yet. Measurement comes after the species question is settled.
Behavioural concerns	Cannot be assessed without species. Note that a 20-gallon can be an excellent tank for one well-chosen community and a poor one for another of the same total length.
Possible safe approaches	Reframe the question: choose one or two species you actually want, research their adult size, group needs and minimum tank length, and build the stocking plan outward from that. This produces a better outcome than any number could.
Common mistakes	Answering with a number. Any figure given without species is either meaninglessly conservative or actively dangerous. Also: assuming 20 gallons is 20 gallons — it is typically nearer 17–18 in service.

Scenario B — “My stocking calculator says 120 percent. Is that bad?”

Aspect	Assessment
What we know	A tool has produced a number above its own threshold. The calculator has flagged something.
What we do not know	What model the calculator uses, which is generally undisclosed. Whether the excess is in the filtration term, the volume term or a species warning. Actual feeding and maintenance. Aquascape. Plant mass.
Follow-up questions to ask	What specifically did it flag — a percentage, or a named warning about a species? Are any of the species below their listed minimum group size? Is any species listed as needing a longer tank? What are your actual nitrate readings over a water-change cycle?
Likely limiting factor	Unknown from the percentage alone. The named warnings are far more informative than the number.
What to measure	Nitrate immediately after a water change and again immediately before the next one. The rise is the real capacity signal. Also ammonia and nitrite to confirm nitrification is keeping pace.
Behavioural concerns	Check whether any social species is below its group threshold, since calculators often flag this and keepers often read past it to look at the percentage.
Possible safe approaches	Treat the percentage as a prompt to look more closely, not as a verdict. Read the specific warnings, verify adult sizes and minimum tank lengths independently, and assess the tank against the twelve constraints in Section 27.
Common mistakes	Treating 100 percent as a hard boundary. Treating 120 percent as either definitely fine or definitely fatal. Ignoring the species-specific warnings, which are the most reliable part of the output, in favour of the percentage, which is the least.

Scenario C — “My filter is rated for 40 gallons and my tank is 20. Can I stock more?”

Aspect	Assessment
What we know	The filter has more processing headroom and probably more circulation than a minimum-specification unit would provide. That is genuinely useful.
What we do not know	What the manufacturer's rating actually assumes, since no standard exists. The proportion of media receiving useful flow. Everything about the tank other than its filtration.
Follow-up questions to ask	More of what? More individuals of a species already present, or a new species? What are the adult sizes? Is the tank's footprint adequate? Is oxygen comfortable in the early morning?
Likely limiting factor	Almost certainly not filtration. Once filtration is oversized, the binding constraint moves elsewhere — usually footprint, oxygen or territory.
What to measure	Nitrate trend across a water-change cycle. KH, to check whether nitrification is depleting buffer. Early-morning behavioural observation for oxygen.
Behavioural concerns	Oversized filtration sometimes means strong flow, which is a stressor for some species and can restrict usable space.
Possible safe approaches	Enjoy the headroom as a safety margin rather than spending it. If more livestock is genuinely wanted, evaluate the other eleven constraints first — the filter has already been established as not being the limit.
Common mistakes	Reading a filter rating as a livestock allowance. A 40-gallon-rated filter does not give a 20-gallon tank 40 gallons of swimming room, oxygen reserve or dilution.

Scenario D — “My nitrates are always under 10 ppm. Can I add more fish?”

Aspect	Assessment
What we know	One measurement, at one point in the water-change cycle, of one parameter.
What we do not know	Why nitrate is low — there are at least six possible reasons, covered in Section 25.3. Whether the reading is accurate. What the trend looks like. Everything non-chemical.
Follow-up questions to ask	When did you last change water before testing? How much plant mass do you have? What does nitrate read immediately before your next change? How much are you feeding? Is your test kit shaken properly and in date?
Likely limiting factor	Not nitrogen processing. Low nitrate does suggest nitrogen input is being handled — but that is one of twelve constraints, and the others are untested.
What to measure	Nitrate immediately post-change and immediately pre-change, for the trend. Ammonia and nitrite. Early-morning fish behaviour for oxygen. KH trend.
Behavioural concerns	Entirely unassessed by a nitrate reading. Group sizes, territory and space all need separate evaluation.
Possible safe approaches	Use the nitrate trend as one input among several. If the rise between changes is slow and every other constraint has headroom, a modest addition is reasonable.
Common mistakes	Treating a single low nitrate reading as a general stocking clearance. In a heavily planted tank, low nitrate can coexist with a substantial nitrogen input and tells you about the plants rather than the fish.

Scenario E — “I only have four tetras. Everyone says add more, but will that overstock my tank?”

Aspect	Assessment
What we know	A social species is being kept in a group that is below the size at which controlled work has found improved welfare indicators. The keeper is reasoning about bioload.
What we do not know	Tank size and footprint. What else is in the tank. Current feeding and maintenance. Filter maturity. Plant mass.

Aspect	Assessment
Follow-up questions to ask	What are the tank's dimensions? What else lives there? How long has it been running? What is your nitrate trend? Are the four tetras currently shoaling, hiding or showing aggression toward each other?
Likely limiting factor	Very likely the social constraint, not the biological one. Six more neon tetras add roughly one and a half to two grams of biomass — a small proportional load increase in most established tanks.
What to measure	Ammonia and nitrite before and for several days after the addition. Nitrate trend beforehand to confirm existing headroom.
Behavioural concerns	This is the case where the advice to add fish is most likely to be correct. Saxby et al. found the largest group tested — ten — produced the best welfare score for neon tetras, with aggression highest in groups of two and five.
Possible safe approaches	If the tank is established and has spare capacity, adding to complete the group is usually the right call. Consider adding them as one batch after quarantine, holding feeding steady during the transition, and testing for several days.
Common mistakes	Assuming that more fish automatically means worse water. For small shoaling species the biomass increase is small and the welfare gain is substantial. The opposite mistake also exists: adding fish to a tank that is genuinely too small, which trades one problem for two.

Scenario F — “I want to add eight schooling fish at once. Will that crash my cycle?”

Aspect	Assessment
What we know	The keeper is planning a single addition of a social group and is worried about the nitrogen step.
What we do not know	Species and adult size, which determines the biomass involved. Tank maturity. Existing load. Plant mass. Whether the fish are quarantined.
Follow-up questions to ask	Which species, and what will they weigh at adult size? How long has the tank been established and has the filter been disturbed recently? What is already in the tank? Is it planted? Have they been quarantined together?
Likely limiting factor	Depends almost entirely on the size of the fish. For eight small characins or rasboras in a mature planted tank, unlikely to be a problem. For eight of a large-growing species, a different question entirely.
What to measure	Ammonia and nitrite daily for roughly a week after the addition. Have dechlorinated water ready for a change if anything appears.
Behavioural concerns	Adding the complete group at once has a real social advantage: no undersized-group period and no conspecific incumbency. For small shoaling species this usually outweighs the modest biological argument for staging.
Possible safe approaches	If the tank is mature and the fish are small: reduce feeding slightly for the first week, add the group after quarantine, test daily, and be prepared to change water. If the tank is young: stage the addition or wait until it is more established.
Common mistakes	Treating ‘eight fish’ as inherently a large addition. Eight is a count; the biofilter responds to biomass and feed. Also: assuming a crash is binary. A brief, small, undetectable excursion is the most likely outcome in a mature tank.

Scenario G — “Should I add three fish now and three later?”

Aspect	Assessment
What we know	The keeper is considering staged introduction of a group.
What we do not know	Species social ecology. Whether the species shows incumbency effects. Tank maturity. Whether a quarantine tank is available.
Follow-up questions to ask	Which species? Is it a peaceful shoaling species or a territorial one? How mature is the tank? Do you have a quarantine tank? What is the total group size you are aiming for?
Likely limiting factor	For small shoaling species in a mature tank, the biofilter argument for staging is weak and the social argument against it is real. For territorial species or a young tank, staging may be preferable.

Aspect	Assessment
What to measure	Ammonia and nitrite after each addition.
Behavioural concerns	Staging leaves the first three in an undersized group for the interval, which is exactly the condition the group is being expanded to fix. In territorial species, the first arrivals may also establish residency.
Possible safe approaches	For peaceful shoaling species with quarantine available: add the whole group. Without quarantine: staging gives disease observation, which may be the stronger argument. For territorial species: simultaneous introduction into a rearranged tank is often preferred.
Common mistakes	Assuming staging is always the safer option. It trades a small and manageable biological risk for a real social cost and a real incumbency risk. Also: planning to complete the group later and then not doing it, which is common.

Scenario H — “My tank can process 2 ppm ammonia in 24 hours. How many fish does that equal?”

Aspect	Assessment
What we know	A fishless cycle has demonstrated a specific oxidation capacity under test conditions.
What we do not know	How to convert fish to ppm per day, which requires biomass and feed input that the keeper does not have. Whether the capacity persists. Everything non-chemical.
Follow-up questions to ask	What species and what adult sizes? How long since the last ammonia dose? What are the tank's dimensions? What is your intended feeding regime?
Likely limiting factor	Not the biofilter, at least not initially. The conversion from fish to ammonia production is the missing step, and every other constraint is untested.
What to measure	Continue dosing until fish arrive, so the population does not decline. Then ammonia and nitrite after each addition.
Behavioural concerns	None assessed by this test at all.
Possible safe approaches	Treat the cycle result as confirmation that a functioning nitrifying community exists, which it is. Then stock according to species requirements and the twelve constraints, in stages, testing as you go.
Common mistakes	Stocking fully on the strength of the cycle result. The test does not replicate a real fish load — it omits solids, dissolved organics, continuous production and post-feeding peaks — and it says nothing about oxygen, space, territory or group size. See Section 30.2.

Scenario I — “My fish are still juveniles. Can I upgrade later?”

Aspect	Assessment
What we know	The keeper is aware that adult size will exceed current provision and is proposing a grow-out approach.
What we do not know	Whether the upgrade will actually happen. The timeline for growth. Whether the current tank is adequate for the interim period.
Follow-up questions to ask	What is the adult size and how fast do they grow? What size tank will they need? Do you have space, budget and a date for it? What will you do if the upgrade is delayed?
Likely limiting factor	Time. The plan is sound in principle and fails on execution more often than on biology.
What to measure	Growth rate, by measuring or photographing the fish periodically. Nitrate trend, which will steepen as the fish grow.
Behavioural concerns	Territoriality often appears at maturity, so the behavioural constraint may bind before the biological one. A group that is peaceful as juveniles may not be at six months.
Possible safe approaches	Grow-out is legitimate with a specific date, a specific budget and a specific location for the larger tank. Without all three it is not a plan. A useful test: if the upgrade could not happen next month for any reason, the current arrangement is the real one.
Common mistakes	Treating 'later' as a plan. Also underestimating growth rate, and not accounting for the fact that maintenance requirements rise continuously during the grow-out period rather than jumping at the end.

Scenario J — “My 20-gallon tank is tall rather than long. Does that change stocking?”

Aspect	Assessment
What we know	The tank has the same volume and a smaller footprint and less surface area than a long tank of equal capacity.
What we do not know	Which species are intended. Whether they are open-water swimmers, benthic or column-oriented.
Follow-up questions to ask	What species? Are any of them active horizontal swimmers or bottom dwellers? Is there adequate surface agitation?
Likely limiting factor	Footprint and surface area. A tall 20 has less bottom area for benthic species, less continuous swimming length for active species, and less surface area for gas exchange per unit volume.
What to measure	Early-morning behaviour for oxygen, since the reduced surface area matters most there. Standard nitrogen parameters.
Behavioural concerns	Deep-bodied species such as angelfish genuinely benefit from height. Active shoaling species and benthic groups are penalised.
Possible safe approaches	Choose species that use vertical space. Ensure good surface agitation to compensate for the reduced interface. Avoid benthic groups that need bottom area and avoid continuous swimmers.
Common mistakes	Assuming equal volume means equal suitability. It does for some species and not for others, and the difference is predictable from the species' swimming style.

Scenario K — “I have lots of plants. Can I keep more fish?”

Aspect	Assessment
What we know	The tank has substantial plant mass, which takes up ammonium and nitrate, produces oxygen during the photoperiod and provides cover.
What we do not know	Whether the plants are actively growing. Whether trimmings are removed. Light and carbon availability. Night-time oxygen behaviour. How much open swimming space remains.
Follow-up questions to ask	Are the plants growing visibly, and do you remove trimmings? What types — fast stems and floaters, or slow epiphytes? What does the tank look like at dawn? How much open water is left?
Likely limiting factor	Probably not nitrogen. Possibly night-time oxygen, and possibly open swimming space, both of which planting can make worse.
What to measure	Nitrate trend. Early-morning fish behaviour, which matters more in a densely planted tank than a sparse one. If available, dissolved oxygen before lights on.
Behavioural concerns	Dense planting improves cover and generally reduces stress, but reduces open swimming volume for active species and increases fry survival in livebearer tanks.
Possible safe approaches	A modest increase in stocking is often reasonable in a genuinely well-planted, actively growing tank. Verify the night-time oxygen picture first, since that is the constraint that planting makes worse rather than better.
Common mistakes	Assuming plants cancel out bioload. They intercept nitrogen; they do not remove oxygen demand, territory pressure or space limitations, and they add night respiration. Also: assuming a tank full of anubias and java fern is doing what a tank full of fast-growing stems does.

Scenario L — “I run two filters. Can I keep more fish?”

Aspect	Assessment
What we know	The tank has more media surface area, more mechanical capacity, more circulation and genuine redundancy against a single mechanical failure.

Aspect	Assessment
What we do not know	Whether either filter is a genuine biological contributor or mainly mechanical. Every non-filtration constraint.
Follow-up questions to ask	What types of filter and what media? Is the flow appropriate for your species? What are the tank's dimensions and what species are in it?
Likely limiting factor	Almost certainly not filtration. Two filters means the biological ceiling has been raised; the binding constraint has moved elsewhere.
What to measure	Nitrate trend. KH trend. Early-morning behaviour.
Behavioural concerns	Combined flow from two filters can be excessive for some species and reduce usable space.
Possible safe approaches	Treat the second filter primarily as redundancy and margin, which is its most valuable function. Evaluate the other constraints independently before adding stock.
Common mistakes	Assuming double filtration means double stocking. Filtration capacity and aquarium carrying capacity are different quantities, and only one of them was doubled.

Scenario M — “I have a sump. Does that increase how many fish I can keep?”

Aspect	Assessment
What we know	The system has genuinely more water volume, more media space and usually better gas exchange than the display tank alone.
What we do not know	Sump volume relative to the display. Whether it is well designed with good flow through media. The display footprint, which is unchanged.
Follow-up questions to ask	How much water does the sump actually hold in operation? What species are you considering? What are the display tank's dimensions?
Likely limiting factor	The display footprint, territory and swimming space, none of which the sump affects at all.
What to measure	Standard nitrogen parameters. The larger total volume will make these more stable, which is a real benefit.
Behavioural concerns	Fish live in the display, not the sump. Every behavioural constraint is set by the display tank's dimensions.
Possible safe approaches	Count the sump toward biological and chemical capacity, where it genuinely helps, and not at all toward spatial or behavioural capacity. Whether that permits more fish depends on which ceiling was binding.
Common mistakes	Treating total system volume as though it were display volume. A 75-gallon display with a 30-gallon sump is a 75-gallon tank for every purpose the fish care about spatially, and a 105-gallon system chemically.

Scenario N — “My parameters are perfect but the fish are fighting.”

Aspect	Assessment
What we know	The chemical constraints are satisfied. The behavioural ones are not. This is a textbook case of the taxonomy in Section 28.
What we do not know	Which fish, which behaviour, and in what context. Whether it is territorial, breeding-related, hierarchical or predatory. Tank layout.
Follow-up questions to ask	Which individuals, and is it one fish or several? Has anything changed recently — new arrivals, rearrangement, maturity, spawning? What does the layout look like, and how many sight breaks are there? What are the species and sex ratios?
Likely limiting factor	Behavioural carrying capacity. No water parameter will change it and no water change will fix it.
What to measure	Nothing chemical is needed here. The relevant observation is behavioural, sustained over time, noting which individuals are involved and when.
Behavioural concerns	This is the entire question. Look for a persistent target; look for breeding behaviour; look for whether one individual holds a contested resource.

Aspect	Assessment
Possible safe approaches	Add sight breaks and structure. Reassess sex ratios. Consider whether group size is wrong in either direction — some species need more conspecifics to distribute aggression, others need fewer. Consider rehoming the aggressor or the target. Rearranging hardscape resets territory claims.
Common mistakes	Assuming perfect parameters mean the stocking is fine. They mean the nitrogen constraint is fine. Also: adding more fish reflexively, which helps in some species and worsens matters in territorial ones.

Scenario O — “My tank is heavily stocked but I change water twice a week.”

Aspect	Assessment
What we know	The keeper is compensating for high nitrogen input with high export, which genuinely works for the constraints water changes address.
What we do not know	Whether the schedule is sustainable over years. The oxygen picture. Whether space and territory are adequate. What happens during a lapse.
Follow-up questions to ask	How long have you actually sustained this schedule? What happens when you travel? What is the tank like at dawn? Are the fish adult, or still growing? Is every individual feeding well?
Likely limiting factor	Likely maintenance sustainability, and possibly oxygen. Water changes do not address either the night-time oxygen minimum or the loss of safety margin.
What to measure	Early-morning behaviour and, if possible, dissolved oxygen before lights on. Nitrate trend, which will be well controlled. KH, which may be depleting.
Behavioural concerns	Space and territory are unaffected by water-change frequency and need separate assessment.
Possible safe approaches	Acknowledge that the schedule works while it is maintained. Then assess the failure modes: what happens during a two-week absence, a power outage, or a heatwave? Section 38 applies directly.
Common mistakes	Believing that maintenance frequency answers the stocking question. It answers part of it. The parts it does not answer — oxygen, space, behaviour, and what happens when the schedule lapses — are the parts most likely to produce a bad day.

Scenario P — “My fish look crowded but ammonia and nitrite are zero.”

Aspect	Assessment
What we know	Nitrification is keeping pace. The keeper's visual impression suggests another constraint may not be.
What we do not know	Whether the impression is accurate. Which constraint, if any, is actually binding.
Follow-up questions to ask	What makes them look crowded — restricted movement, constant near-misses, inability to rest? Are they adult or still growing? What are the tank dimensions? How do they behave at dawn?
Likely limiting factor	Possibly spatial, possibly behavioural, possibly nothing — some shoaling species naturally cluster and look crowded while being entirely comfortable.
What to measure	Nitrate trend, KH trend, and early-morning behavioural observation. Ammonia and nitrite have already told you what they can tell you.
Behavioural concerns	The key question. Are individuals able to move freely, rest undisturbed and feed without exclusion? If yes, the crowded appearance may be normal shoaling. If no, the tank is spatially or behaviourally overstocked.
Possible safe approaches	Trust the observation enough to investigate it properly rather than dismissing it because the test kit is clean. Zero ammonia rules out one of twelve constraints.
Common mistakes	Concluding that zero ammonia means the tank cannot be overstocked. It means it is not biologically overstocked. Section 28 lists five other kinds.

Scenario Q — “My bottom feeders occupy a different area. Do they count toward stocking?”

Aspect	Assessment
What we know	The species use a different zone from the midwater fish, which genuinely reduces spatial and territorial competition.
What we do not know	Which species, and what their adult sizes are. Whether the bottom area is adequate for their group.
Follow-up questions to ask	Which bottom dwellers, and what is their adult size? How many? What is the tank's bottom footprint? What substrate?
Likely limiting factor	For corydoras and similar groups, bottom footprint and group size. For large plecos, adult biomass and waste output — potentially the largest single load in the tank.
What to measure	Standard nitrogen parameters. The bottom dwellers contribute to these fully.
Behavioural concerns	Bottom dwellers compete with each other for foraging area, and some are territorial. They also need their own food, which adds to feed input.
Possible safe approaches	Count them fully toward biomass, oxygen demand and nitrogen output. Count them separately for space, using bottom area rather than volume.
Common mistakes	The folklore that bottom dwellers do not count. Using a different zone reduces spatial competition and reduces nothing else. See Section 22.2.

Scenario R — “Do shrimp count?”

Aspect	Assessment
What we know	Invertebrates contribute less per individual than fish of comparable nominal size, and are more sensitive to water quality than most fish.
What we do not know	Which species, how many, and whether the population is reproducing.
Follow-up questions to ask	Which shrimp, and are they breeding? How large is the colony now, and what was it six months ago? What are you feeding them?
Likely limiting factor	Rarely the shrimp themselves at typical colony sizes. But a large established colony has real aggregate biomass, and shrimp are usually the first casualties when another constraint is exceeded.
What to measure	Standard nitrogen parameters, with the note that shrimp are more sensitive to ammonia and nitrite than most fish. Also copper in any medication or fertiliser.
Behavioural concerns	Shrimp need cover, particularly during moulting. Predation pressure from tankmates affects colony size and behaviour.
Possible safe approaches	Count them, at a small fraction of a fish of similar nominal length, and revisit the assessment as the colony grows. Treat them as livestock with their own requirements rather than as a cleanup mechanism.
Common mistakes	Assuming zero contribution. Also assuming they will control waste — detritivores accelerate the breakdown of organic matter into dissolved nitrogen, which is useful, and remove nothing from the system.

Scenario S — “My livebearers keep having babies.”

Aspect	Assessment
What we know	Population is growing without purchases. Load is rising continuously.
What we do not know	The trajectory. Whether the keeper has an outlet for the offspring. Current sex ratio.
Follow-up questions to ask	How many fish now versus six months ago? What is your sex ratio? Do you have somewhere for the fry to go? How much cover is there?
Likely limiting factor	Population growth, which compounds. Feed input rises with it, and maintenance requirements rise in step.

Aspect	Assessment
What to measure	Nitrate trend over successive months, which will show the trajectory more clearly than a single reading. Feed quantity over time.
Behavioural concerns	Male harassment of females is a separate welfare issue driven by sex ratio rather than by numbers. Fry survival is largely a function of cover, so a heavily planted tank produces a steeper curve.
Possible safe approaches	Decide on an outlet before the population forces the issue: a shop, another keeper, another tank, or a predatory tankmate. Separating sexes or keeping a single-sex group stops the growth entirely. Reducing cover reduces fry survival, which some keepers find uncomfortable and which is how natural populations are regulated.
Common mistakes	Not planning for it at all. A livebearer tank stocked correctly on day one is stocked differently by month six, with no purchase and no decision having been made.

Scenario T — “My school is too small but I do not have room for more.”

Aspect	Assessment
What we know	A genuine conflict between a welfare requirement and a physical constraint. There is no comfortable answer.
What we do not know	How constrained the tank really is. Whether the constraint is footprint, biology, or the keeper's assumption.
Follow-up questions to ask	What are the actual dimensions? What else is in the tank? Which constraint is genuinely binding — is it that you cannot add biomass, or that the footprint is too small, or that you are working from an inch-per-gallon figure that may not apply? Is a larger tank possible at any point?
Likely limiting factor	Depends on which constraint is real. Keepers sometimes discover their tank has more headroom than assumed — particularly for small shoaling fish, where the biomass involved is minimal.
What to measure	Nitrate trend and early-morning behaviour, to establish whether there is genuine biological headroom.
Behavioural concerns	The existing group is experiencing a documented welfare cost. That cost is real and ongoing, and it should be weighed against the cost of any alternative rather than treated as the status quo.
Possible safe approaches	First, verify the constraint is real. For small shoaling fish, a few more may be biologically trivial. If the constraint is genuine: consider rehoming the group to someone who can house it properly and restocking with a species suited to this tank; or upgrading; or, as an acknowledged compromise, maximising cover, planting and water quality to mitigate the stress of the undersized group.
Common mistakes	Two opposite mistakes. One is assuming there is no headroom without checking. The other is adding fish to a tank that genuinely cannot take them, which creates a second problem without solving the first. See Section 15.2.

41. Additional Scenarios Identified During Research

These arose repeatedly during research into hobbyist discussion and are as common as the lettered scenarios above.

Scenario U — “I added fish and now I have nitrite. What do I do?”

Aspect	Assessment
What we know	The addition exceeded the nitrite-oxidising community's current capacity. Nitrite oxidisers commonly lag behind ammonia oxidisers.
What we do not know	How large the excursion is. Whether ammonia is also elevated. What was added and how much.
Follow-up questions to ask	What are the actual readings? What did you add and when? Has the filter been disturbed recently? What is your temperature and pH?
Likely limiting factor	Biofilter capacity, specifically the second nitrification step.
What to measure	Ammonia and nitrite daily until both return to undetectable. Watch fish behaviour for signs of respiratory distress.
Behavioural concerns	Nitrite impairs oxygen transport, so affected fish may show respiratory signs even at adequate dissolved oxygen.

Aspect	Assessment
Possible safe approaches	Water changes to dilute; reduce or suspend feeding to lower input; do not clean the filter; do not add more livestock; increase aeration. This usually resolves within days as the population catches up.
Common mistakes	Cleaning the filter in response, which removes the very population that needs to grow. Also continuing to feed normally, which sustains the input that caused the problem.

Scenario V — “How long should I wait between adding fish?”

Aspect	Assessment
What we know	The keeper wants an interval. Section 11 established that no universal constant is supportable.
What we do not know	Tank maturity, existing load, the size of each addition, temperature, pH, plant mass, and whether quarantine is available.
Follow-up questions to ask	How mature is the tank? How large is each addition relative to what is already there? Do you have a quarantine tank? What is your nitrate trend?
Likely limiting factor	Varies. In a mature, planted, well-buffered tank the biofilter interval may be short; the disease-observation interval may be the real constraint.
What to measure	Ammonia and nitrite after each addition. The test result, not the calendar, is the signal that the system has accommodated the change.
Behavioural concerns	For social species, long intervals mean long periods of undersized groups and increased incumbency effects.
Possible safe approaches	Test-driven rather than calendar-driven: add, test daily, and when readings have been stable for several days with no excursion, the biofilter has accommodated it. Separately, allow the quarantine period the disease question requires.
Common mistakes	Applying a fixed interval regardless of context. Also assuming the interval serves only the biofilter, when for most keepers the disease-observation function is more valuable.

Scenario W — “Can I keep a single fish of a schooling species if I have no room for a group?”

Aspect	Assessment
What we know	A social species would be kept in the condition that controlled work identifies as producing the worst welfare indicators.
What we do not know	Which species, and how strong its social requirement is.
Follow-up questions to ask	Which species? What are the tank's dimensions? What is the actual constraint preventing a group?
Likely limiting factor	Social requirement. In the Saxby study, single fish and very small groups of shoaling species showed elevated aggression and darting and reduced shoaling; in zebrafish, the lowest density produced the highest cortisol and aggression.
What to measure	Nothing chemical. This is entirely a behavioural question.
Behavioural concerns	The core issue. A solitary shoaling fish cannot express its primary social behaviour.
Possible safe approaches	Choose a species suited to the space instead. Many attractive species are genuinely suited to small tanks or to being kept singly. This is a species-selection problem, not a compromise problem.
Common mistakes	Assuming a single fish is the low-impact option. For a shoaling species it is the highest-impact option available on the welfare axis, even though it is the lowest on the bioload axis.

Scenario X — “My tank has been fine for three years at this stocking level.”

Aspect	Assessment
What we know	The system has demonstrated stability under normal operation for a meaningful period. That is real evidence and should not be dismissed.
What we do not know	What 'fine' means — survival, or normal behaviour and condition. What the margin looks like. Whether the fish are adult. Whether anything has been quietly suboptimal.
Follow-up questions to ask	Are the fish fully grown? Is every individual in good condition and feeding? What is their lifespan and how far through it are they? What happened the last time the power went out or you went on holiday?
Likely limiting factor	Possibly nothing. Possibly safety margin, which is invisible until it is tested.
What to measure	Early-morning behaviour. Nitrate trend. Honest condition assessment of each individual.
Behavioural concerns	'Fine' assessed by mortality is a different standard from 'fine' assessed by behaviour and lifespan. Chronic effects such as suppressed growth or shortened lifespan are not visible as events.
Possible safe approaches	Take the evidence seriously — three years is real. Then ask the 'should' question from Section 39 alongside the 'can' question, and separately assess what would happen during a disruption.
Common mistakes	Both directions. Dismissing three years of successful keeping is unreasonable; so is treating survival under normal conditions as proof that all twelve constraints are satisfied.

Scenario Y — “The shop said this fish would be fine in my tank.”

Aspect	Assessment
What we know	A retail recommendation has been given. Its quality is unknown and highly variable.
What we do not know	Whether the advice accounted for adult size, group needs, footprint or compatibility.
Follow-up questions to ask	What species exactly, including the scientific name if possible? What adult size did they quote? Did they ask about your tank dimensions and existing stock?
Likely limiting factor	Frequently adult size. Species sold small that grow large are a persistent problem across the trade.
What to measure	Nothing yet. Verify the species and its adult size independently first.
Behavioural concerns	Compatibility with existing stock is frequently the second issue.
Possible safe approaches	Verify the species identity and adult size from an independent reference before purchase. Good retailers ask about tank size and existing stock before recommending; that question is a reliable quality signal.
Common mistakes	Assuming retail advice is either always reliable or always wrong. Quality varies enormously between shops and between individual staff.

Scenario Z — “Can I stock more heavily if I use a canister filter rated well above my tank plus an air stone plus lots of plants?”

Aspect	Assessment
What we know	The keeper has raised several ceilings at once: biological processing, gas exchange and nitrogen uptake. Those are genuine improvements.
What we do not know	The footprint, which is unchanged. The territorial capacity, which is unchanged. The species involved and their adult sizes.
Follow-up questions to ask	What species, and what are their adult sizes and group requirements? What are the tank's dimensions? What is your realistic maintenance schedule?
Likely limiting factor	Almost certainly footprint, territory or maintenance — the constraints that equipment cannot raise.
What to measure	Nitrate trend, KH trend, early-morning behaviour.
Behavioural concerns	Heavy equipment provision does not change how much room a fish has to swim or how many defensible sites exist.

Aspect	Assessment
Possible safe approaches	Accept that the biological ceiling has genuinely risen and evaluate whether it was ever the binding one. In most home aquariums with adequate filtration, it was not.
Common mistakes	Stacking improvements to the same ceiling and assuming the total capacity rises proportionally. Three improvements to the nitrogen constraint do not move the spatial one at all.

42. Myths, Rules and Controversies

Each claim is classified and explained. The classification describes what kind of claim it is, not how popular it is. Several claims classified as oversimplified or unsupported nonetheless produce acceptable outcomes for beginners, and that is noted where it applies.

42.1 — Counting rules

Claim	Evidence classification	Assessment and better explanation
"One inch of fish per gallon."	Useful rule of thumb within narrow limits; misleading outside them	Works acceptably for small, slender, similarly shaped community fish in a conventionally proportioned tank with moderate feeding. Fails badly for deep-bodied fish, large-growing fish, territorial fish and heavily fed tanks, because mass scales with roughly the cube of length (Froese 2006) and because it ignores behaviour entirely. Better explanation: total body length is a crude proxy for biomass only within a narrow class of similarly shaped fish.
"One fish per gallon."	Unsupported aquarium folklore	Discards body size, which is the only variable the inch rule was tracking. Only defensible if 'fish' silently means a one-inch tetra, which is exactly the assumption a beginner cannot make. Better explanation: there is no meaningful sense in which a fish is a unit of load.
"Small fish barely count."	Plausible but oversimplified	Individually true — a neon tetra weighs on the order of 0.2 to 0.3 g. But mass-specific metabolic rate is higher in smaller fish (Clarke & Johnston 1999), so ten one-gram fish have a higher combined resting oxygen demand than one ten-gram fish. Large numbers of small fish also have real spatial and social requirements. Better explanation: small fish have low load and non-trivial requirements, and these are different axes.
"Bottom dwellers do not count."	Unsupported aquarium folklore	They respire, feed, excrete nitrogen and contribute biomass exactly as any other fish does. Some are territorial, and a common pleco is among the highest-load fish sold to beginners. Better explanation: occupying a different zone reduces spatial and territorial competition and reduces nothing else.
"Shrimp do not count."	Plausible but oversimplified	Individual dwarf shrimp contribute very little. Established colonies of several hundred do not. Shrimp are also more sensitive than most fish to ammonia, nitrite and low oxygen, so they are often the first indicator that another constraint has been exceeded. Better explanation: low individual load, real aggregate load, high sensitivity.
"Snails do not count."	Misleading as stated	Highly species-dependent. Adult mystery snails are substantial animals with real feeding and waste output. Nerites are moderate and do not reproduce in freshwater. Pest snail populations are usually a symptom of overfeeding rather than an independent load. Better explanation: count snails by size and by whether the population reproduces.

42.2 — Equipment and plants

Claim	Evidence classification	Assessment and better explanation
"Plants cancel out bioload."	Plausible but oversimplified	Plants take up ammonium directly and often preferentially, take up nitrate, and add oxygen during the photoperiod — all real. They also respire at night, reduce open swimming volume, and do nothing for territory, footprint or adult size. Uptake requires active growth, so slow-growing epiphytes contribute far less than fast stems and floaters. Better explanation: plants raise the nitrogen ceiling, make the night-time oxygen picture worse, and leave the spatial and behavioural ceilings untouched.
"A bigger filter means more fish."	Plausible but oversimplified	A bigger filter raises ammonia processing capacity, solids removal and circulation. It does not raise oxygen saturation, swimming space, territory, or dilution against nitrate accumulation. Better explanation: it raises one of twelve ceilings, and only helps if that ceiling was the binding one.

Claim	Evidence classification	Assessment and better explanation
"Double filtration means double stocking."	Misleading	Even the biological capacity does not double reliably, since flow distribution and media utilisation are not additive in a simple way. And the constraint most likely to be binding in a well-filtered home aquarium is not filtration. Better explanation: the second filter is most valuable as redundancy and margin.
"A sump lets you stock like a bigger display tank."	Misleading	A sump adds genuine system water volume, media space and gas exchange — all real and all valuable. It adds zero display space, zero footprint and zero territory. Better explanation: total system volume and display volume are different quantities and the fish only experience one of them.
"A filter rated for 50 gallons lets me stock a 20-gallon like a 50."	Unsupported; the rating is a manufacturer recommendation with no independent standard	Filter ratings are not standardised and do not specify assumed fish biomass, feeding rate, temperature or maintenance interval. Better explanation: a filter rating rates the filter, not the livestock.

42.3 — Water testing and stocking

Claim	Evidence classification	Assessment and better explanation
"Low nitrate means you can add more fish."	Misleading	Nitrate can be low because of heavy planting, frequent water changes, recent testing timing, low feeding, denitrification, or test kit error — as well as because of low stocking. And it says nothing about oxygen, space, territory or group size. Better explanation: the rate of nitrate rise between water changes is informative; a single value is close to uninformative.
"If ammonia and nitrite are zero, the tank is not overstocked."	Misleading	Zero ammonia and nitrite establish that nitrification is keeping pace with nitrogen input. That is one of six distinct forms of overstocking in Section 28. Better explanation: the standard test kit measures the constraint most likely to already be fine and none of the constraints most likely to be binding.
"Overstocking is only a water-quality problem."	Unsupported	Spatial, territorial, social, oxygen and maintenance constraints are all independent of water chemistry, and several of them cannot be improved by any water treatment. Better explanation: overstocking is at least six different conditions and only one of them is chemical.
"Perfect parameters mean perfect stocking."	Unsupported	As above. This is the single most common reasoning error in hobby stocking discussion. Better explanation: parameters describe the water, not the fish's experience of the space.
"TDS tells you how heavily stocked a tank is."	Contested and generally over-claimed	TDS is non-specific and cannot distinguish waste-derived accumulation from source water hardness, fertiliser, or mineral concentration from evaporation top-offs. Useful as a trend within one known system, weak as a general stocking indicator.

42.4 — Group size and social claims

Claim	Evidence classification	Assessment and better explanation
"Schooling fish should always be kept in groups of six."	Useful rule of thumb; unsupported as a biological threshold	No source establishing six as a threshold was located. Controlled work on ornamental species tested 1, 2, 5 and 10 and found 10 best for the small shoaling species examined (Saxby et al. 2010) — six is not a data point. Better explanation: six is a sensible conservative floor for beginner guidance and should not be presented as a finish line or as a measured threshold.
"Six is the magic number."	Unsupported aquarium folklore	As above, with the additional problem that framing it as magic converts a useful convention into a claim about biology. Better explanation: group-size requirements are species-specific and the evidence that exists does not identify a universal number.

Claim	Evidence classification	Assessment and better explanation
"More schooling fish always means happier fish."	Plausible but oversimplified	Well supported at the low end: undersized groups show elevated aggression, elevated cortisol and reduced shoaling. Not supported without limit: larger groups increase resource competition, and the zebrafish density literature found costs at high densities as well as low ones. Better explanation: there is a species-specific window, and both ends of it matter.
"Crowding reduces aggression."	Contested claim	Genuinely observed in some systems: Nile tilapia at high density were less aggressive than at low density. But the same study found those fish were darker in skin colour (a stress proxy), less active, and showed gene expression consistent with a reactive stress-coping style and activation of the stress axis. Better explanation: reduced aggression is not necessarily improved welfare, and a suppressed fish is not a content one.
"Cichlid tanks should be overstocked."	Contested claim; common aquarium practice	The practice of controlled crowding to distribute aggression in rift lake cichlids is widespread and does reduce visible violence. Whether it improves welfare is a genuinely open question given the tilapia findings above, and it demands filtration, oxygenation and maintenance that most keepers underestimate. Better explanation: it is a technique with a real mechanism, real costs and a contested welfare balance — not a general licence.
"A tank can be understocked and that is a problem."	True in one sense, false in the other	Biologically, low stocking is never a problem — it is a safety margin. Socially, a schooling species below its functional group size is a genuine and documented welfare problem. Better explanation: separate the two senses explicitly, because they lead to opposite advice.

42.5 — Adding fish

Claim	Evidence classification	Assessment and better explanation
"Never add a full school at once."	Common aquarium practice; overstated as an absolute	The biological step is a function of added biomass and feed, not of animal count. Eight small tetras add roughly two grams. For a mature, planted tank the biological argument is weak and the social argument for a complete group is real. Better explanation: the risk scales with proportional load increase and with tank maturity, not with the number of animals.
"Only add two or three fish at a time."	Useful rule of thumb; the specific numbers are unsupported	No evidence base was located for two, three, one week or two weeks. The underlying principle — smaller step changes are more easily absorbed — is sound. Better explanation: keep additions small relative to existing load, and note that the interval is doing disease-observation work as well as biofilter work.
"Wait two weeks between every addition."	Common aquarium practice; unsupported as a constant	Adjustment time depends on step size, existing population, temperature, pH, alkalinity, surface area, oxygen and plant uptake. Better explanation: test-driven rather than calendar-driven. When ammonia and nitrite have been stable for several days, the system has accommodated the change.
"You should always add fish slowly."	Plausible but oversimplified	Staged addition has real benefits (smaller steps, disease observation) and real costs (prolonged undersized social groups, territorial incumbency favouring early arrivals, repeated disturbance, and the risk the plan is never completed). Better explanation: the right approach depends on species social ecology, tank maturity and whether quarantine is available.
"A cycled tank can handle any reasonable addition."	Plausible but oversimplified	'Cycled' means a nitrifying community exists sized to the load it has been receiving. A lightly loaded filter has a small population and correspondingly small absolute reserve. Better explanation: reserve capacity is proportional to established load, not to the fact of being cycled.
"If your tank was fishless-cycled to 2 ppm ammonia, you can fully stock immediately."	Contested claim; the strong version is not supportable	The test does not replicate a real fish load, the keeper cannot convert fish to ppm per day, populations decline without continued substrate, and ammonia capacity is one of twelve constraints. Better explanation: the cycle result confirms a working nitrifying community, which is valuable, and is not a stocking permit. See Section 30.2.

Claim	Evidence classification	Assessment and better explanation
"Adding fish always causes an ammonia spike."	Unsupported as an absolute	Mature biofilters operate below saturation and typically absorb modest additions with no detectable excursion. Better explanation: whether an excursion is measurable depends on the size of the step relative to spare capacity, and hobby test kits have limited resolution at low concentrations.

42.6 — Tools and tank format

Claim	Evidence classification	Assessment and better explanation
"Stocking calculators are accurate."	Misleading	They are good at species knowledge — adult size, minimum tank length, temperament, group thresholds — and weak at system modelling. They cannot observe feeding, maintenance, aquascape, filter maturity, individual temperament or local water chemistry. No independent validation studies were located. Better explanation: a screening and warning tool, not a measurement.
"Stocking calculators are useless."	Misleading, in the opposite direction	The species database is genuinely valuable and encodes information a beginner does not have. Dismissing the tool wholesale discards its most useful function. Better explanation: use the warnings, treat the percentage as a prompt.
"Tall tanks hold as many fish as long tanks of equal volume."	Misleading	Equal volume, unequal footprint, unequal surface area and unequal swimming length. Tall tanks favour deep-bodied species and penalise active swimmers and benthic groups. Better explanation: footprint predicts suitability better than volume for most freshwater community stocking.
"Goldfish need 20 gallons each."	Useful rule of thumb without a traceable derivation	No research derivation for the specific figure was located. The reasoning behind it — large adult size, very high waste, high oxygen demand, decade-scale longevity and sociality — is sound. Better explanation: present the reasoning as evidence-based and the figure as a convention.
"A labelled 10-gallon tank holds 10 gallons."	Misleading	Substrate, hardscape, equipment and fill level typically reduce it to roughly 8.5 to 9 gallons in service. Better explanation: it matters modestly, mostly in small tanks, and mostly for dilution and thermal stability rather than for the constraints that usually bind.

42.7 Additional claims identified during research

Claim	Evidence classification	Assessment and better explanation
"Overstocking is fine if you do enough water changes."	Plausible but oversimplified	Water changes address nitrate, dissolved organics, alkalinity and phosphate. They do not address the night-time oxygen minimum, footprint, territory, group size, or what happens during a lapse in the schedule. Better explanation: maintenance raises the chemical ceiling only, and the schedule has to survive holidays and illness.
"Bottom feeders will clean up the waste."	Unsupported aquarium folklore	Detritivores accelerate the breakdown of particulate organic matter into dissolved nitrogen. That is useful for cycling material through the system and removes nothing. Most commonly sold bottom dwellers also require their own dedicated food. Better explanation: nothing removes nitrogen from a closed system except export.
"Fish only grow to the size of their tank."	Unsupported aquarium folklore	Growth suppression under poor conditions is real but is a sign of stress and poor welfare, not a benign adaptation, and skeletal growth often continues while organ development is compromised. Better explanation: stunting is a welfare problem being described as a feature.
"Air stones increase how many fish you can keep."	Plausible but oversimplified	Surface agitation genuinely improves gas exchange and raises the oxygen ceiling toward saturation. It cannot exceed saturation, which falls with temperature, and it does nothing for space, territory or nitrogen accumulation. Better explanation: it can move a tank from oxygen-limited to not oxygen-limited, and no further.

Claim	Evidence classification	Assessment and better explanation
"My fish look happy, so the stocking is fine."	Anecdotal observation	Behavioural observation is genuinely valuable and is the only tool available for several constraints. But chronic effects such as suppressed growth, shortened lifespan and elevated baseline stress are not visible as events. Better explanation: observation is necessary and not sufficient, and it should be structured — is every individual feeding, resting and moving normally?
"Wild fish live in huge rivers, so any tank is a compromise."	True but not action-guiding	Correct and unhelpful as stated. Many aquarium species occupy small seasonal pools, dense vegetation or restricted territories in the wild, and habitat volume is a poor guide to requirements. Better explanation: what matters is whether the animal can perform its normal behavioural repertoire, not how much water its ancestors had access to.
"You can tell a tank is overstocked by looking at it."	Anecdotal observation with some validity	Experienced keepers do detect real problems visually, and the impression should be investigated rather than dismissed. But it is unreliable for oxygen (invisible), maintenance sustainability (a fact about the future) and shoaling species that naturally cluster. Better explanation: a prompt to investigate, not a diagnosis.
"Adding fish slowly builds up the biofilter gradually."	Reasonable inference; correct as far as it goes	This one is broadly right. Nitrifier populations do expand in response to sustained substrate supply, so incremental increases are absorbed more readily than a single large step. The overstatement is in the precision of the intervals, not in the principle.

43. Current Hobbyist Questions and Pain Points

Recurring questions observed across aquarium forums, community discussion, retail guidance and consumer-facing aquarium publications. Community discussion is evidence of where people are confused. It is not scientific evidence of what is true, and nothing in this section should be read as such.

A caveat on method: this survey is qualitative. It reflects the recurring shape of public aquarium discussion rather than a systematic sample, and no frequency claims are made.

Table 43.1 — Recurring hobbyist pain points

The question	Why hobbyists are confused	The conflicting advice they encounter	What the evidence suggests	What remains uncertain
How many fish can I keep?	The question has no answer as asked, but it is always answered anyway, which teaches people that a number exists.	Inch-per-gallon; fish-per-gallon; calculator percentages; 'it depends' with no follow-up; confident species-specific numbers from experienced keepers who are implicitly assuming their own tank format and maintenance.	Capacity is set by whichever of roughly twelve constraints is lowest, and that varies between tanks. Species identity and tank footprint determine more than volume does.	There is no validated framework for home aquaria, and no consensus reference values even for common species.
Is my tank overstocked?	The word means six different things (Section 28) and the asker and responder are often discussing different ones.	'Your parameters are fine, you are fine' versus 'that is way overstocked' on the same thread, both from experienced keepers.	Both can be correct about different constraints. Naming which form of overstocking is being discussed resolves most of these disagreements.	Where the boundary sits between acceptable and unacceptable is partly a values judgement and should be labelled as one.
Can I add more fish?	The keeper usually offers one piece of evidence — a nitrate reading, a filter rating, or a calculator output — and expects it to settle the question.	'Your nitrates are low, go ahead' versus 'you are already overstocked.'	No single measurement answers it. The eight-point assessment in Section 30.3 is a more complete approach.	How much headroom is enough is not quantified anywhere for home tanks.
Should I add more schooling fish?	It appears to conflict directly with the advice not to add too many fish at once, and both pieces of advice come from credible sources.	'Never add a whole school at once' versus 'your school is too small and that is cruel.'	They address different variables: rate of nitrogen change versus final social composition. For small shoaling fish in mature tanks they usually do not conflict at all.	Species-specific thresholds are largely unknown outside the handful of species that have been studied.
How many schooling fish should I keep?	'Six' is repeated so consistently that it sounds like a finding rather than a convention.	Six as a minimum; six as sufficient; 10, 12 or 20-plus from more experienced keepers; retail advice that sometimes recommends fewer.	Group size genuinely affects welfare and the effect is species-specific. The best-supported aquarium study tested up to ten and found ten best for the small shoaling species examined.	Where benefits plateau, for almost every species. Six has no traceable derivation.
Can I add the whole school at once?	The biological risk is framed in units of fish count, which is the wrong unit.	Absolute prohibitions versus experienced keepers who routinely do it.	Risk scales with added biomass and feed relative to existing load, and with tank maturity. Quarantining the group together removes the strongest argument against.	No controlled study of aquarium introduction strategy was located.

The question	Why hobbyists are confused	The conflicting advice they encounter	What the evidence suggests	What remains uncertain
How long between adding fish?	Multiple confidently stated intervals circulate, none of them derived.	One week, two weeks, a month, 'until parameters are stable.'	No universal constant is supportable. Test-driven is better than calendar-driven. The interval also serves disease observation, which may matter more.	Adjustment time is genuinely system-specific and spans a wide range.
Do bottom feeders count?	The visible logic — they are in a different place — is conflated with biological load.	'They barely count' versus 'a pleco is the biggest load in your tank.'	They count fully for biomass, oxygen and nitrogen. They count differently for space, using bottom area rather than volume.	Little uncertainty here. This is one of the clearer cases in the document.
Do shrimp and snails count?	The animals are small and their contribution is genuinely low per individual, which makes the aggregate easy to overlook.	'They do not count' versus 'count them like fish.'	Low individual load, real aggregate load in established colonies, and high sensitivity to water quality. Mystery snails are substantial animals.	Quantitative contribution figures for ornamental invertebrates are largely unavailable.
Does filtration increase stocking?	'Capacity' is used for two different things — ammonia processing and aquarium carrying capacity.	'Overfilter and you can stock heavily' versus 'filtration is not the limit.'	It raises the biological ceiling and no other. Whether that permits more fish depends entirely on which ceiling was binding.	Filter ratings are unstandardised, so even the biological gain cannot be quantified from the packaging.
Does a planted tank allow more fish?	Plants visibly improve water quality metrics, which is read as general capacity.	'Plants cancel bioload' versus 'plants make no difference.'	Plants take up ammonium directly and often preferentially, and export nitrogen when trimmings are removed. They also respire at night and reduce open volume.	Aquarium-scale uptake rates are not quantified in any usable way for hobbyists.
Is my stocking calculator right?	The output is a single confident-looking number produced by an undisclosed model.	'Trust the calculator' versus 'calculators are garbage.'	Strong on species knowledge, weak on system modelling. Best used as a screening and warning tool.	No independent validation of any aquarium stocking calculator was located.

43.1 The pattern underneath the pain points

Nearly all twelve questions share a structure: the keeper has one piece of information and wants to know whether it settles a question that actually requires several. The nitrate reading, the filter rating, the calculator percentage, the fish count — each is a real measurement of a real thing, and each is being asked to answer a question about a different thing.

That observation may be more useful for content development than any of the individual answers, because it suggests the most valuable thing to teach is not a rule but a habit: **ask which constraint the evidence actually speaks to.**

EVIDENCE: ANECDOTAL OBSERVATION OF COMMUNITY DISCUSSION; THE PATTERN ITSELF IS AN INFERENCE

44. Search-Intent Findings

How people phrase these questions when they type them, and what the phrasing reveals about what they actually want. No search volume figures are given, because reliable volume data was not available and inventing it would be worse than omitting it.

Table 44.1 — Query families and underlying intent

Query family	Representative phrasings	Underlying intent	What a good answer must do
Capacity by tank size	how many fish can I put in my tank; how many fish in a 10 gallon tank; how many fish in a 20 gallon tank; how many fish in a 29 gallon tank; how many fish in a 40 gallon tank	Wants a number. Almost always has already bought or chosen the tank, and often has specific species in mind but has not said so.	Must give something actionable rather than only 'it depends.' The most useful move is to reframe around species and footprint while still providing concrete worked examples for that tank size.
The rules themselves	fish per gallon rule; one inch per gallon rule; is the inch per gallon rule true	Has encountered the rule and is checking it. Often already suspects it is unreliable.	Should explain where it came from and where it works, not just declare it a myth. The historical framing is more persuasive and more useful than dismissal.
Tools	aquarium stocking calculator; best aquarium stocking calculator; is AqAdvisor accurate	Wants a tool, or wants validation of a result a tool produced.	Should explain what calculators do well and badly rather than endorsing or attacking one.
Diagnosis	is my aquarium overstocked; how do I know if my fish tank is overstocked; signs of overstocked aquarium	Suspects a problem. Often has been told something by a community and is checking.	Should distinguish the six forms of overstocking, since the person may be looking for the wrong signs entirely.
Adding livestock	can I add more fish to my tank; how many fish can I add at once; how long between adding fish to aquarium; will adding fish crash my cycle; can I add 10 fish at once	Ready to buy. Wants permission or a constraint.	Should give a decision procedure rather than a number, and should distinguish the biofilter question from the disease question.
Equipment claims	does a bigger filter mean more fish; do plants increase aquarium stocking capacity; does a sump increase stocking	Has made or is considering an equipment investment and wants to know what it buys.	Should separate the ceiling the equipment raises from the ceilings it does not.
What counts	do shrimp count toward bioload; do snails count toward bioload; do bottom feeders count toward stocking	Trying to complete a mental calculation. The framing assumes a single additive quantity exists.	Should answer the literal question and then question the additive model, because that model is the deeper problem.
Group size	how many tetras should be kept together; how many corydoras should be kept together; minimum school size aquarium fish; do tetras need to be in groups	Wants a species-specific number and will accept one.	Should give a defensible range with the reasoning, and should be honest that six is a convention rather than a finding.

44.1 What the phrasing reveals

- **Tank-size queries dominate and are the least answerable.** People search by the thing they have already bought. Any content that refuses to engage with 'how many fish in a 20 gallon' loses the audience before it can reframe the question.
- **'Can I' queries are permission-seeking.** The person has usually decided and is looking for support or a stop signal. That means the answer needs to be clear about which it is.
- **'Do X count' queries reveal the additive mental model.** The person believes stocking is a single quantity to which each animal contributes a share. Correcting that model is more valuable than answering the specific question.

- **Rule-checking queries indicate existing doubt.** Someone searching 'is the inch per gallon rule true' has already encountered a contradiction. They are receptive to nuance in a way that a first-time searcher is not.
- **Group-size queries are species-anchored.** People search 'how many corydoras' rather than 'minimum school size,' which suggests species-specific content is more useful than general principles here.

EVIDENCE: ANECDOTAL OBSERVATION OF PUBLIC SEARCH PHRASING; NO VOLUME DATA CLAIMED

45. Additional Questions and Topics Identified During Research

Questions the original research brief did not include, which came up repeatedly and which the eventual episode may want to address.

Table 45.1 — Questions the brief did not include

Question	Why it matters	What the evidence suggests	Uncertainty
Does stocking level affect fish lifespan?	Most stocking discussion is framed around survival and water quality. Lifespan is the outcome most keepers actually care about and it is almost never discussed.	Chronic stress is associated with metabolic cost and immune modulation across vertebrates, and aquaculture literature links poor conditions to impaired growth. A link to shortened lifespan is plausible.	Direct evidence for ornamental species in home aquaria is essentially absent. This should be presented as a plausible mechanism rather than a finding.
Does stocking level affect growth rate and adult size?	Growth suppression under crowding is well documented in aquaculture and is the mechanism behind the ‘fish grow to their tank’ folklore.	Density effects on growth are documented in cultured species. Suppressed growth is a stress response, not a benign adaptation.	Magnitude in home aquaria is unquantified. Whether stunted fish suffer specific pathologies is discussed in the hobby more confidently than the evidence supports.
Does alkalinity depletion set a stocking limit in soft-water areas?	Nitrification consumes alkalinity. In areas with very soft tap water, KH can be drawn down to the point where pH becomes unstable and nitrification itself is impaired.	The chemistry is established. The practical consequence is that soft-water keepers have a lower stocking ceiling than hard-water keepers at identical everything else, which is rarely stated.	How commonly this is the binding constraint in practice is unknown.
Does carbon dioxide injection interact with stocking?	Injected carbon dioxide depresses pH, competes with the need for surface agitation, and high dissolved carbon dioxide impairs fish oxygen uptake and ammonia excretion.	The mechanisms are established. A heavily stocked, carbon-injected planted tank is balancing several constraints simultaneously in a narrow window.	No aquarium-specific guidance combining stocking and carbon injection was located.
How does stocking affect the keeper's experience over time?	Heavily stocked tanks require more work, offer less margin, and produce more crises. Burnout and hobby abandonment are real outcomes with real welfare consequences for the animals.	No formal evidence located. But maintenance capacity is a documented constraint in this document, and a tank the keeper stops enjoying is a tank at risk.	Entirely uninvestigated. Possibly a valuable angle precisely because it is uninvestigated.
Does adding fish disrupt an established hierarchy?	Introduction disturbs existing social structure, not just water chemistry, and this is discussed far less than the cycle question.	Prior-residency effects are well documented, and territorial incumbency is real. Section 31.4 covers what is known.	Species-specific magnitude is largely unknown for ornamental species.
Should stocking change with the seasons?	Room temperature varies. A tank at the edge of its oxygen ceiling in winter is over it in summer.	The physics is unambiguous (Section 19). Reducing feeding and increasing agitation seasonally is a reasonable practice.	Not discussed as a stocking question in any source located, only as a heatwave emergency response.
Does tank shape affect stocking more than any other single variable?	Footprint determines swimming distance, bottom area, surface area and territory distribution simultaneously.	The argument in Section 33 suggests footprint may be the single most informative physical variable, more so than volume.	No formal comparison of tank geometry against stocking outcomes was located. This is an inference from mechanism.
How does stocking interact with medication?	Many medications reduce dissolved oxygen, harm nitrifying bacteria, or are lethal to invertebrates. A heavily stocked tank has less margin for any of these.	Well established that some treatments affect oxygen and biofiltration. The stocking interaction follows from Section 38.	Not quantified. Worth raising because a keeper is most likely to medicate at exactly the moment their margin is already reduced.

Question	Why it matters	What the evidence suggests	Uncertainty
Is there a difference between stocking a display tank and a breeding or grow-out tank?	Purpose-built temporary systems operate under different assumptions, much closer to aquaculture practice.	Higher densities are routine and defensible in short-term, closely monitored grow-out contexts. Applying those densities to a display tank is the error.	The boundary between the two is not well articulated anywhere in hobby literature.

45.1 Adjacent topics worth flagging

- **The retail supply chain.** Stocking decisions are made at the point of sale, frequently on the basis of what is available and what is recommended in the moment. Any content aimed at improving stocking outcomes competes with a purchasing environment that is optimised differently.
- **The upgrade path as a hobby norm.** The grow-out-then-upgrade pattern is extremely common and rarely discussed honestly. It works when planned and fails silently when not.
- **Rehoming infrastructure.** Several answers in this document depend on a keeper being able to rehome fish. Whether that is realistically available varies enormously by location, and advice that assumes it is less useful than advice that acknowledges it may not be.
- **The gap between the 'can' and 'should' communities.** Sections of the hobby operate on explicitly different standards, and each finds the other unreasonable. Naming this explicitly may be more productive than arguing within it.
- **Measurement access.** The single largest practical gap identified in this research is that hobbyists cannot measure dissolved oxygen, which is plausibly the most frequently binding constraint. Inexpensive dissolved oxygen measurement would change hobby stocking practice more than any amount of better advice.

46. Glossary

Term	Definition
Allometric scaling	A relationship in which one quantity changes as a power of another rather than in direct proportion. Fish mass scales with roughly the cube of length; metabolic rate scales with roughly mass to the power 0.79.
Ammoniotelic	Excreting nitrogenous waste primarily as ammonia. Most teleost fish are ammoniotelic.
Behavioural carrying capacity	The number of individuals a system can hold without chronic behavioural conflict, independent of its chemical capacity.
Biofilter / biological filtration	The nitrifying microbial community that oxidises ammonia to nitrite and nitrite to nitrate. Present on all wetted surfaces, not only in the filter.
Bioload	Hobby term for the total waste-generating burden of livestock and their food. Not a precisely defined scientific quantity.
Biomass	Total wet mass of living animals in the system.
Carrying capacity	The maximum load a specific system can sustain indefinitely under a specific management regime.
Comammox	Complete ammonia oxidation performed by a single organism, carrying out both nitrification steps. Comammox <i>Nitrospira</i> have been found to be abundant in established freshwater aquaculture biofilters.
Dilution effect	The reduction in an individual's predation risk as group size increases, because the probability of being the one taken falls.
Dither fish	Aquarium term for active, visible fish added to signal environmental safety to shyer species.
Dissolved organic carbon (DOC)	Soluble organic material from waste, food and decay. Drives heterotrophic bacterial oxygen consumption and is not measured by standard hobby test kits.
Heterotrophic bacteria	Bacteria that consume organic carbon. Faster-growing than nitrifiers and often the largest oxygen consumers in a heavily fed tank.
Incumbency (prior residency)	The advantage a resident individual holds in a territorial contest against a newcomer.
Isometric growth	Growth in which body proportions remain constant, corresponding to a length-weight exponent of exactly 3.
Nitrification	The aerobic microbial oxidation of ammonia to nitrite and nitrite to nitrate. Consumes oxygen and alkalinity.
Q_{10}	The factor by which a biological rate changes for a 10 degree C temperature rise. A Q_{10} of 2 means the rate doubles.
RAS (recirculating aquaculture system)	An engineered closed or semi-closed fish production system with continuous filtration, oxygenation and monitoring.
Rh glycoproteins	Rhesus family proteins in the gill epithelium that facilitate ammonia movement, central to the current model of branchial ammonia excretion.
Schooling	A shoal in which individuals are polarised — aligned and moving with coordinated speed and heading.
Shoaling	Any social aggregation of fish for social reasons, without a requirement for alignment or coordination.
Social facilitation	The tendency of an individual's behaviour to be influenced by the presence and behaviour of others, including using their calm as evidence of safety.
Standard / resting metabolic rate	The baseline metabolic rate of a resting, unfed, unstressed fish at a given temperature. Actual demand in an aquarium is always higher.
Stocking density	Livestock per unit volume or footprint. Expressed as kg per cubic metre in aquaculture and as fish per litre in laboratory husbandry.
TAN (total ammonia nitrogen)	The combined concentration of un-ionised ammonia and ammonium ion. Standard aquaculture unit; corresponds to what hobby ammonia tests measure.
TDS (total dissolved solids)	Total dissolved ionic content, usually estimated from conductivity. Non-specific.
Un-ionised ammonia (NH_3)	The acutely toxic form of ammonia. Its proportion of TAN rises with pH and temperature.

47. Key Evidence Summary

The load-bearing findings in this document, with their classification, for rapid retrieval.

Finding	Source	Classification
Fish mass scales with approximately the cube of length. Median length-weight exponent across 3,929 relationships covering 1,773 species is 3.03.	Froese 2006, J. Appl. Ichthyol. 22:241–253	Strong evidence
Resting metabolic rate in teleost fish scales with body mass to a mean exponent of 0.79 (SE 0.11) across 69 species; overall between-species exponent near 0.80.	Clarke & Johnston 1999, J. Anim. Ecol. 68:893–905	Strong evidence
A typical tropical fish at 30 deg C requires approximately six times the resting oxygen of a polar fish at 0 deg C. Mean Q_{10} about 1.83 across species; within-species acclimation median about 2.40.	Clarke & Johnston 1999	Strong evidence
Group size affects welfare in ornamental species in a species-specific way. Neon tetras and white cloud mountain minnows showed reduced aggression and darting and more shoaling in larger groups, with 10 the best of the sizes tested. Angelfish showed no clear link. Small tiger barb treatments were terminated on ethical grounds.	Saxby et al. 2010, Appl. Anim. Behav. Sci. 125:195–205	Strong evidence for species tested
Species composition changes group-size effects: tiger barbs were less aggressive toward smaller tankmates when angelfish were present.	Sloman et al. 2011, Appl. Anim. Behav. Sci.	Strong evidence for the assemblage tested
Zebrafish held at 1 fish per litre showed higher aggression, higher water cortisol and altered vertical distribution than fish at 4 to 16 fish per litre.	Zebrafish holding density study, 2022	Strong evidence
Zebrafish reared at 1 fish per litre showed significantly higher aggression and cortisol than at 3 or 6 fish per litre.	Frontiers in Behavioral Neuroscience 2023	Strong evidence
Holding density affects zebrafish growth, reproduction, stress response and behaviour; evidence across studies is partly inconsistent.	Andersson & Kettunen 2021, Zebrafish, systematic review	Strong evidence with acknowledged inconsistency
Nile tilapia reared at high density were less aggressive but darker (stress proxy), less active, and showed gene expression consistent with a reactive stress-coping style and stress-axis activation.	Transcriptomic study of aquaculture intensification in Nile tilapia, 2019	Strong evidence; central caveat on the crowding argument
Approximately 1 kg of 32 percent crude protein feed generates about 30 g of ammonia nitrogen entering the water.	Ebeling et al. 2006, widely applied in RAS design	Aquaculture practice
In a documented commercial RAS, about 35 percent of feed nitrogen was assimilated into fish flesh and about 65 percent was excreted or unassimilated, at roughly 62 g excreted N per kg fish produced.	Commercial RAS nitrogen budget, IntechOpen 2013	Aquaculture practice
Teleosts excrete nitrogen predominantly as ammonia across the gills. In juvenile rainbow trout: 53–68 percent ammonia-N, 6–10 percent urea-N, 4–10 percent amino acids, 3–11 percent protein.	Rainbow trout nitrogen partitioning study, J. Exp. Biol. 2004	Established principle
Branchial ammonia excretion involves facilitated diffusion through Rh glycoproteins whose expression is induced under ammonia loading.	Wright & Wood; Weihrauch et al. 2009, J. Exp. Biol. 212	Established principle
In RAS biofilter start-up, ammonia oxidisers became detectable over weeks and nitrite-oxidising <i>Nitrospira</i> developed more slowly.	Roalkvam et al. 2021, Aquaculture Research	Strong evidence
Established freshwater RAS biofilters host complex communities including ammonia-oxidising archaea and comammox <i>Nitrospira</i> , not only <i>Nitrosomonas</i> .	Bartelme et al. 2017, Front. Microbiol. 8:101	Strong evidence
Oxygen solubility in freshwater falls from about 14.6 mg/L at 0 deg C to about 9.1 at 20 deg C, 8.2–8.3 at 25 deg C and about 7.0 at 35 deg C.	FAO water quality reference; standard environmental references	Established principle

Finding	Source	Classification
Culture systems should generally not be allowed to fall below 50 percent of oxygen saturation; response is governed by the overnight minimum, with monitoring recommended between midnight and dawn.	Boyd, Global Seafood Alliance / Responsible Seafood Advocate	Aquaculture practice
Larger shoals detect an approaching predator sooner and sustain foraging longer; foraging latency falls and foraging rate rises with shoal size even without a predator.	Magurran et al. 1985; Godin et al. 1988; Morgan 1988; Pitcher et al. 1982	Strong evidence
Individual predation risk declines with school size in accordance with the dilution hypothesis.	Godin & Morgan 1985, Behav. Ecol. Sociobiol. 16	Strong evidence
Wild bronze corydoras live in mixed-age, mixed-size groups, observed as small as three, with very low aggression and tactile communication.	Riley et al. 2019, Ethology; Corydoras developmental social experience study 2024	Strong evidence
Male Betta splendens in larger tanks were significantly more active, swimming on average 92 seconds more during observation periods than those in 1.5 L jars. Authors recommend a minimum of 5.6 L for retail display and larger for home keeping, with gravel, live plants and refuges.	Clark-Shen et al. 2024, Animal Welfare	Strong evidence
Male bettas in larger aquaria performed fewer approaches and aggressive displays and more bouts of foraging.	Animal Welfare 2024, betta space and enrichment study	Strong evidence
Environmental structure can reduce aggression by delimiting territories and can increase it by raising resource value; effects documented in mice, zebrafish and cichlids.	Cichlid enrichment and social dominance literature	Contested
Aquatic plants take up ammonium directly and generally preferentially over nitrate, with leaf uptake from the water column a significant route.	Aquatic macrophyte nitrogen uptake literature	Established principle
Surface-area stocking rules of roughly 12 square inches per inch of slender fish and 20-plus for deep-bodied fish appear in aquarium reference literature.	Scott, The Complete Aquarium, as applied in aquarium stocking references	Common aquarium practice
No traceable evidential derivation was located for: the inch-per-gallon rule; six as a minimum group size; two-week addition intervals; or 20 gallons per goldfish.	This document's own search	Unsupported as measurements; several remain useful rules of thumb

48. Bibliography

Sources consulted. Where a claim in this document could not be traced to a reliable source, the document states so explicitly rather than supplying a citation. No citations have been fabricated. Where a source was consulted through a secondary reference or an abstract rather than the full text, that is noted.

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48.10 Claims for which no source could be found

Stated explicitly, in line with the brief's instruction not to fabricate citations.

- **The origin of the one-inch-per-gallon rule.** No primary source deriving it from data was located. It appears in mid-twentieth-century beginner literature and retail guidance without derivation.
- **Six as a minimum group size.** No scientific source establishing six as a threshold for any species was located.
- **Specific addition intervals.** No evidence base was located for two fish, three fish, 10 percent, 20 percent, one week or two weeks.
- **Twenty gallons per goldfish.** No research derivation for the specific figure was located, though the reasoning behind conventions of this kind is well supported.
- **Filter rating standards.** No independent standard defining what a 'rated for X gallons' claim assumes was located.
- **Validation of aquarium stocking calculators.** No independent study assessing the accuracy of any aquarium stocking calculator was located.
- **Quantitative bioload figures for ornamental invertebrates.** No usable published figures for shrimp or snail waste production in home aquaria were located.

Note on source quality for downstream use

The strongest evidence in this document comes from three places: fish physiology (allometry, metabolism, nitrogen excretion), aquaculture engineering (feed-based nitrogen budgets, biofilter behaviour, dissolved oxygen management), and behavioural ecology and welfare science (group size, density, aggression, enrichment).

The weakest evidence concerns the things hobbyists most want numbers for: how many fish fit in a specific tank, how long to wait between additions, and what group size a specific species needs. Those gaps are real and should be represented as gaps rather than filled with confident-sounding conventions.

The most defensible position available to downstream content is therefore: **be precise about the mechanisms, honest about the numbers, and explicit about which constraint any given piece of advice addresses.**